IDEA for Teachers: Tutorial

Kolibri, Nextcloud, Wikipedia offline, Files Disk and Backup Disk

For Justice and the teachers he trains. Version 1.0, 13 October 2026

IDEA - offline school computing.
No real passwords are included. Demo passwords come from your IDEA contact.

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Part A: For teachers

1. What IDEA is

IDEA (Initiative for Digital Education in Africa) is a charity based in Belgium. It puts offline computer systems in rural schools. Everything works from a small computer in the school, without internet.

> **No internet needed.** Everything in this guide works inside the school. Nothing you do is sent to the internet.

1.1 Words used in this guide

Word	Meaning
IDEA box	The small Raspberry Pi computer in the school that runs the apps. (On the demo it is named `wizardly-hugle`; your school's box has its own name.)
Console	The web page that lists the apps and opens them.
IDEA shortcut	The one shortcut on a classroom computer's desktop. It finds the IDEA box and opens the Console (chapter 2).
System Disk	The IDEA box's own disk. The apps live on it.
IDEA Disk	A USB disk or stick that the Console has prepared. It can become a **Files Disk** or a **Backup Disk**.
Spare disk	An IDEA Disk that is still empty.
Files Disk	An IDEA Disk for your own files. It appears as the folder **School Files** in Nextcloud (chapter 9).
Backup Disk	An IDEA Disk that keeps copies of the apps' data (chapter 10).
USB stick	Your own USB stick (FAT32 or exFAT), filled on any computer. IDEA only reads it and never changes it (chapter 5).
Channel builder	The IDEA app that turns a folder of your videos and PDFs into a Kolibri channel (chapter 5).
Operator	The person at the school who looks after the IDEA box, with the IDEA contact on the phone (Part B).

1.2 The apps

App	What you use it for	Chapters
Kolibri	Ready-made lessons, videos and exercises (for example Khan Academy). You make classes, lessons and quizzes and see each learner's progress. With the **channel builder** you turn your own files into a Kolibri channel.	3, 4, 5
Nextcloud	Your own files: share worksheets with a class, write documents together in **Nextcloud Office**, collect homework.	6, 7, 9

App	What you use it for	Chapters
Kiwix	Wikipedia and other reference books, offline.	8

1.3 How to use this guide

- **Part A (chapters 1 to 10)** is for teachers.
- **Part B (chapters 11 to 13)** is for the school operator.
- Each chapter has numbered steps with screenshots. The screenshots come from the IDEA demo box and use the demo school **IDEA Demo School** with class **Grade 5** and the demo accounts in the appendix (`admin`, `teacher1`, `learner1` to `learner3`; teachers and the admin have a password: the account name followed by `911!`; learners sign in in Kolibri with only their username). Your school's names will be different.

2. Getting connected and opening an app

All apps open from one web page: the **Console**. How you get to the Console depends on the computer you use.

2.1 Connect to the Wi-Fi

The classroom computers are already connected. Do this only on your own laptop, tablet or phone.

The IDEA router at your school makes two Wi-Fi networks. Their names and first passwords are the same at every IDEA school:

Network	Password (until your school changes it)	Who joins it	What it reaches
IDEA	`idea911!`	Students, and the classroom computers	Only the IDEA box: the Console and its apps. No internet.
IDEA-Teachers	`idea-teachers911!`	Teachers only	The IDEA box, and the internet. Internet is **on**; your IDEA contact can switch it off when the head teacher asks.

- Give students only the **IDEA** password. Keep the **IDEA-Teachers** password among the teachers.
- **Change the passwords later.** These first passwords are easy on purpose, but anyone who knows them can join. Once IDEA is running at your school, the head teacher chooses new passwords and asks the IDEA contact to set them on the router. After that, every phone and laptop must join again with the new password.

To connect:

1. Open the Wi-Fi settings on your device.
2. Choose **IDEA-Teachers** (teachers) or **IDEA** (students).
3. Type the password and join.

2.2 Open the Console

There are two ways to open the Console. Use the one that fits the device in front of you.

2.2.1 On a classroom computer

The classroom computers have **one** shortcut on the desktop, named **IDEA**. There are no separate icons for Kolibri, Nextcloud or the other apps.

1. Double-click **IDEA** on the desktop. You can also find it in the app menu. It opens in Chrome.

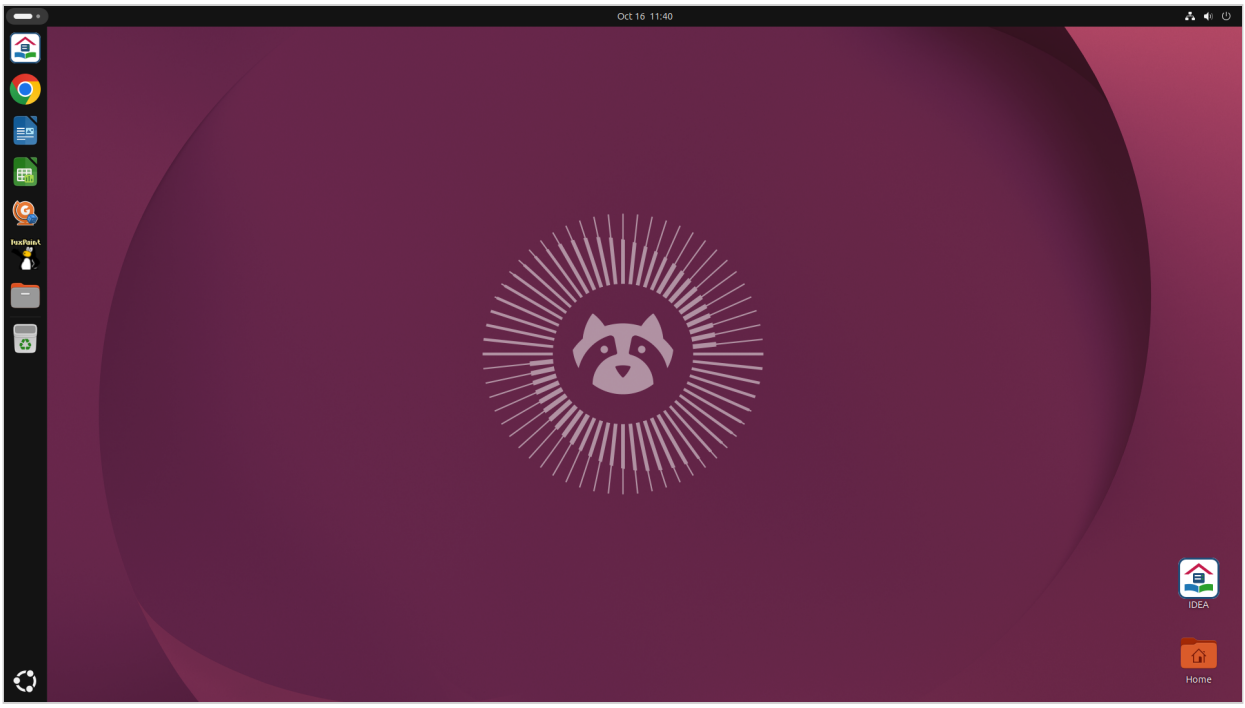


Figure 1. The classroom computer desktop with the one IDEA shortcut

2. A start page titled **IDEA** appears for a moment and says **Looking for the school server…**.

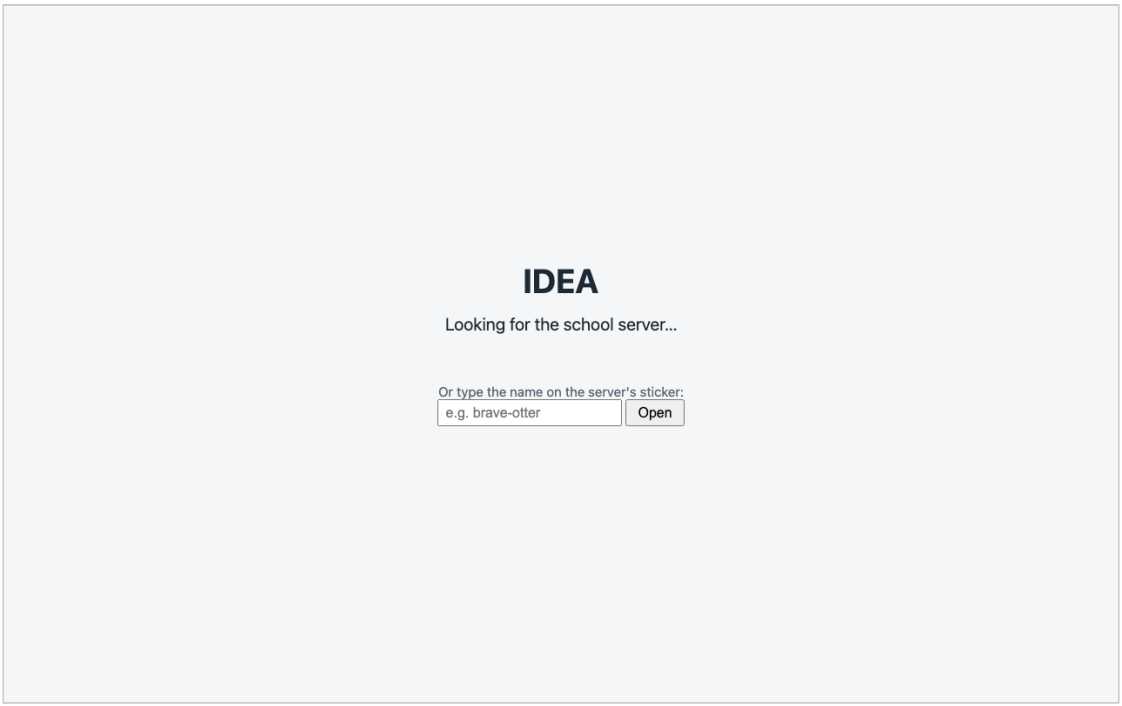


Figure 2. The IDEA start page: Looking for the school server…

3. The Console opens by itself.

If the start page says **No school server was found**, check that the computer is connected to the school network. Then type the first line of the box's sticker, for example `brave-otter`, in the field **Or type the name on the server's sticker**. Use small letters, with `-` between the words. Press **Open**. The start page remembers that IDEA box.

2.2.2 On your own laptop, tablet or phone

1. Connect to the Wi-Fi (2.1).
2. Open the browser.
3. Type `http://idea.local` in the address bar and press Enter.
4. The Console opens.

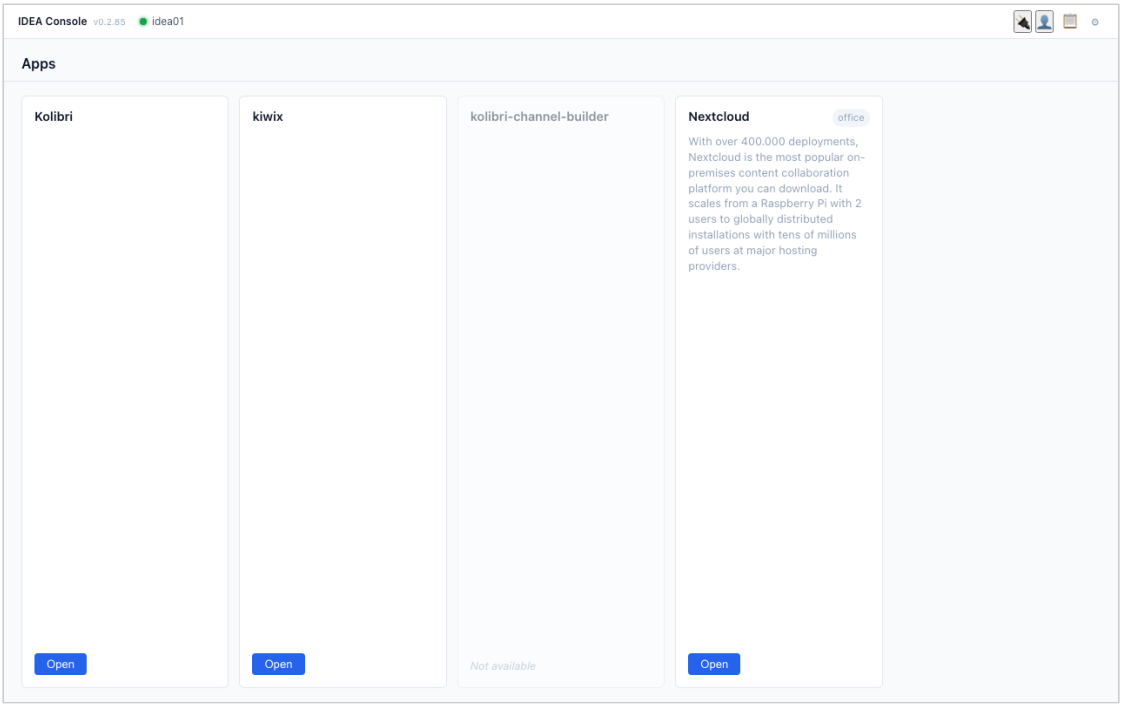


Figure 3. The Console in the browser

5. Bookmark it, so you can open it again with one click. In Chrome, press **Ctrl+D** (on a Mac ⌘+**D**), then **Done**.

If the Console does not open, type the address on the second line of the box's sticker instead, for example `http://brave-otter.local`.

> **Two IDEA boxes at your school?** The Console shows the apps of all of them.

2.3 First steps in the Console

1. The Console opens on the **Apps** page: one card per app, each with a short description and a blue **Open** button. The green dot at the top shows the Console is connected to the IDEA box.

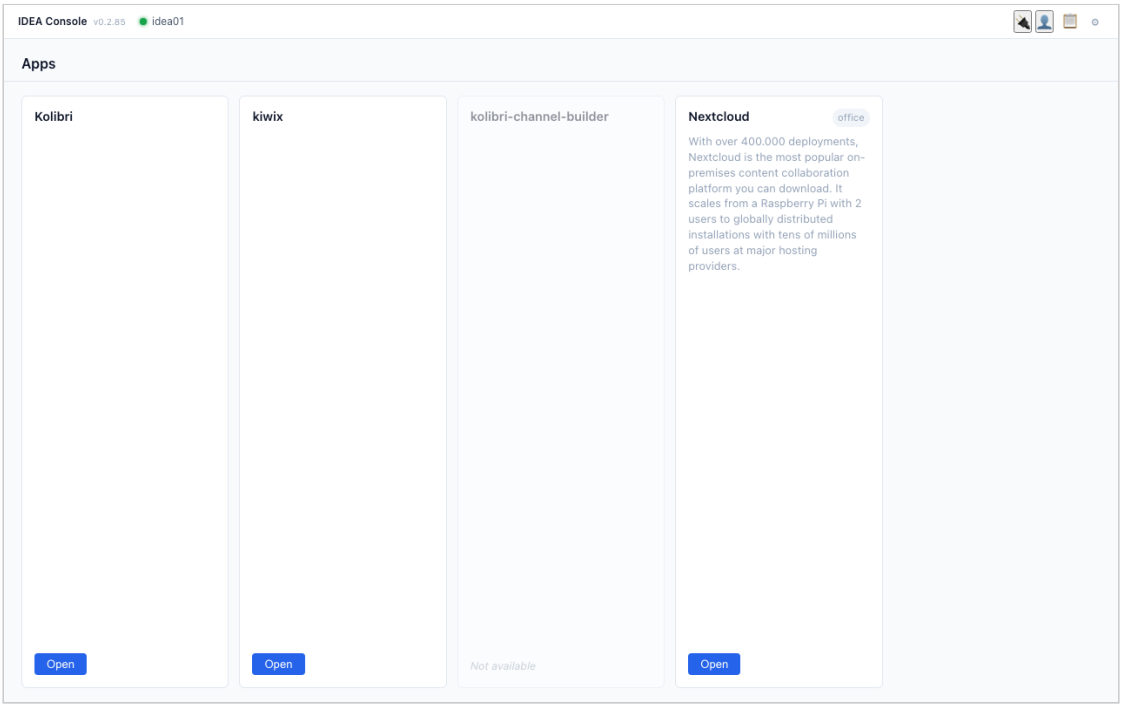


Figure 4. Console Apps page (two of the cards)

2. Click the blue **Open** button on the card of the app you need. The app opens in a new tab. Here it is Kolibri's sign-in page.

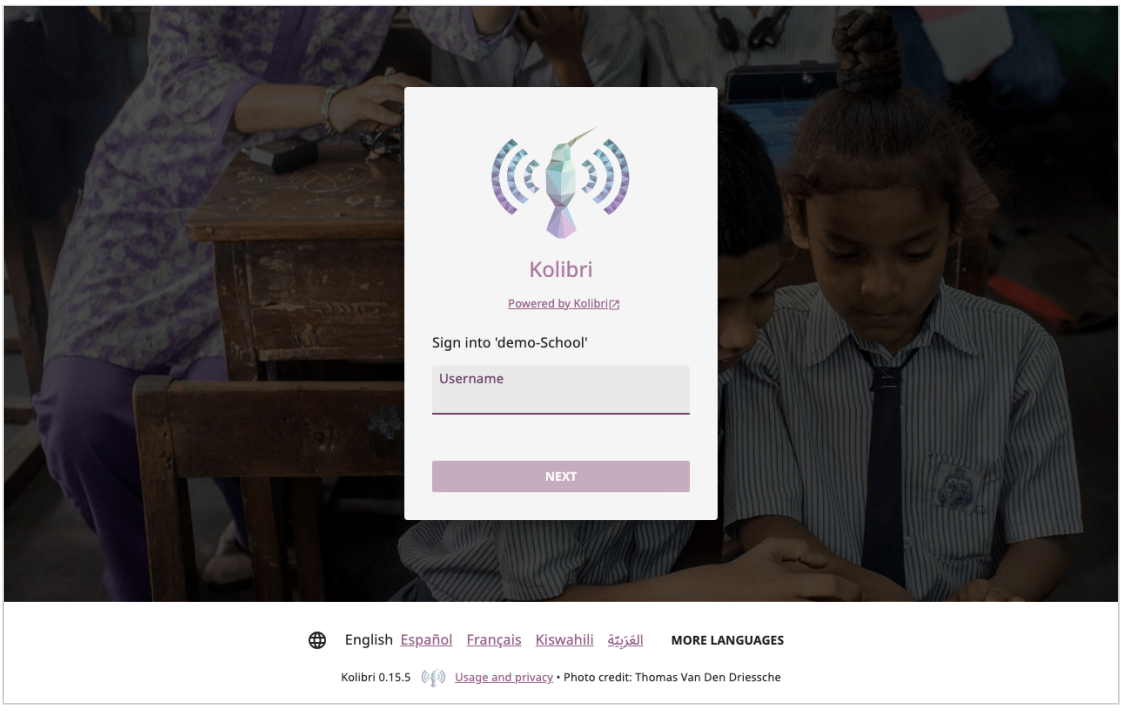


Figure 5. Kolibri opened from the Console

Good to know:

- A card that is greyed out and says **Not available** is not running. Do not try to fix it: tell your school operator.
- Do not type numbers (addresses or ports) into the address bar. Always start from the Console and use **Open**.
- A student computer that no longer works properly is not repaired: the operator resets its card (chapter 13). Students lose only what they saved on the computer itself, so let them save their work in Nextcloud (chapter 7).
- The person icon at the top right (**Operator Login**) is for the operator (Part B). Teachers do not need it.

> **Tip.** You can bookmark the page that opens. If the bookmark stops working, go back to the Console and use **Open** again.

3. Kolibri for teachers

About the examples in this chapter. The screens in this chapter come from the IDEA demo setup. That setup is switched on (activated) on the demo IDEA box. It creates a Kolibri school named **IDEA Demo School**, with the class **Grade 5** and the demo accounts. The steps work the same at your school, but your school, class and account names will be different. How to turn on the demo setup is described in Appendix: activating the demo setup at the end of this tutorial. The demo accounts are listed in Appendix: demo accounts at the end of this tutorial.

In the demo, you sign in as `teacher1`. The demo school has three channels from the IDEA-DEMO disk: **Maths**, **Grade 5 Science** and **Early Reading**. The demo lesson uses **Maths > Fractions**. The quiz in 3.5 uses **Khan Academy (English - US curriculum) > Math by grade > 5th grade > Add and subtract fractions**.

3.1 Signing in

Open Kolibri from the Console (2.3). Type your username and click **NEXT**, then type your password.

3.2 Create a class

Creating classes needs a Kolibri admin account (in the demo school, `admin`, password `admin911!`; `teacher1` is the coach of Grade 5).

1. Open **Facility > CLASSES**. You see the list of classes.

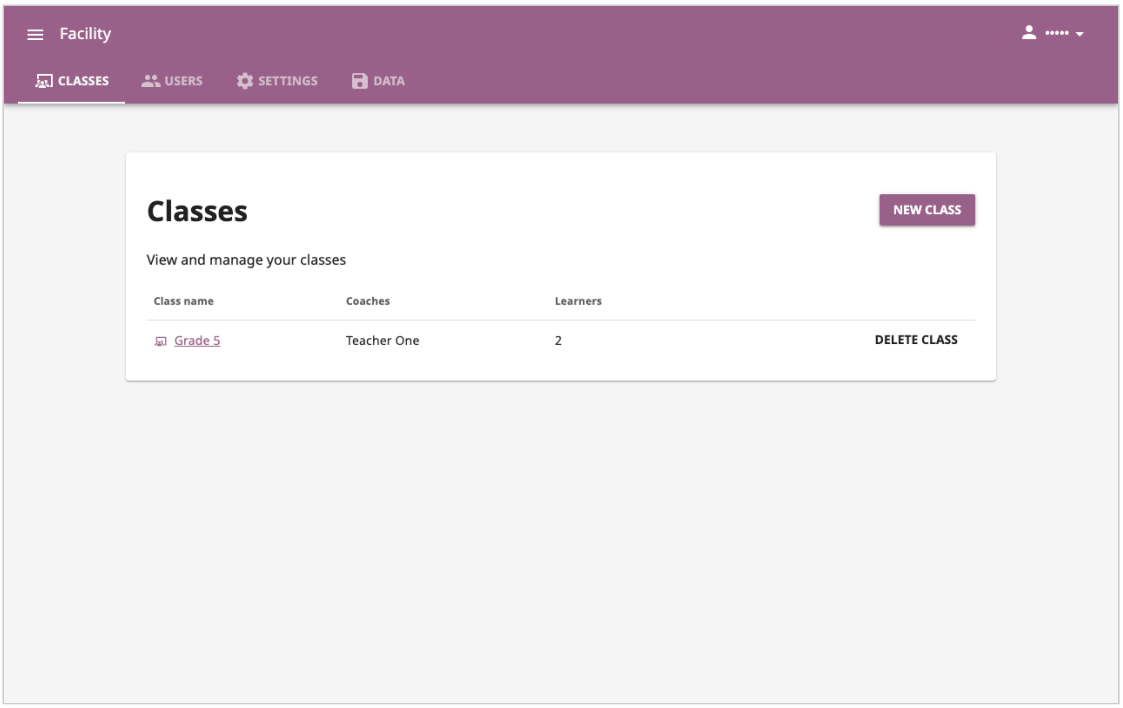


Figure 6. Classes list

2. Click **NEW CLASS**, type the class name, click **SAVE**. (In the demo we pressed **CANCEL** because `Grade 5` already exists.)

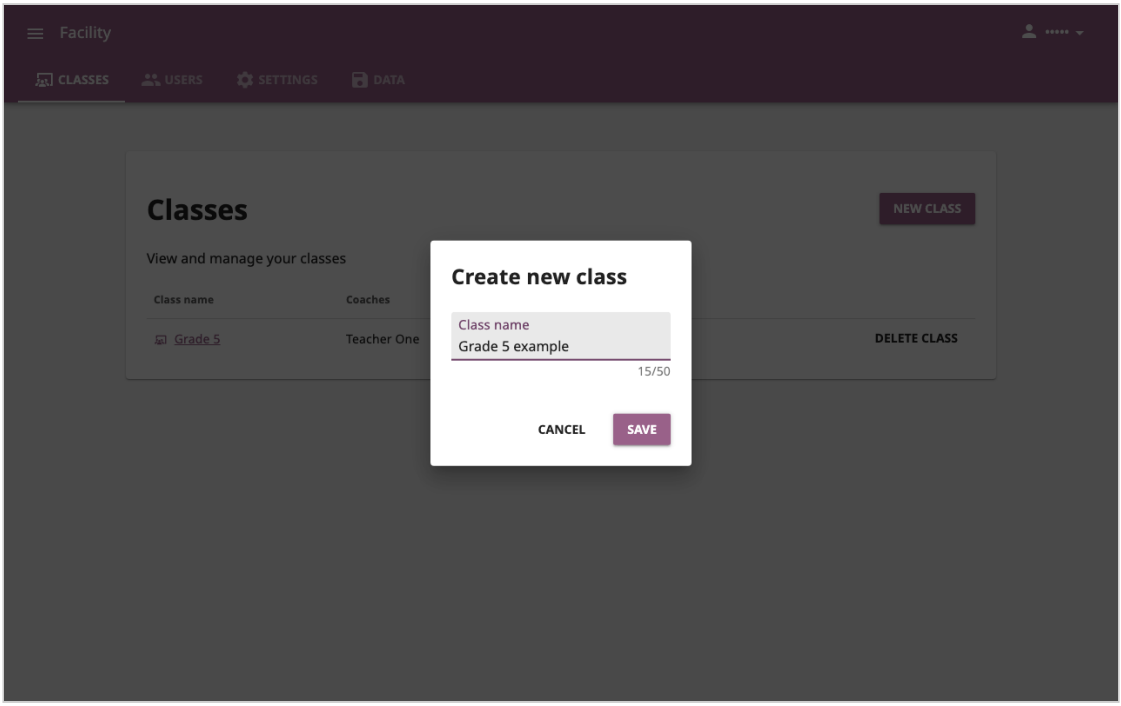


Figure 7. Create new class

3.3 Enroll learners

1. Click the class name. On the class page (with **Coaches** and **Learners**), click **ENROLL LEARNERS**. The **Enroll learners** page opens.

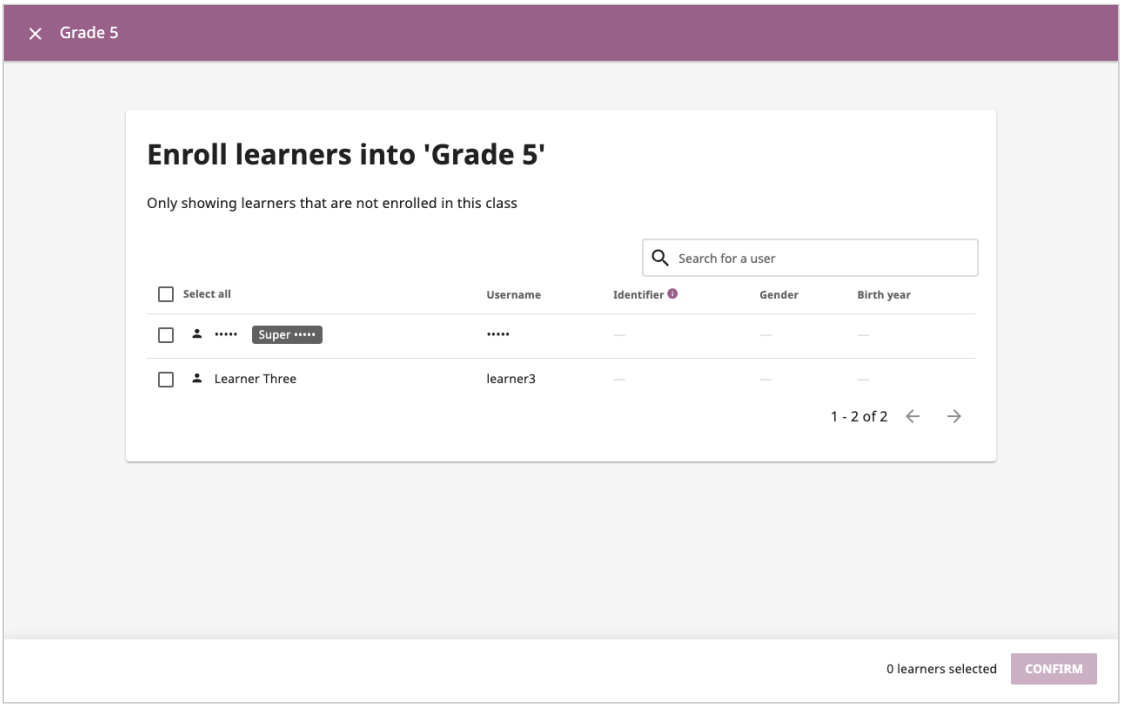


Figure 8. Enroll learners page

2. The list only shows learners who are not yet in the class. Tick each learner (here: Learner Three), then click **CONFIRM**.

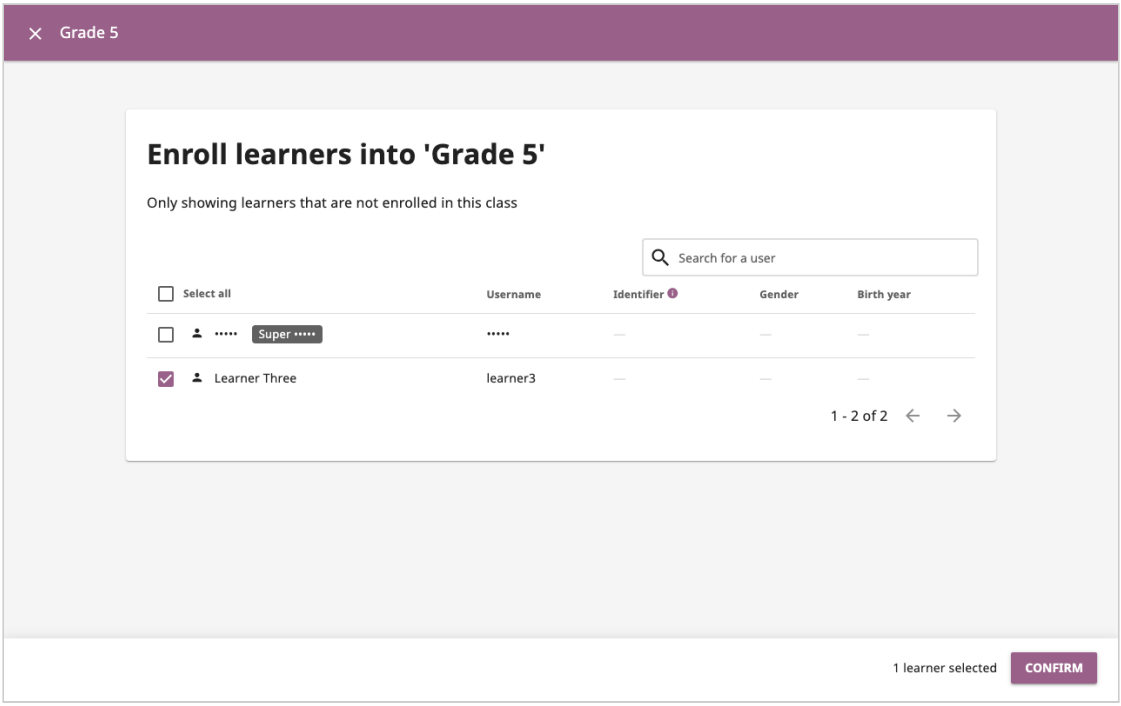


Figure 9. Learner ticked

3. The learner now appears under Learners.

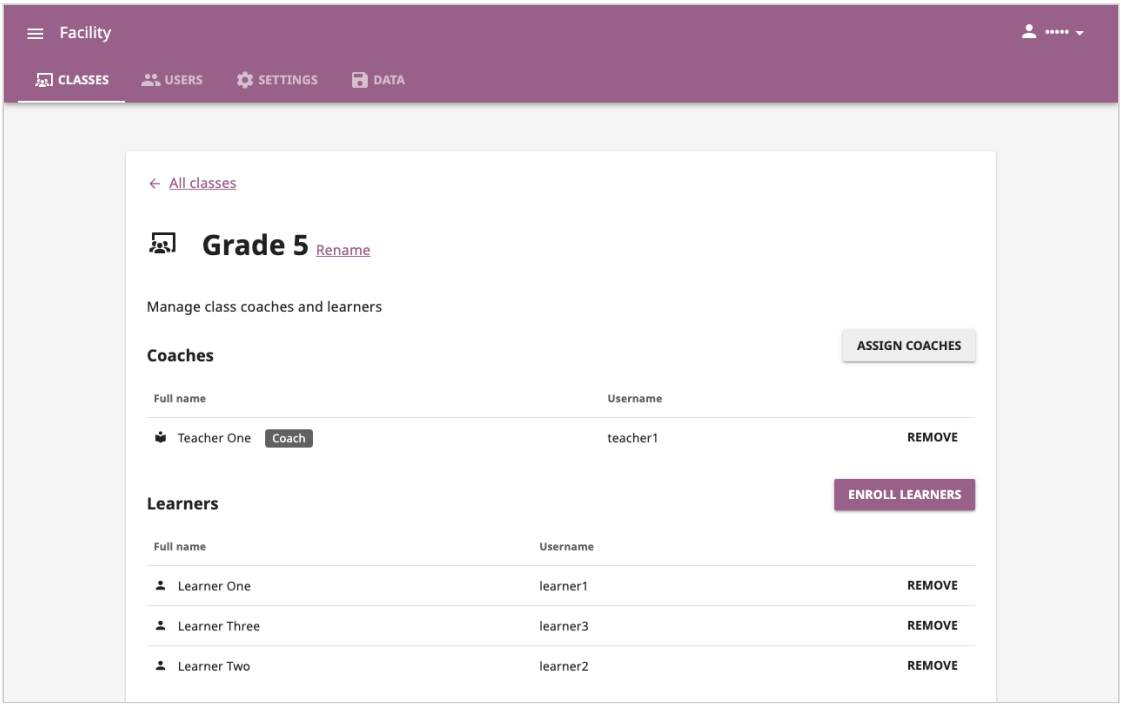


Figure 10. Learner enrolled

3.4 Make a lesson and show it to the class

A good lesson mixes resources that explain (videos, activities) with resources that let learners practise (exercises).

1. Open **Coach**, choose the class, click **PLAN > LESSONS > NEW LESSON**.

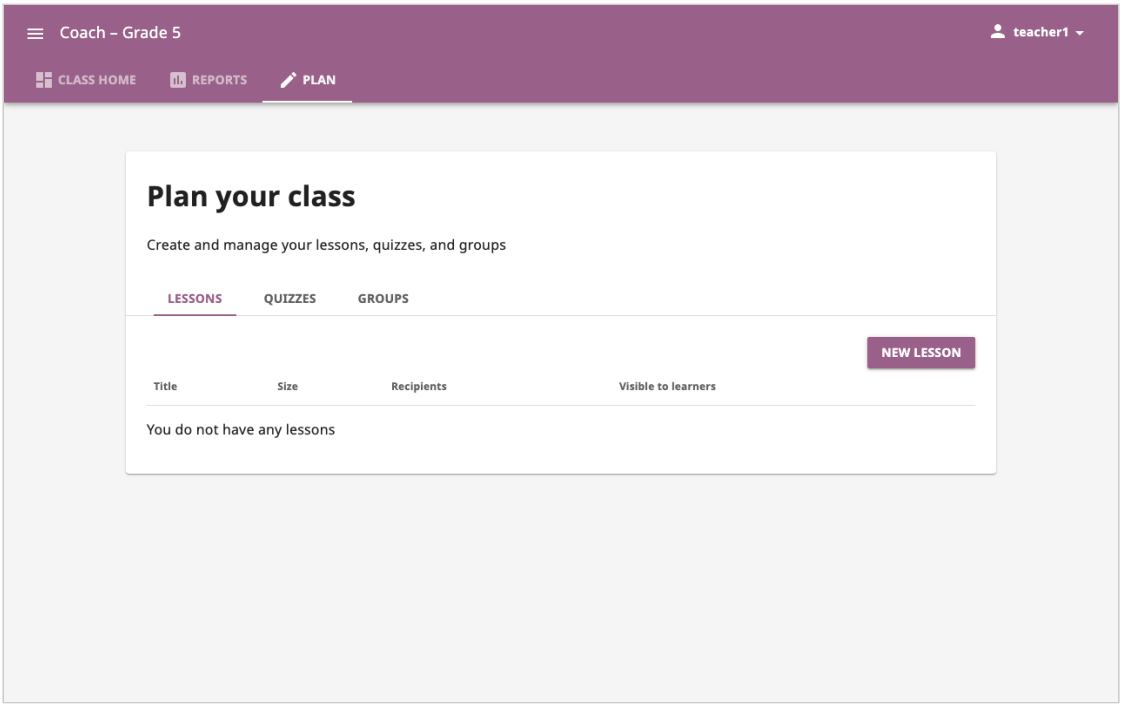


Figure 11. Plan your class

2. Type a title (`Fractions lesson`). Under **Recipients** keep **Entire class**. Click **SAVE CHANGES**.

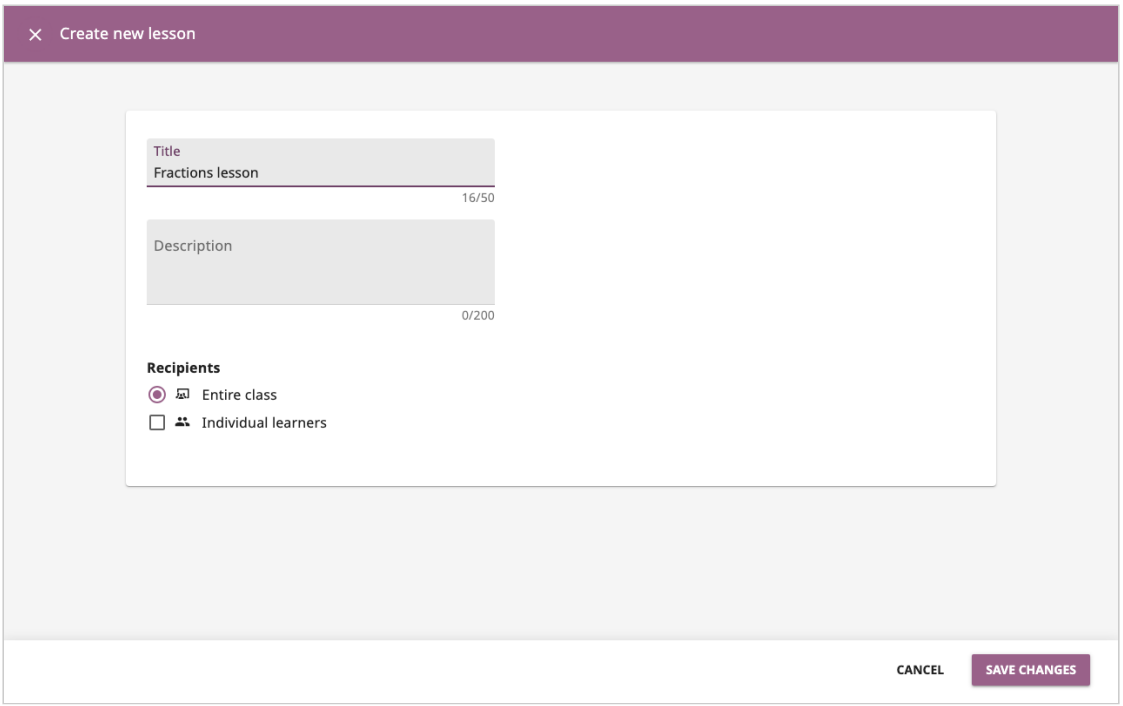


Figure 12. New lesson

3. On the lesson page, click **MANAGE RESOURCES**.

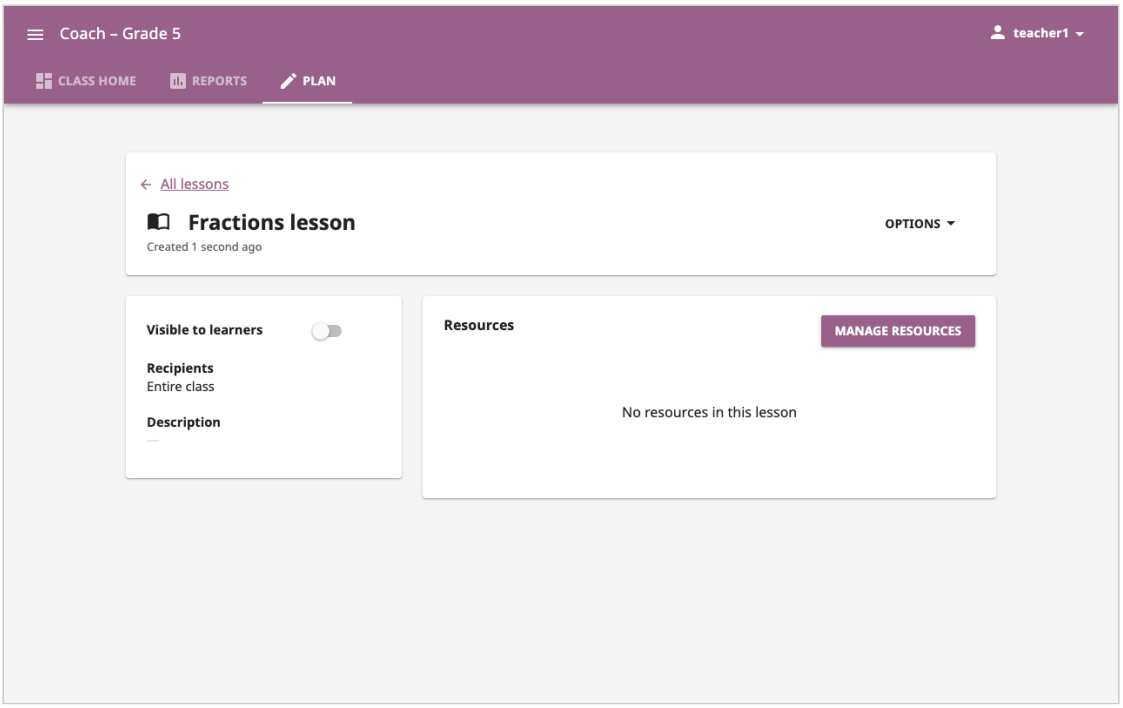


Figure 13. Empty lesson

4. Choose the channel.

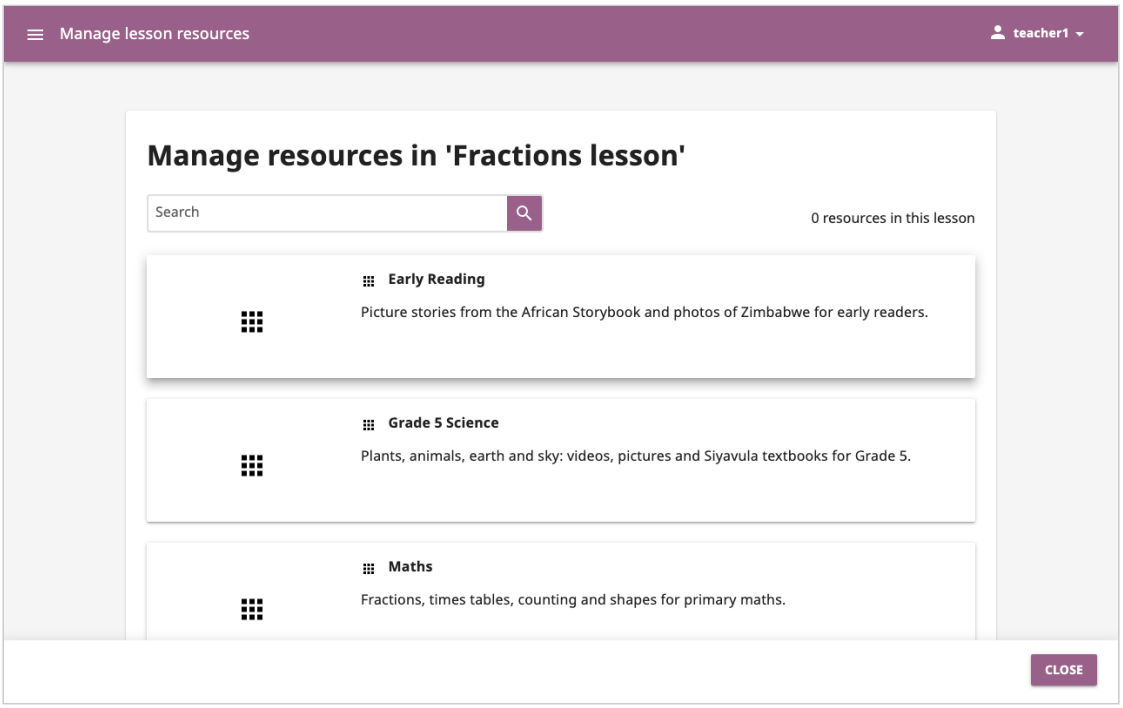


Figure 14. Channels

5. Click the channel (**Maths**). You see its topics.

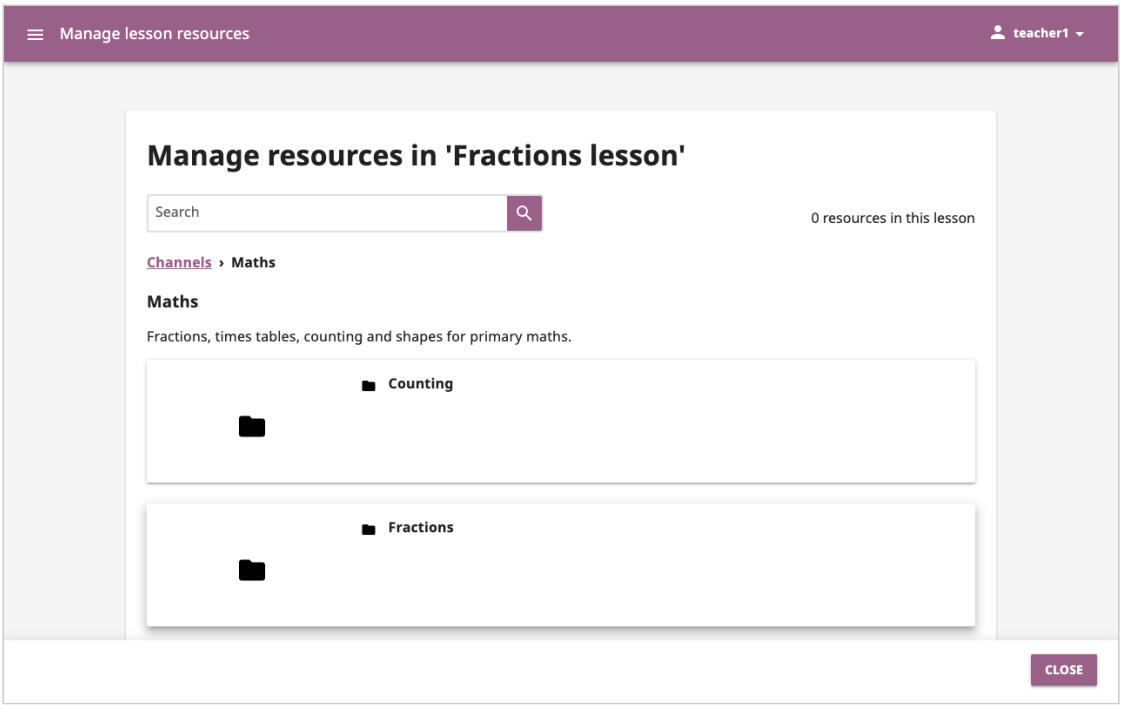


Figure 15. Topic

6. Open the topic **Fractions** and tick the resources you want (here all 3: Cake in quarters, Fraction circles, Paper fraction pies). Each tick is saved straight away; the count at the top right goes up. Click **SAVE CHANGES** (or **CLOSE**) when done.

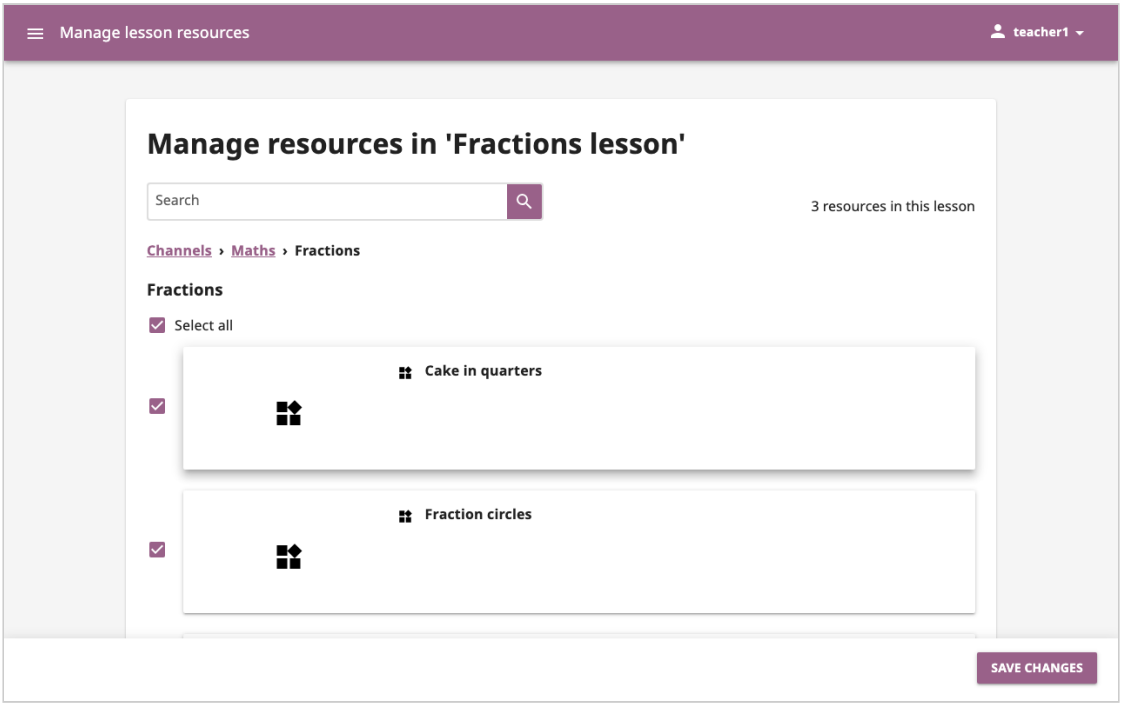


Figure 16. Resources ticked

7. Back on the lesson page, check the resources and switch on **Visible to learners**.

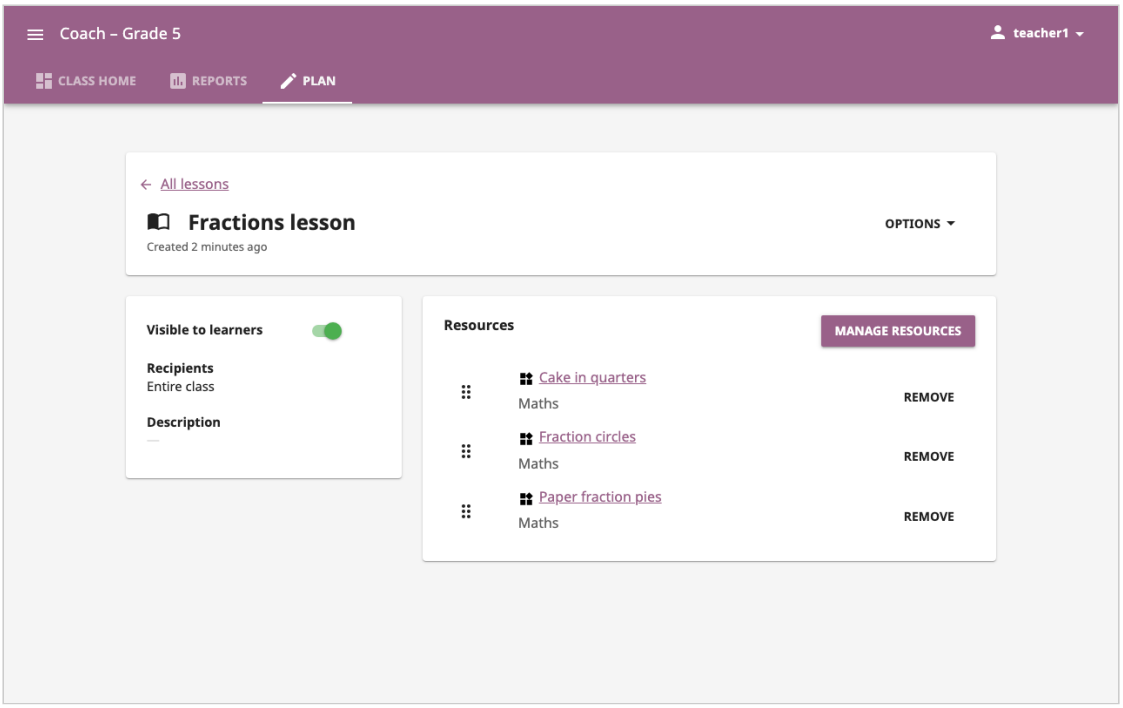


Figure 17. Lesson with its resources, visible

3.5 Make a quiz

Quizzes are built from exercises only.

1. Click **PLAN > QUIZZES > NEW QUIZ**.

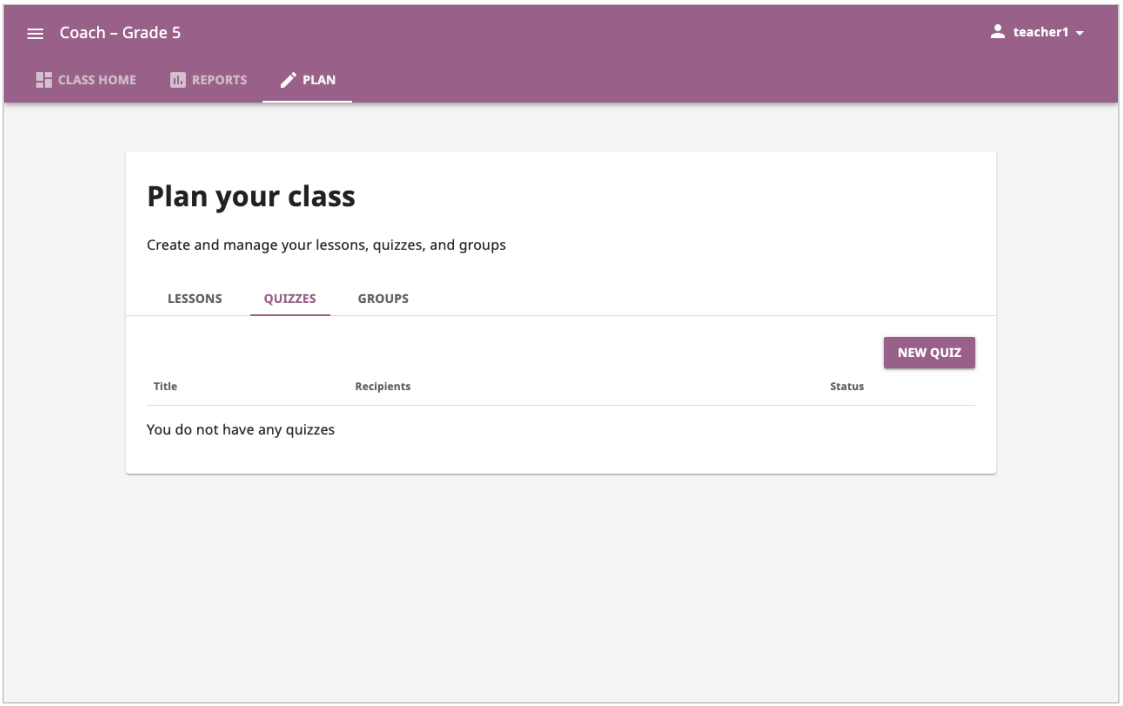


Figure 18. Quizzes tab

2. Type a title (`Fractions quiz`) and the **Number of questions** (6). Browse to **Khan Academy (English - US curriculum) > Math by grade > 5th grade > Add and subtract fractions** and tick a folder (here **Adding and subtracting fractions with unlike denominators**, 3 exercises) or single exercises. Click **CONTINUE**.

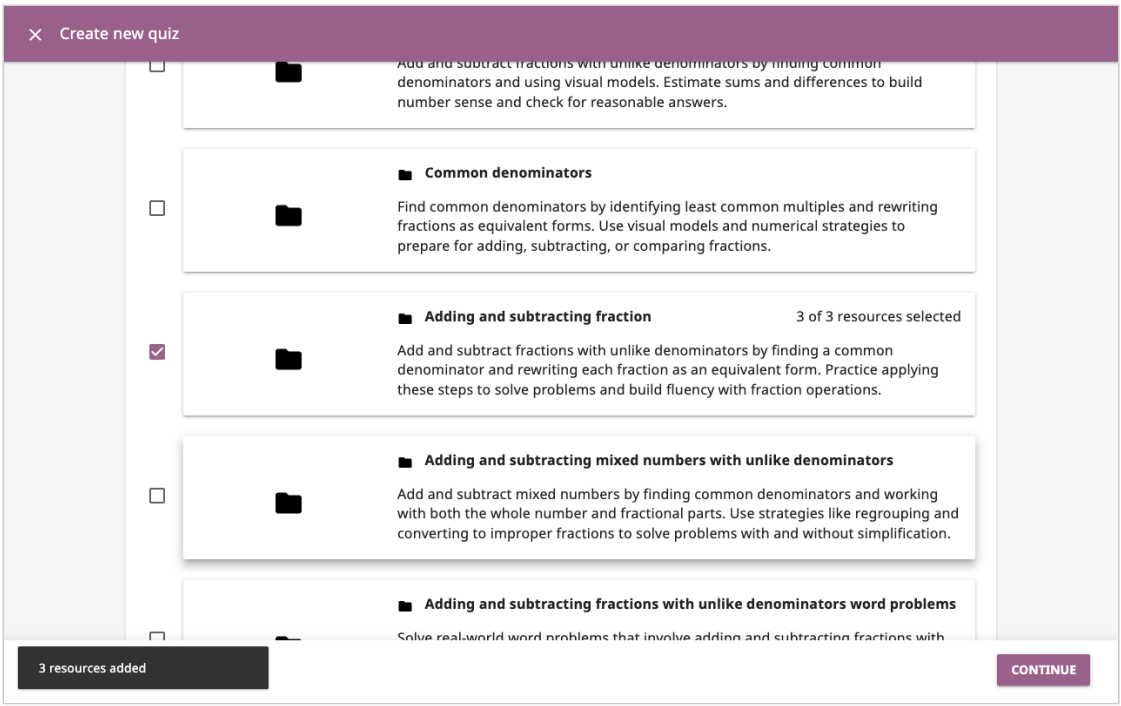


Figure 19. Create new quiz

3. **Preview quiz**: check the questions, choose **Randomized** or **Fixed** order, click **FINISH**.

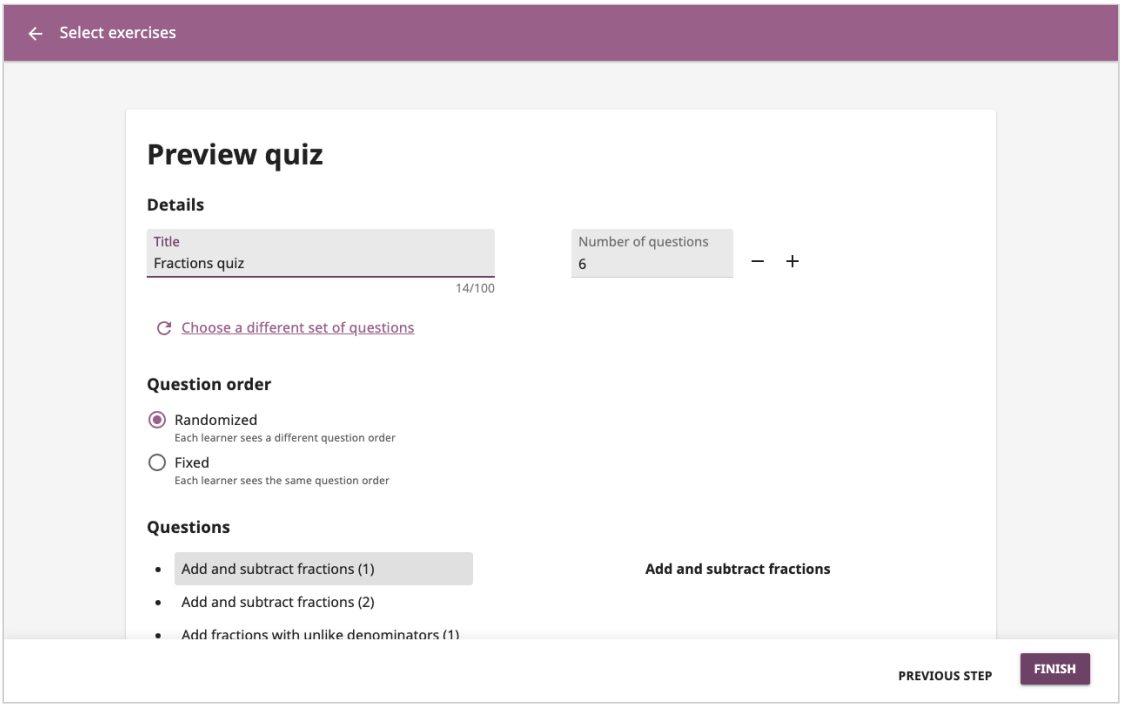


Figure 20. Preview quiz

4. The quiz is not open yet. When the class is ready, click **START QUIZ** and then **CONTINUE**. When the class has finished, open the quiz in **Coach** again and end it there, so no more answers come in.

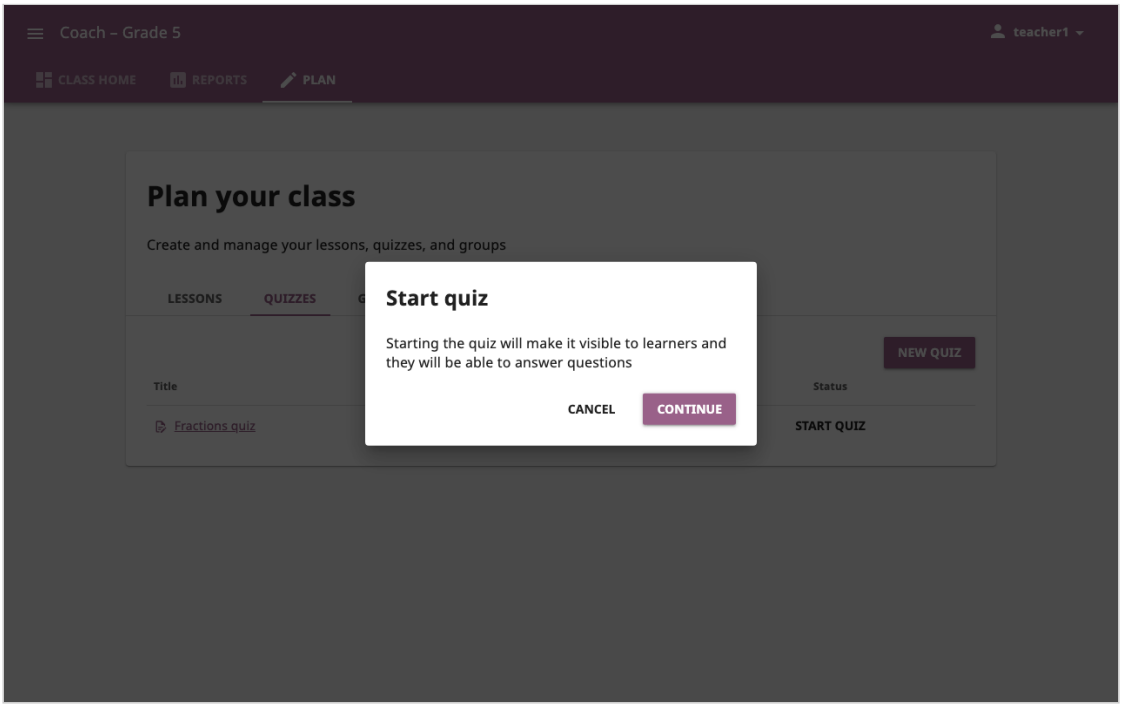


Figure 21. Start quiz

3.6 See progress (Reports)

Do this after learners have worked (chapter 4).

1. Click **REPORTS**. Tabs: **LESSONS**, **QUIZZES**, **GROUPS**, **LEARNERS**.

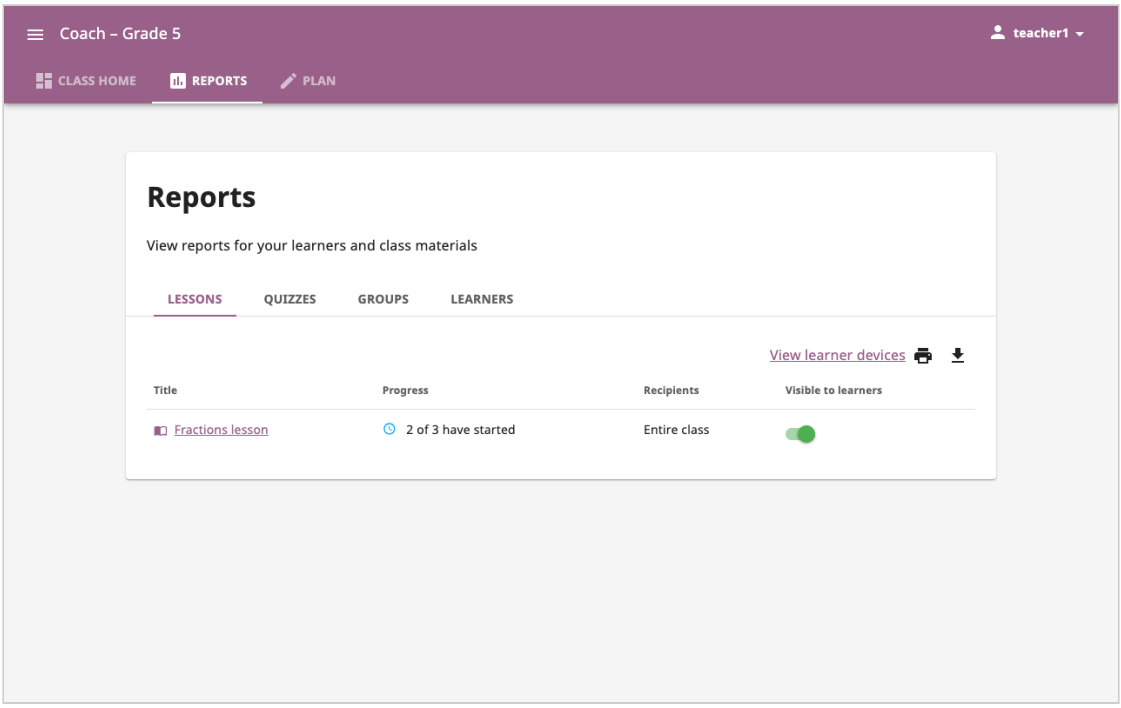


Figure 22. Reports

2. **LESSONS > (the lesson) > LEARNERS**: who completed, started or did not start.

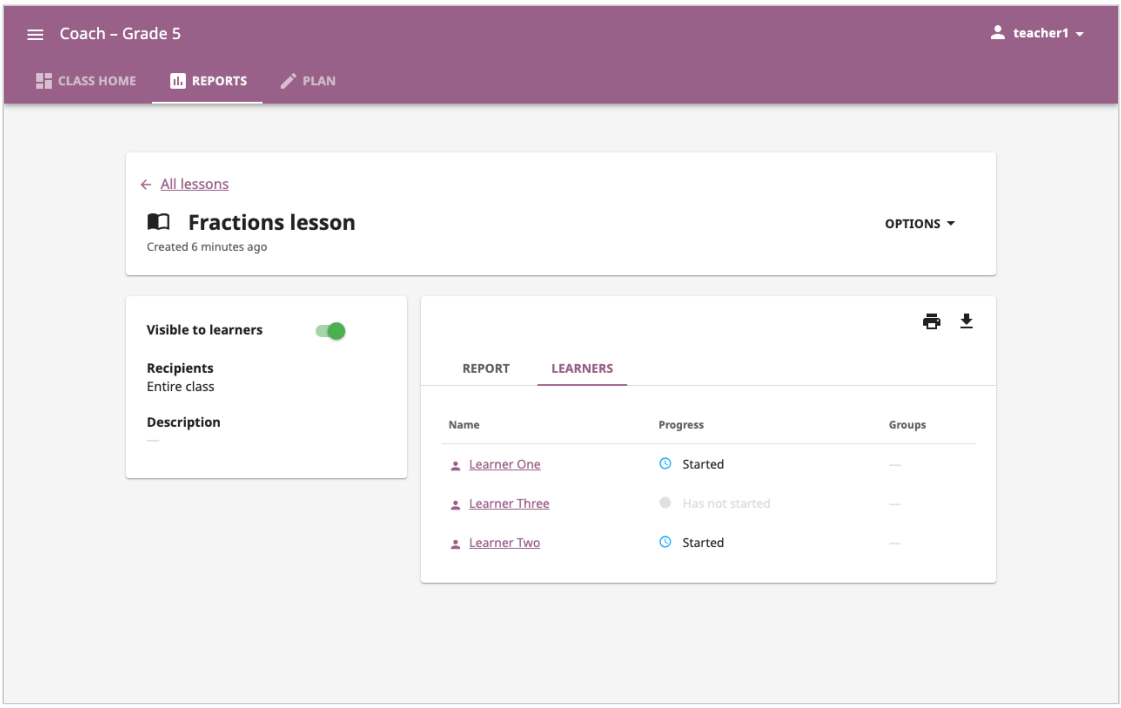


Figure 23. Lesson report

3. **QUIZZES > (the quiz)**: status and score per learner; **DIFFICULT QUESTIONS** shows what to re-teach.

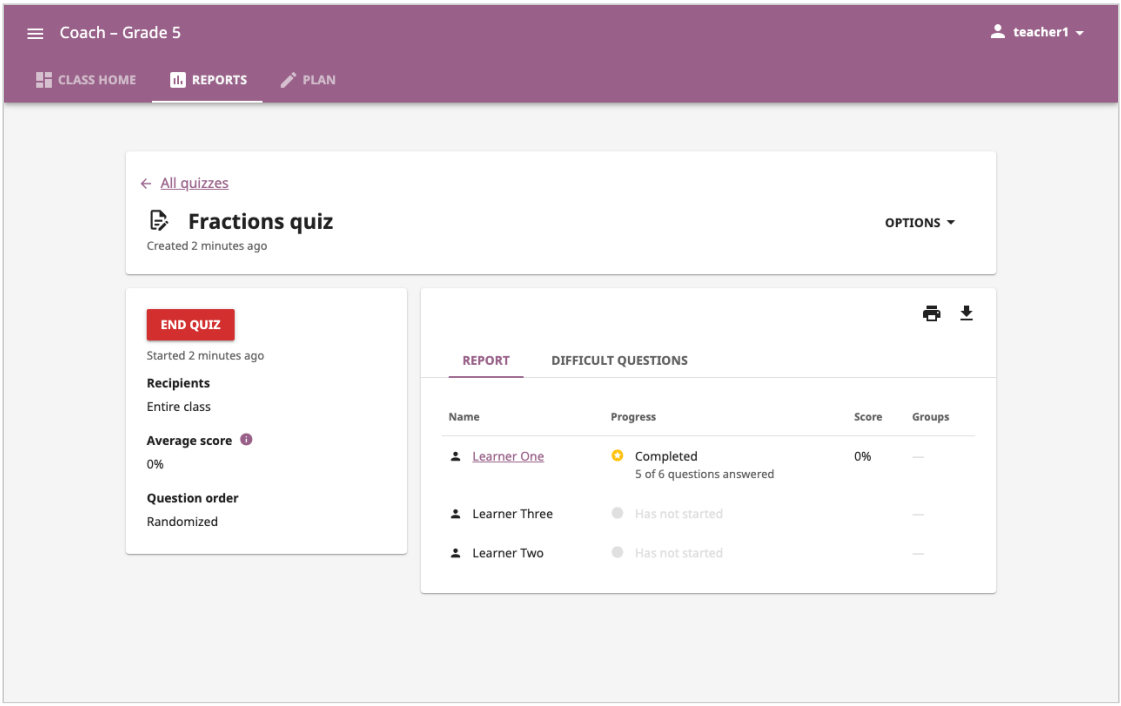


Figure 24. Quiz report

4. **LEARNERS > (a name)**: that learner's lessons and quizzes.

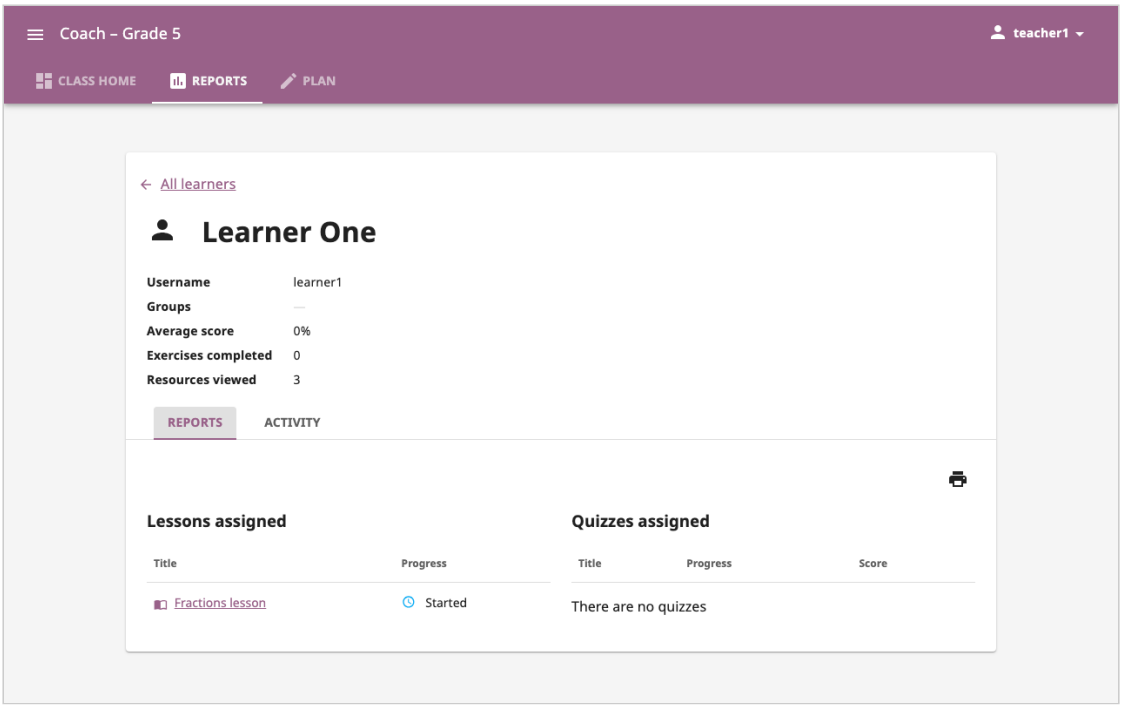


Figure 25. Learner report

Reports can be printed or exported as CSV (icons top right).

> **Tip.** Do not chase 100%. Look for the question most learners got wrong and teach it again.

4. Kolibri for learners (try it yourself)

Your learners open Kolibri the same way you do: on a classroom computer they double-click **IDEA** on the desktop and click **Open** on the Kolibri card in the Console (2.2.1). Show them this once.

Sign in as a learner in a second browser window (private window) to see exactly what your class sees. Learners have no password in Kolibri. They sign in with only their username, for example `learner1`.

> **At an existing school.** A teacher with admin rights (or the Kolibri admin) turns this on in Kolibri: open **Facility > SETTINGS**, untick **Require password for learners**, then click **SAVE CHANGES**.

4.1 Do a lesson

1. Sign in as `learner1`. **HOME** shows **Your classes**, **Recent lessons** and **Recent quizzes**.

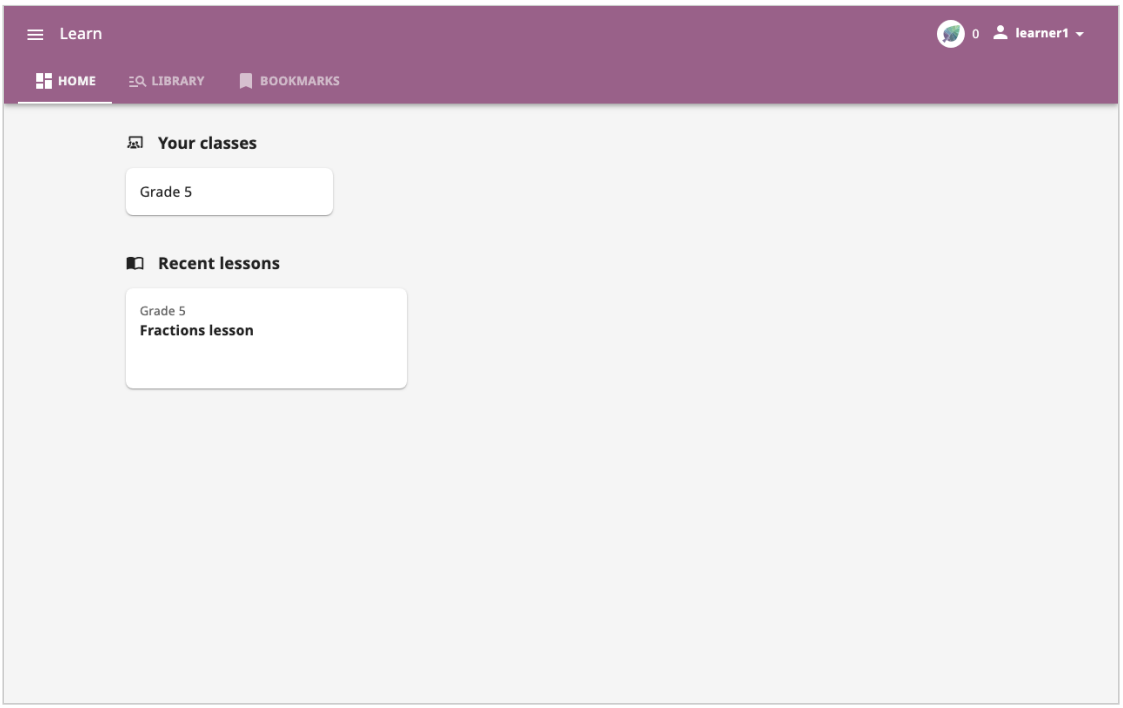


Figure 26. Learner home

2. Click **Fractions lesson**. Its items are listed in order.

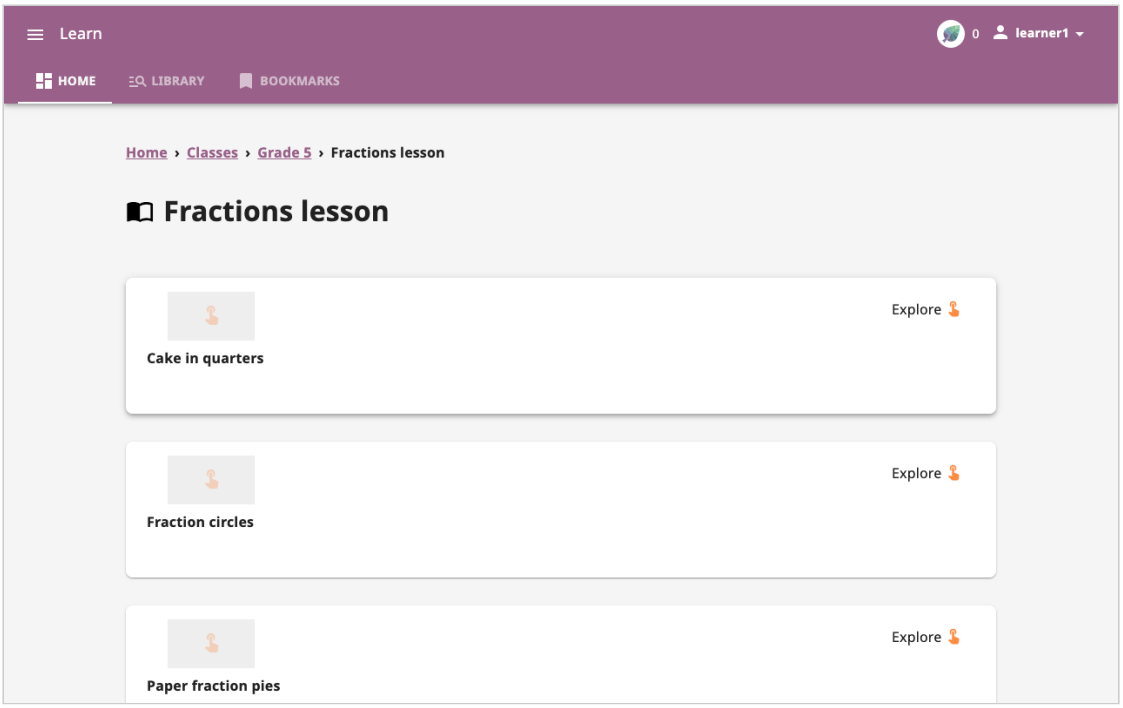


Figure 27. Lesson

3. Open the first item (**Cake in quarters**, a picture activity). Videos open the same way.

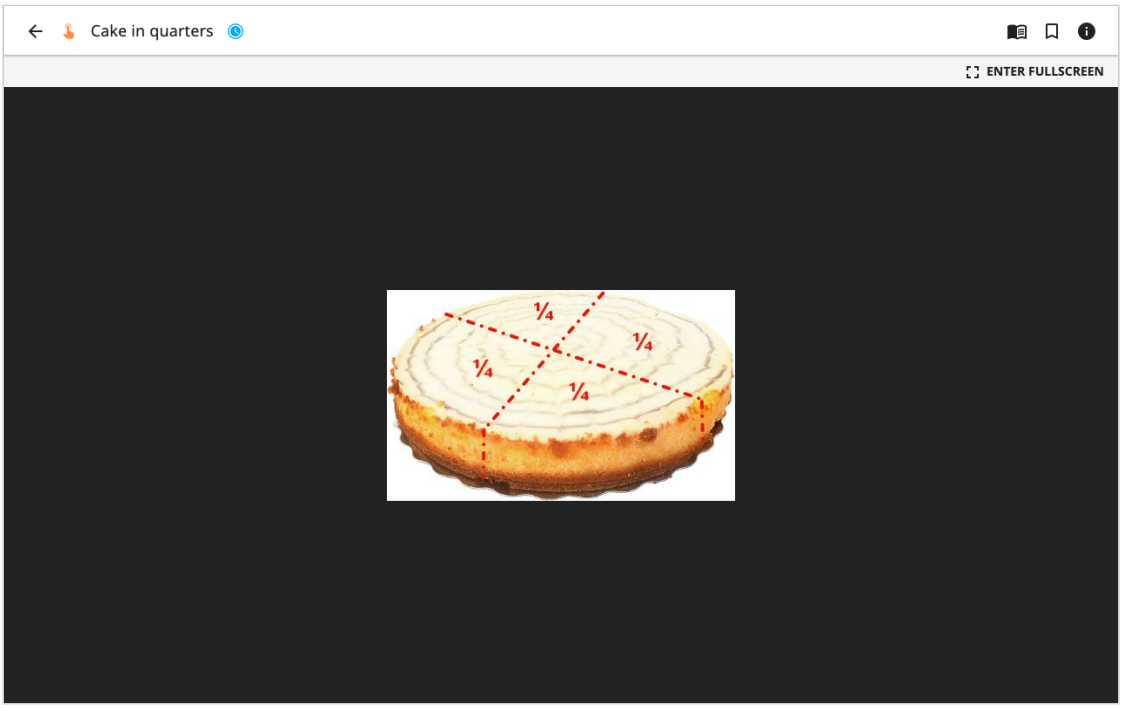


Figure 28. Video

4. Exercises work like this (here the Khan Academy exercise **Add fractions with unlike denominators**): type an answer, click **CHECK**. Keep going until the goal at the top ("Get 5 correct") is reached.

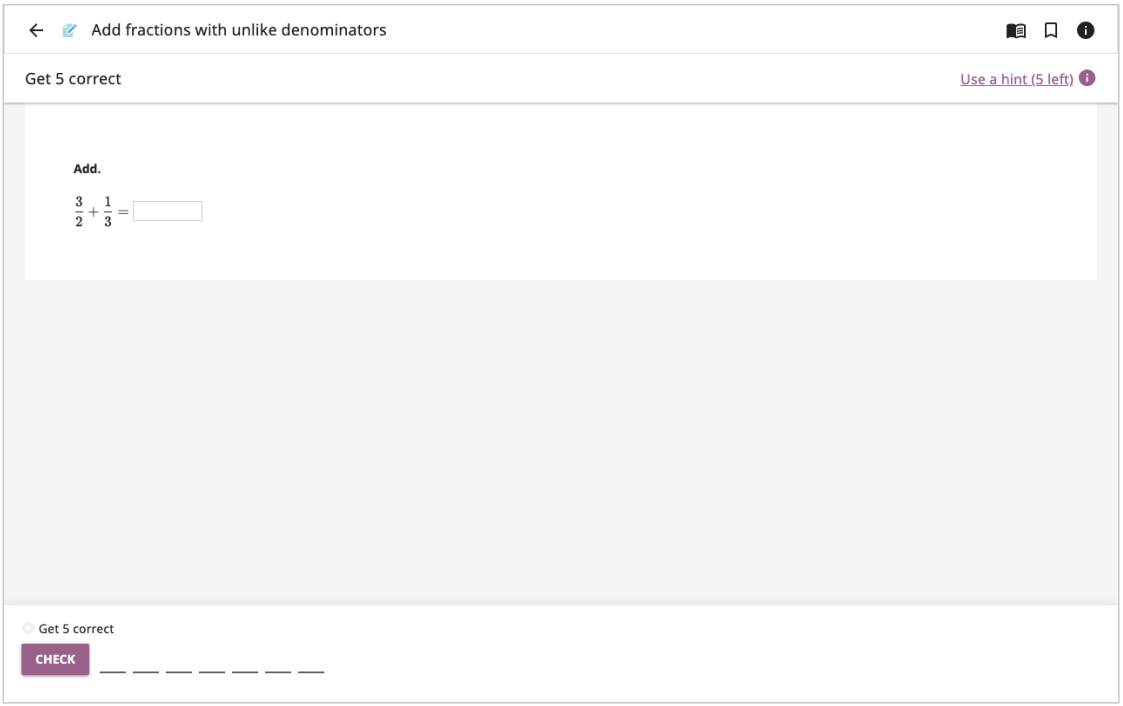


Figure 29. Exercise

5. Sign out. Sign in as `learner2` and open only one item, so the report shows a difference.

4.2 Take a quiz

1. Open the quiz from **Recent quizzes** (or the class page).

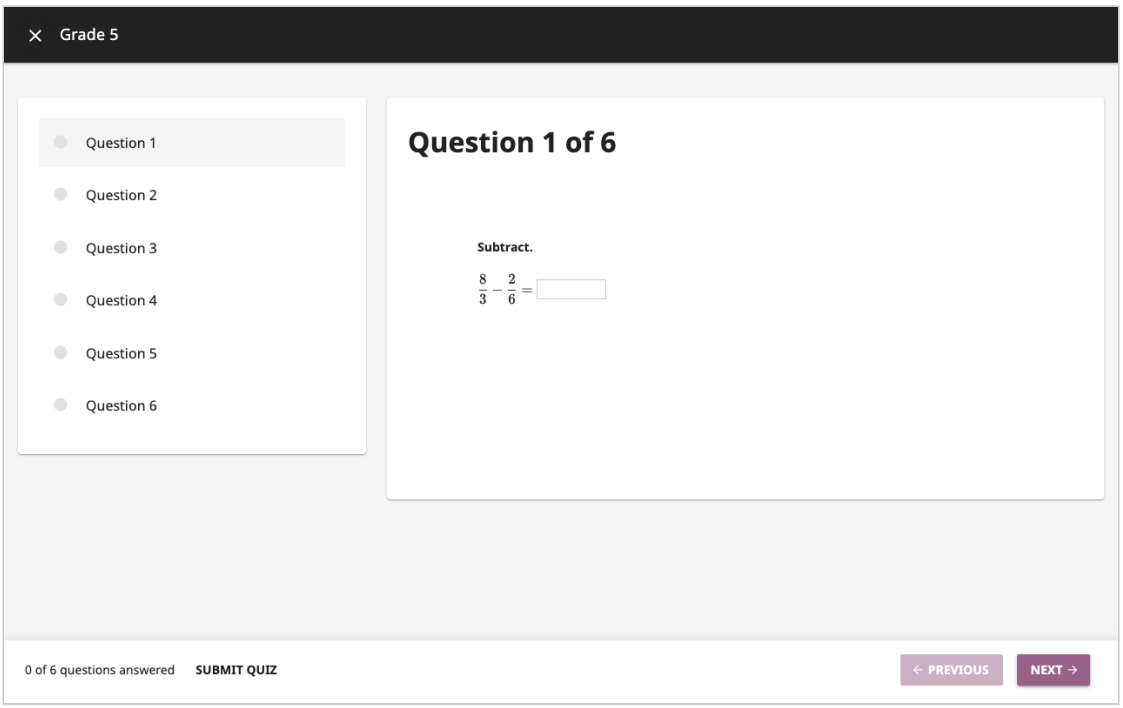


Figure 30. Quiz

2. Answer each question and use **NEXT** / **PREVIOUS**. The list on the left shows answered questions.

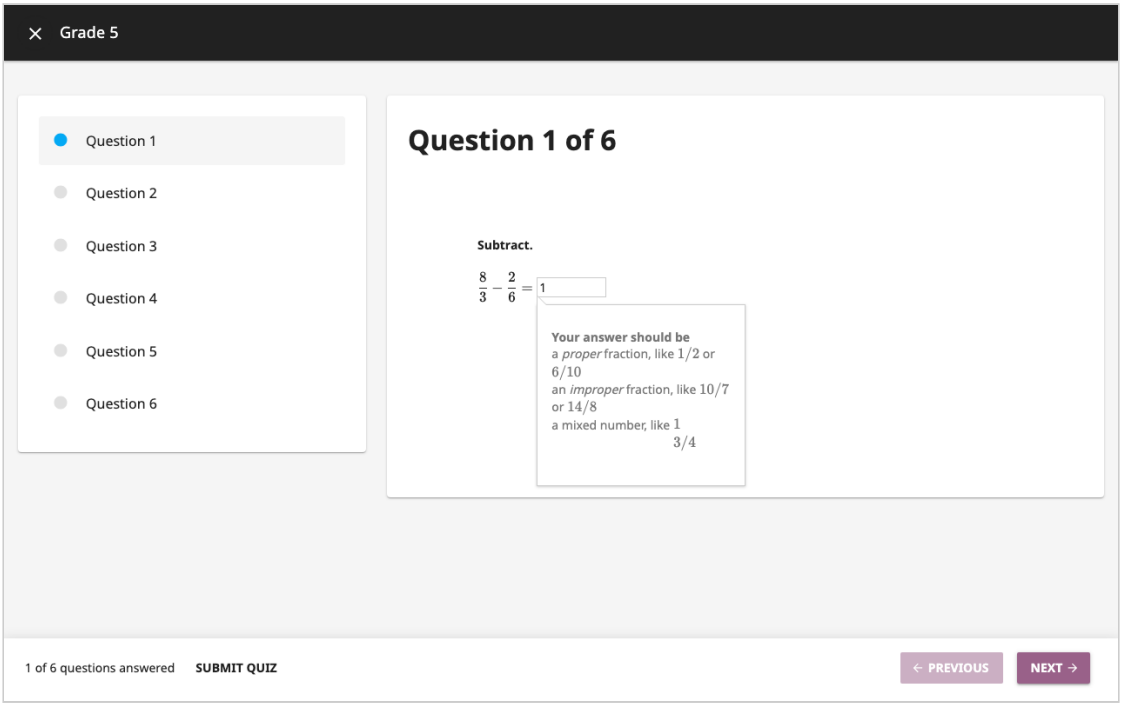


Figure 31. Question

3. Click **SUBMIT QUIZ** and confirm. After submitting, answers cannot be changed.

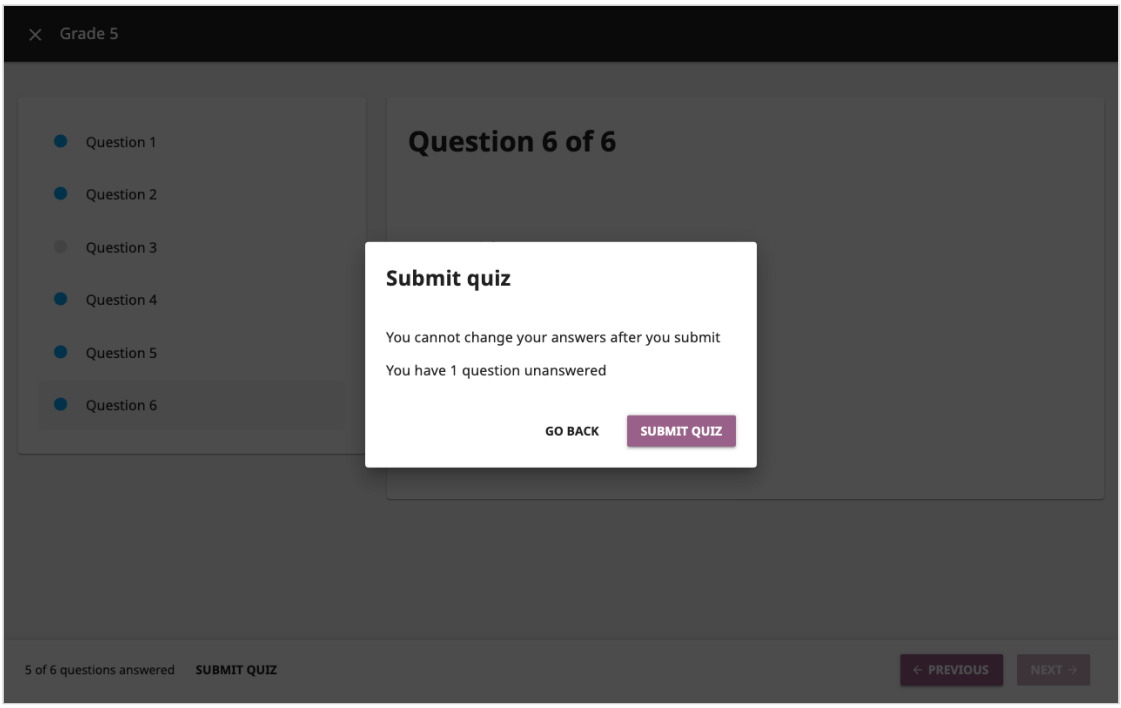


Figure 32. Submit

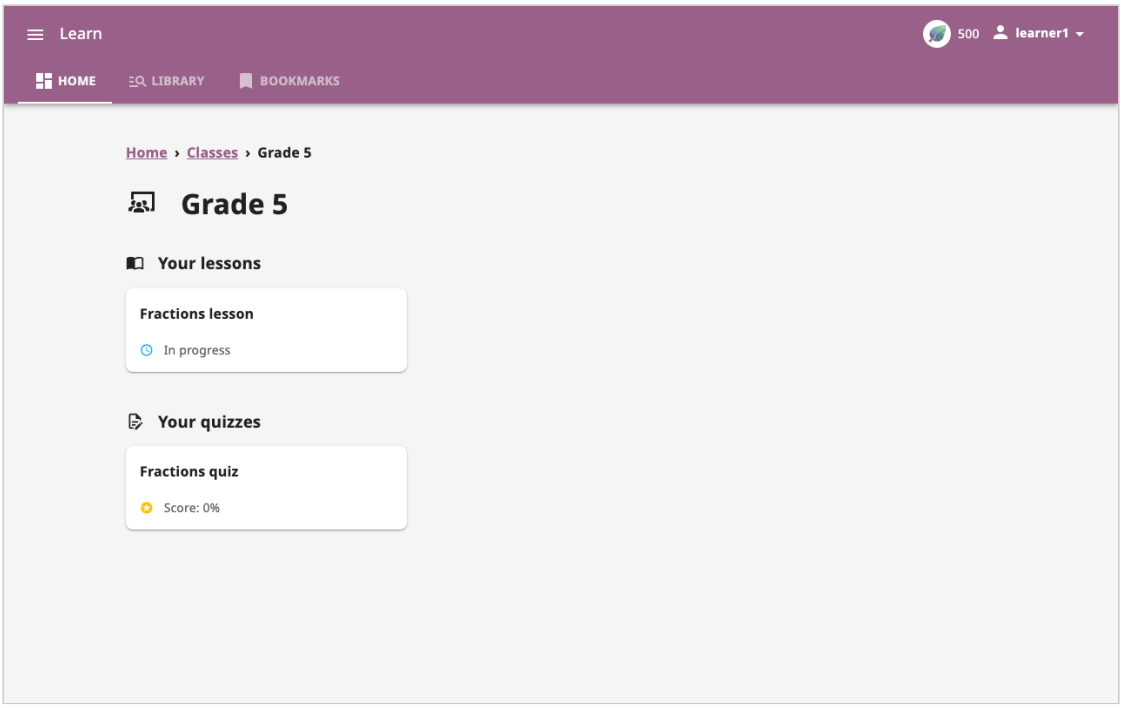


Figure 33. After submit

Now go back to the teacher window and look at Reports (3.6).

5. Make your own Kolibri channel: the channel builder and your USB stick

This is the most useful chapter for teachers. The **channel builder** turns your own videos, PDFs and pictures into a **Kolibri channel**. Your learners then find it in Kolibri next to the Khan Academy lessons, and you can put it in lessons and quizzes (chapter 3).

You do not need internet or an account. Bring your files on a normal **USB stick**, filled at home, in town or from a colleague.

Good to know:

- **IDEA only reads your stick.** It never writes to it, so nothing on the stick changes, and the stick still works in any Windows laptop afterwards. You do not have to prepare or erase it.
- The stick must be **FAT32 or exFAT**. Almost all sticks are, unless someone formatted them as NTFS.
- **Building is not publishing.** A channel reaches learners only after someone imports it in Kolibri (5.2).

5.1 From your USB stick to a channel (automatic)

At home or in town (any computer):

1. Open the stick and make a folder named **`kolibri`** at the top level of the stick.
2. Inside `kolibri`, make **one folder per channel**, for example `Grade 5 Science`. The folder name becomes the channel name.
3. Inside the channel folder, make **one folder per topic**, for example `Plants` and `Animals`, and copy your files into them. Each file becomes one resource. The builder takes PDF, EPUB, MP4/WebM video, MP3 and pictures.

> **Optional: `channel.json`.** To set the channel's name, author, description or licence yourself, put a small text file named `channel.json` into the channel folder, for example: `{"name":"Plants and Water","author":"Mary Okello","description":"Term 2","license":"CC BY","license_description":""}` Without it (or if the file has a mistake), the channel gets the folder name, the stick's name as author, and the licence All Rights Reserved.

At school:

4. Plug the stick into a free USB port of the IDEA box.
5. The IDEA box starts the channel builder by itself, if the channel builder is on this box. It then builds **one channel for each folder inside `kolibri`**, one after the other. You do not have to click anything.
6. To watch it, open the Console. Under the IDEA box you see the row **USB stick: (name) (read-only, size)** with the line Visible in Nextcloud for teachers on this Pi and the link **Open the channel builder**. Click the link. You can also open the app **Kolibri Channel Builder** in the app list with **Open**.

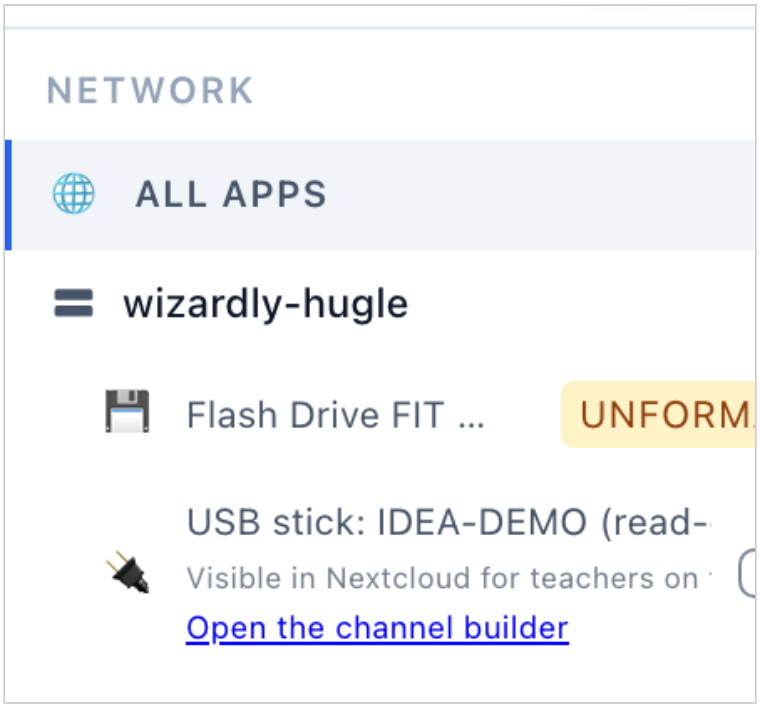


Figure 34. The USB stick row in the Console (here IDEA-DEMO), with Open the channel builder

The same stick is also listed once more, by its make and size, as **UNFORMATTED**. That is normal: leave it alone (chapter 12).

7. On the builder's page, the box **From USB sticks** shows each channel. A channel waiting for its turn says **Queued (number N in line)**, a channel being built says **Building N/M files**, and a finished channel says **Built version V: R resources.**

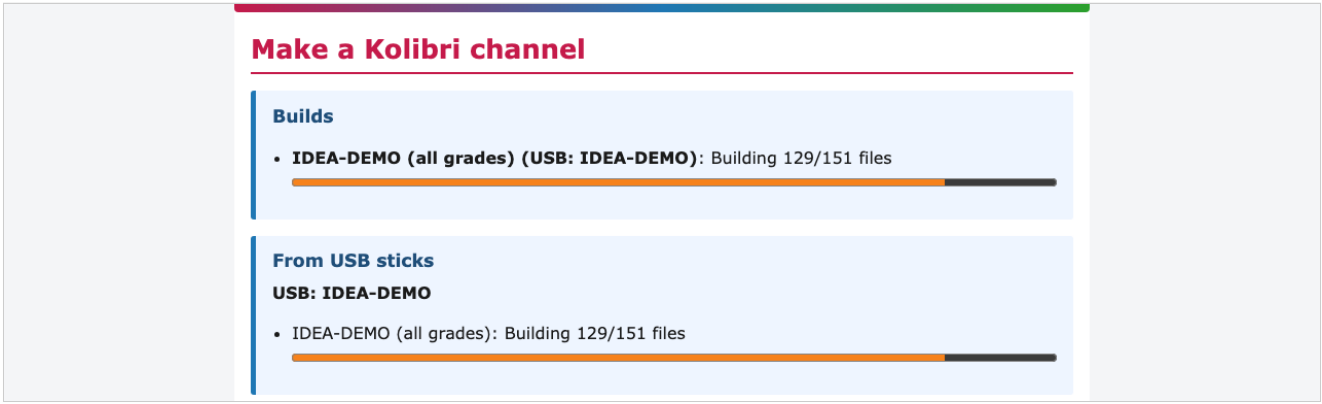


Figure 35. Channel builder, From USB sticks box while a channel is Building N/M files

8. When the stick's line says **Built: you can pull the stick out.**, you can take your stick out. The channels stay on the IDEA box.

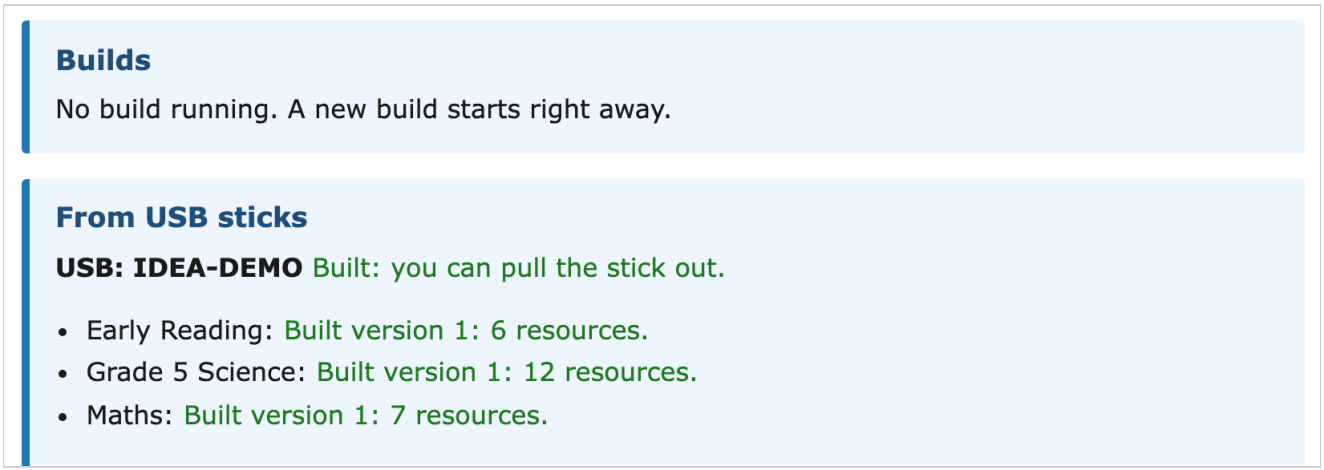


Figure 36. From USB sticks: all three channels of the stick IDEA-DEMO built

Don't pull the stick out while it is building. If you do, nothing breaks: that build stops, and the last good version of the channel stays. Plug the stick in again and the build starts over.

Large sticks. If the `kolibri` folder holds more than 8 GB, the builder does not build it by itself. It says Too large for an automatic build: use the manual build below. Build it by hand (5.4).

5.2 Import the channel into Kolibri

Your channel is built, but learners cannot see it yet. Someone who may manage channels on the device in Kolibri (on the demo school: `admin`) imports it once.

1. Open Kolibri from the Console and sign in.
2. Open **Device**. The **CHANNELS** tab lists the channels on the IDEA box. Click **IMPORT**.

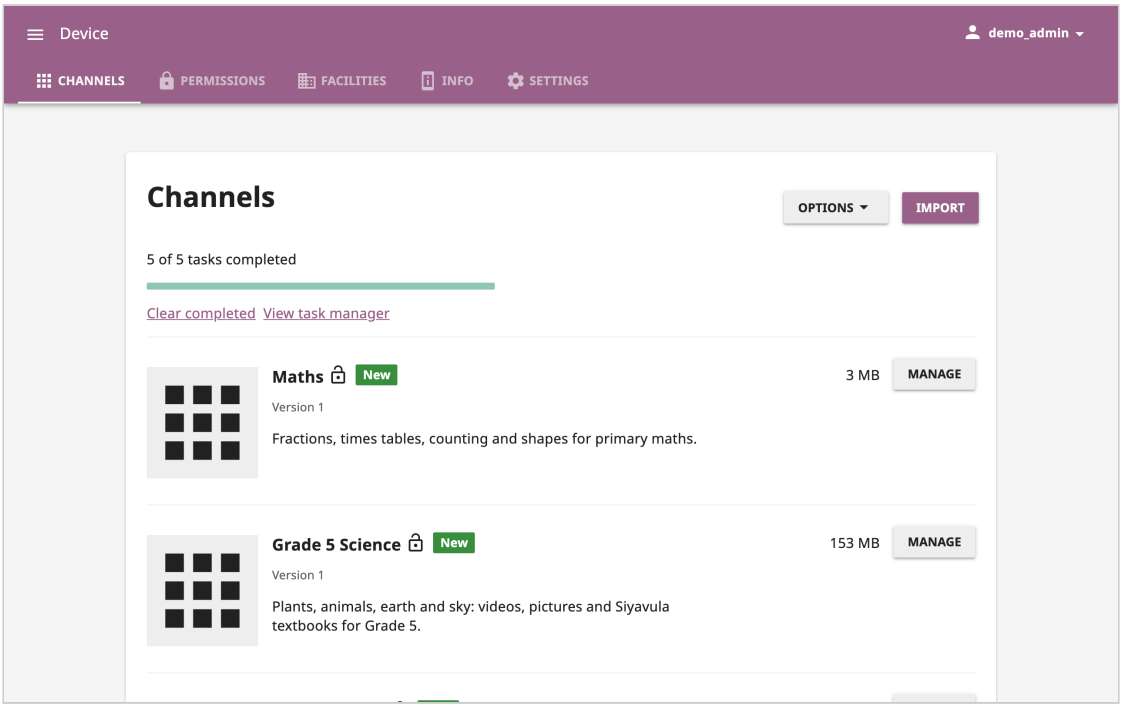


Figure 37. Device > Channels in Kolibri, with the IMPORT button. The three channels from the stick are already imported here

3. In **Select a source**, choose **Local network or internet** and click **CONTINUE**.

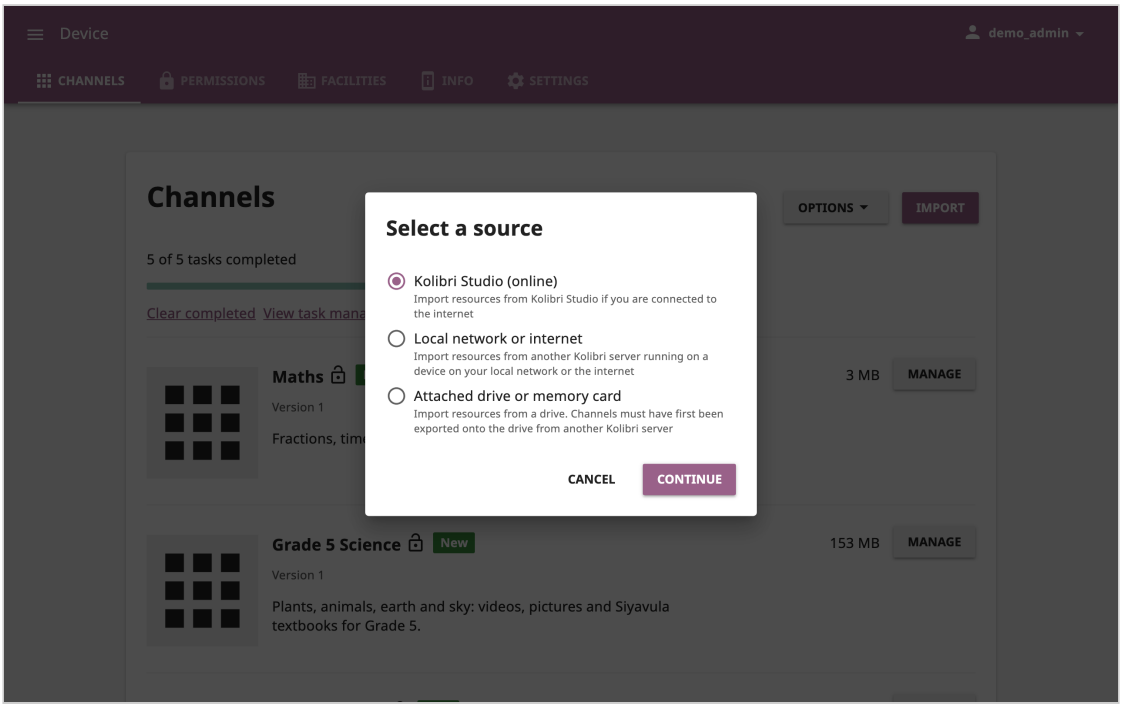


Figure 38. Select a source: Local network or internet

4. In **Select network address**, pick the channel builder and click **CONTINUE**. The first time, click **Add new address** and type the address shown at the bottom of the builder's page (To import into Kolibri…). Kolibri then lists it as **Channel builder (name of your IDEA box)**.

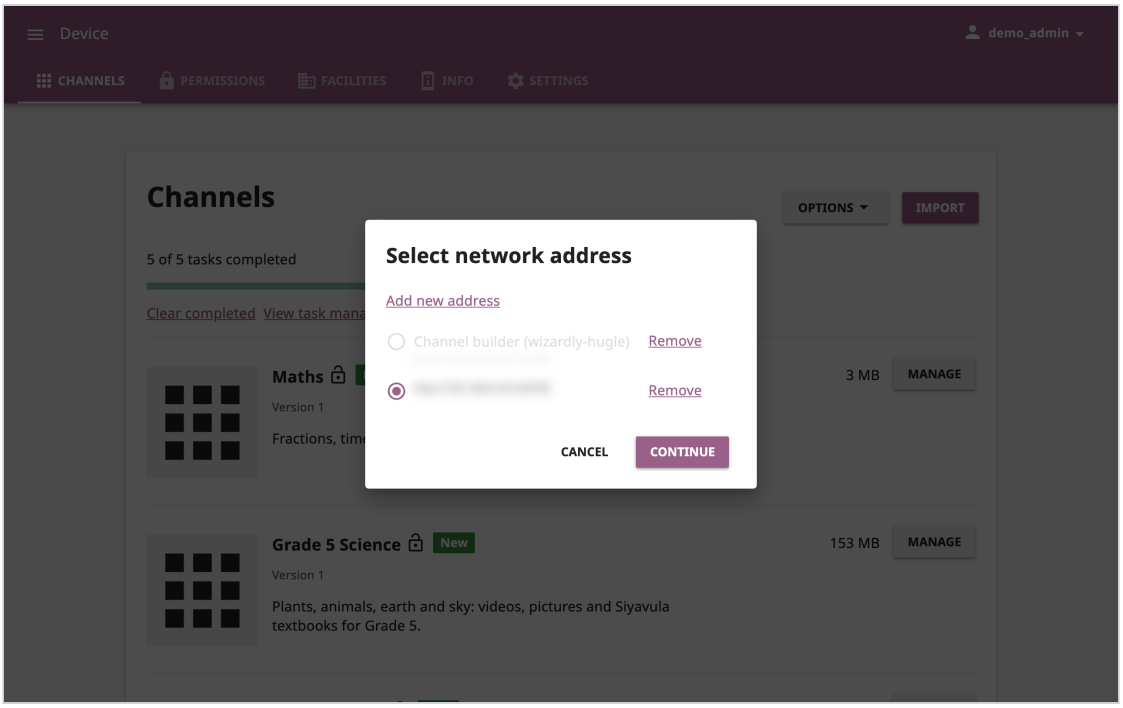


Figure 39. Select network address. If an address is greyed out, it no longer answers: pick the other one

5. Kolibri lists the builder's channels. Next to your channel (for example **Grade 5 Science**) click **SELECT RESOURCES**.

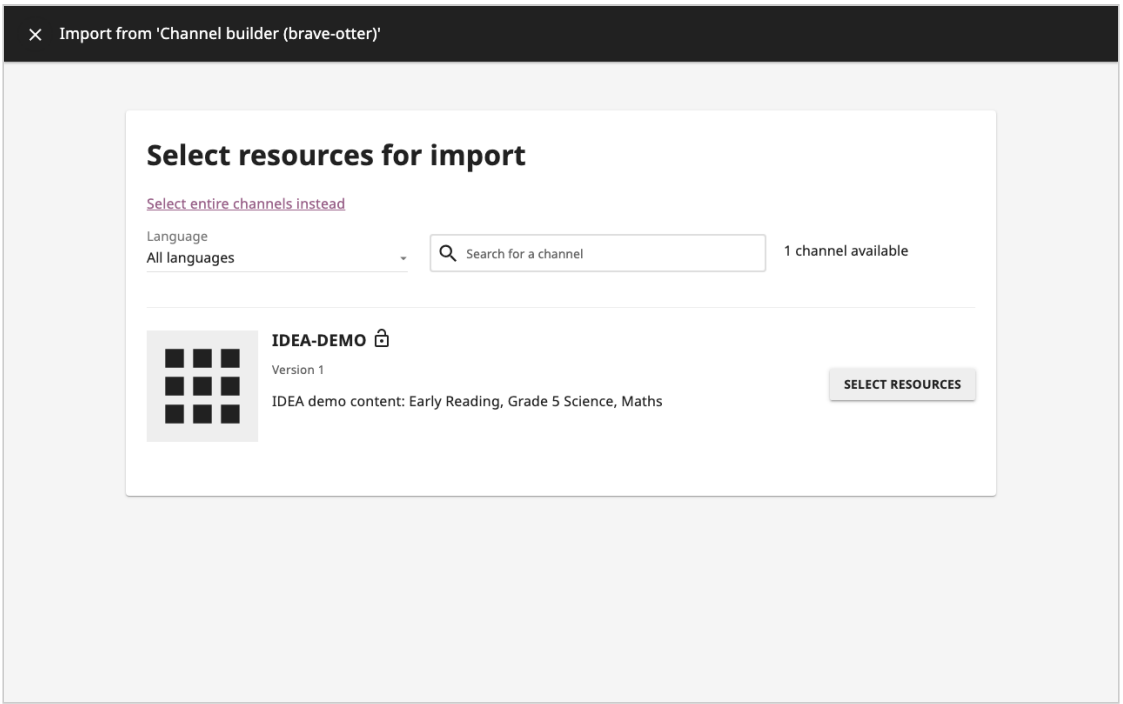


Figure 40. Import from 'Channel builder (brave-otter)': the builder's channels, here IDEA-DEMO version 1, each with SELECT RESOURCES

6. Choose the topics (or everything) and click **IMPORT**. When the task is finished, the channel is in the **Channels** list (Kolibri shows the label **New** until you click **Clear completed** in the task bar).
7. Now use it like any other channel: add its resources to a lesson (3.4) or a quiz (3.5).

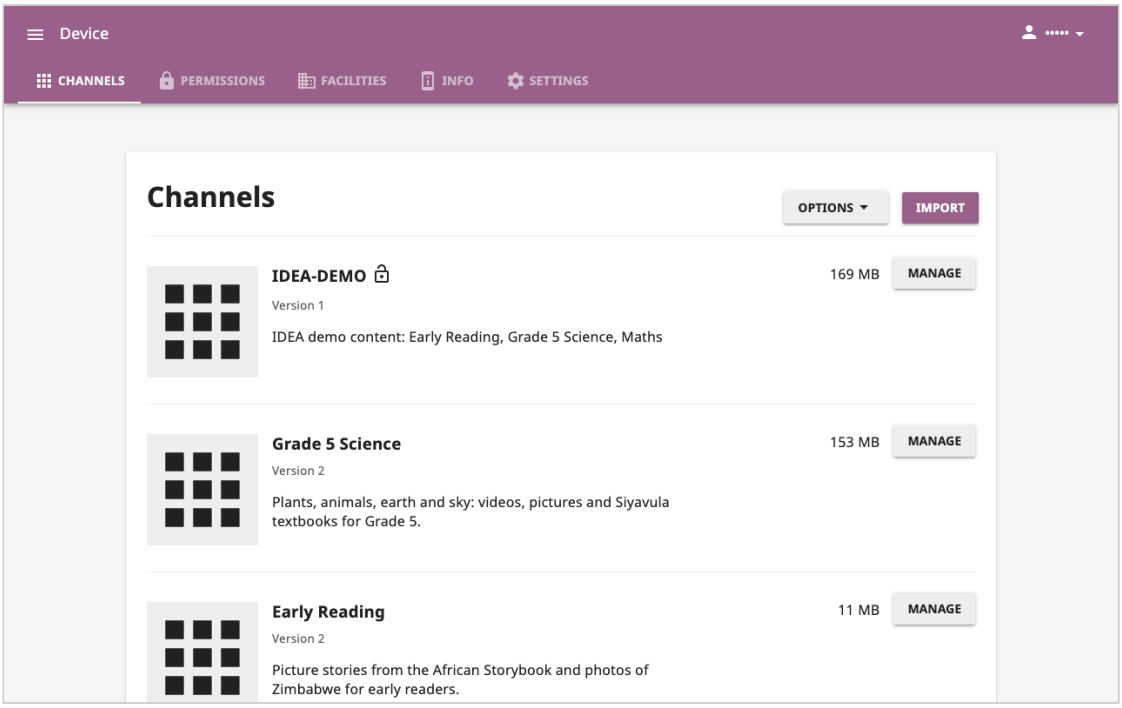


Figure 41. Device > Channels: the new channel (here IDEA-DEMO, version 1) above the other channels

5.3 Update the channel

1. Next term, add or change files **in the same folder** on your stick (for example a new topic folder `Soil`).
2. Plug the stick in again. The builder sees what changed and builds a new version of the **same channel**. If nothing changed, it does nothing.
3. In Kolibri, open **Device > Channels** and import the channel again from the builder, as in 5.2, to get the new version.

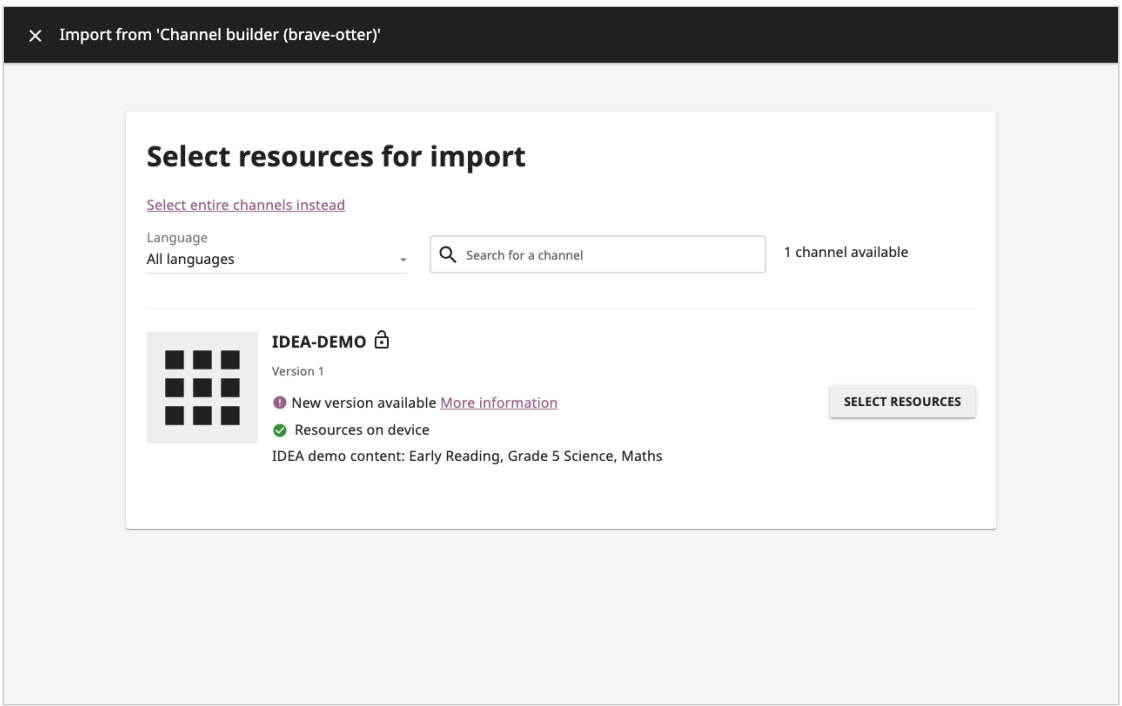


Figure 42. Importing again from the builder: Kolibri shows New version available for the channel already on the device

5.4 Build a channel by hand

Use this when your stick has **no `kolibri` folder**, when it is too large for an automatic build, or when your files are in Nextcloud.

1. Open the channel builder from the Console.
2. Under **Where are the files?** choose your stick (**USB: (name)**) or **Nextcloud**. Then choose the folder, for example under **Folder on a USB stick**.
3. Type the **Channel name**, a **Description** and the **Author** (your name), pick the **Licence**, and click **Build channel**.

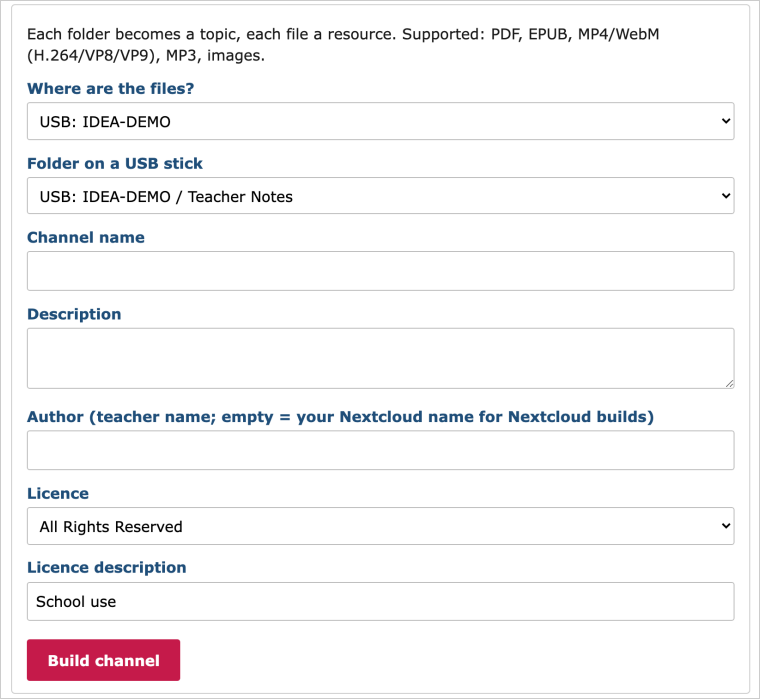


Figure 43. The builder's form: Where are the files?, the folder, the channel name, author and licence

4. When it is done, the box **Builds** says **Built (name) version V: R resources.**, and the channel is listed under **Built channels**. Then import it in Kolibri (5.2).

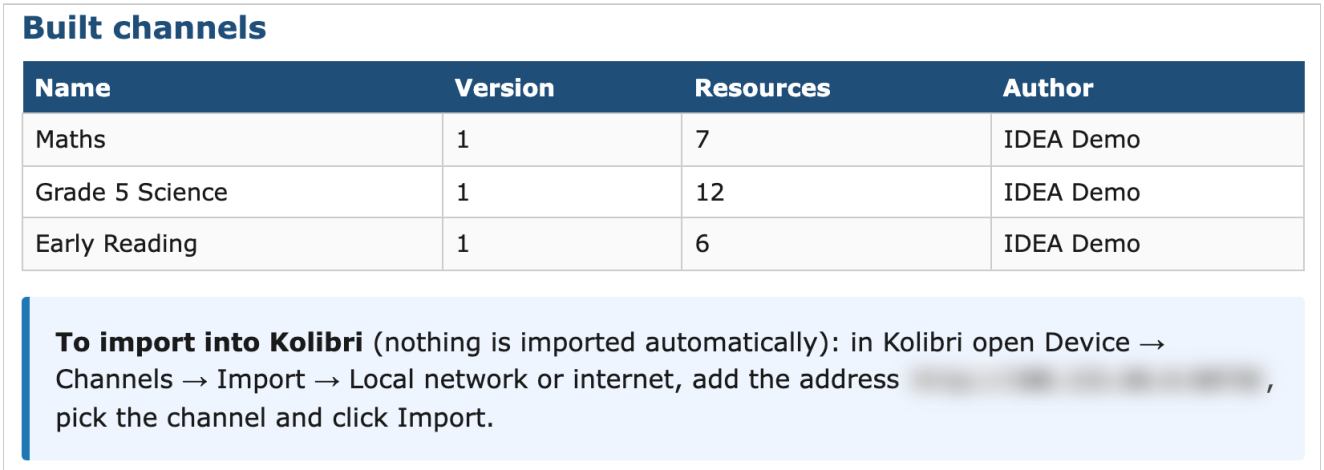


Figure 44. Built channels, and how to import them into Kolibri

Building from the same folder again makes a new version of the same channel (5.3).

5.5 Use your stick in Nextcloud

Your stick also appears in Nextcloud, so you can open its files or copy them into your own folders, without building a channel.

1. Plug the stick in. Open Nextcloud and sign in as a teacher. In **All files** you see a folder **USB: (your stick's name)**. Nextcloud shows its size as Pending: that is normal.

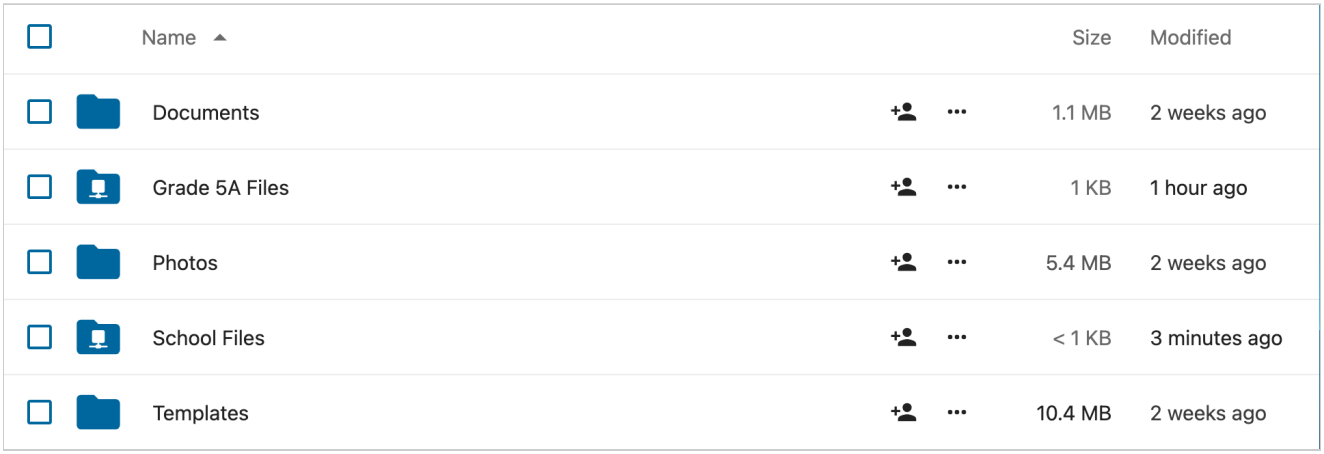


Figure 45. Folders from outside Nextcloud have a folder icon with a small screen, here School Files and a class folder. Your USB: folder shows the same way

2. Open the folder to view or play a file. You can't add or change anything in it (Nextcloud says You do not have permission to upload or create files here).

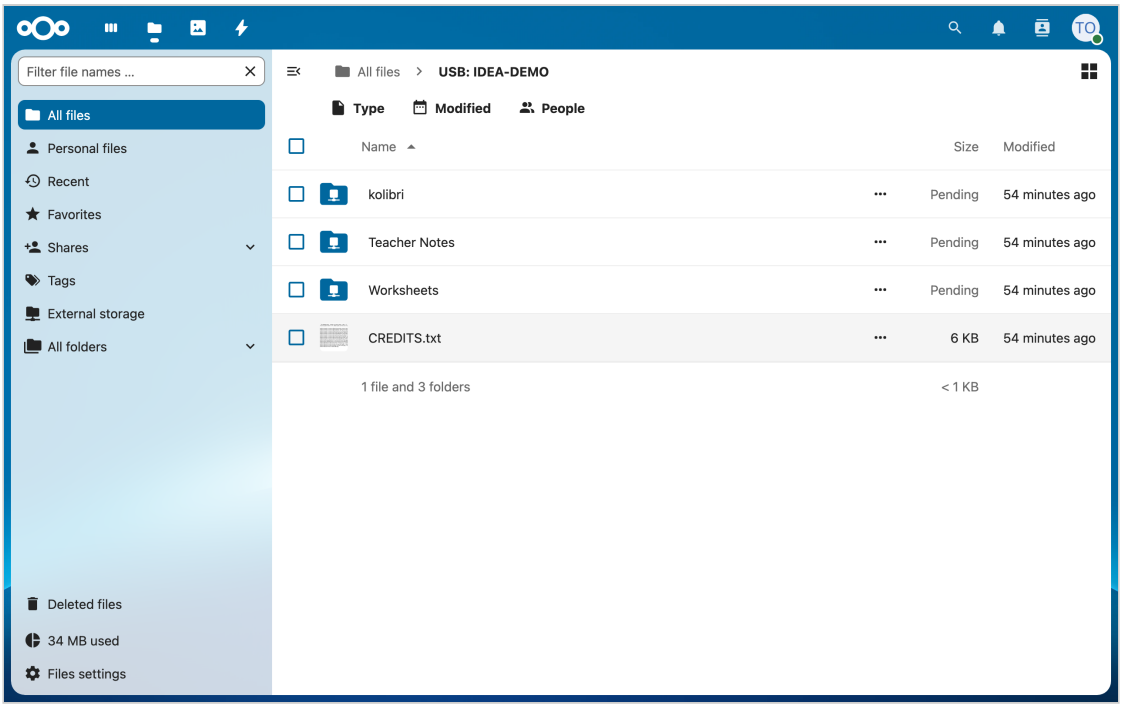


Figure 46. Inside the folder USB: IDEA-DEMO: the folders kolibri, Teacher Notes and Worksheets, and the file CREDITS.txt

3. To keep a file, click ··· on its row, choose **Copy**, and copy it into one of your own folders (for example a course folder, or **School Files**).
4. Pull the stick out when you are done. The **USB:** folder disappears; the files you copied stay in Nextcloud.

Only teachers (and the Nextcloud admin) see the **USB:** folder. Learners do not.

5.6 Questions

Question	Answer
My stick doesn't show up.	Is it plugged into **this** IDEA box? A stick only shows on the box it is plugged into. Is it NTFS? NTFS sticks are not read; copy the files onto a FAT32 or exFAT stick, or ask the operator.
Nothing was built.	Check that the stick has a folder named `kolibri` at the top, with one folder per channel inside it. Check that the channel builder is on this IDEA box and was not stopped by the operator: the Console row then says Channel builder was stopped by an operator; start it to build from this stick. Once the operator starts it, it builds the stick without unplugging. Over 8 GB: build it by hand (5.4).
I saved a file, but it's not on my stick.	IDEA never writes to your stick. Download the file from Nextcloud onto a laptop and copy it to the stick there.
My learners can't see my channel.	Building is not publishing. Import the channel in Kolibri first (5.2), then add it to a lesson.
I changed my files, but the channel is the same.	Plug the stick in again, wait for **Built**, then import the channel again in Kolibri (5.3).

6. Nextcloud for teachers

Demo: `teacher1`, groups `Grade 5` (class) and `teachers`. Office documents open in **Nextcloud Office**.

Open Nextcloud from the Console (chapter 2) and sign in. Click **Files** at the top.

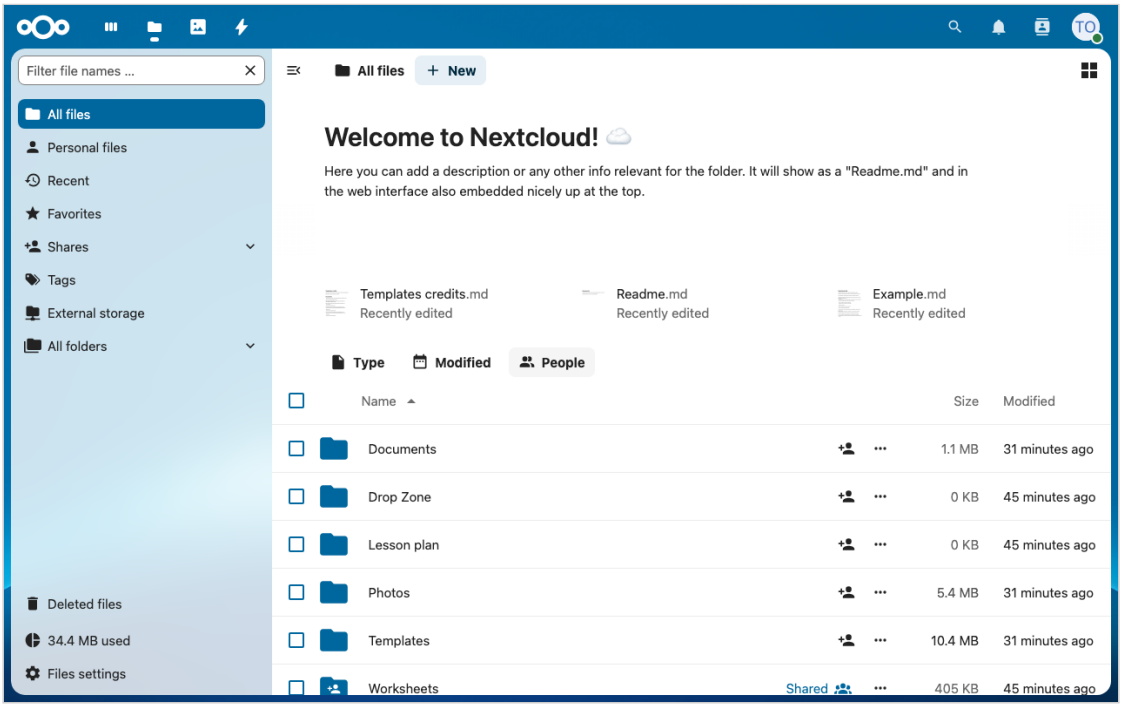


Figure 47. Files

6.1 Share worksheets with the class (view only)

1. Next to the folder `Worksheets`, click the share icon (**Show sharing options**). The right panel opens on **Sharing**.

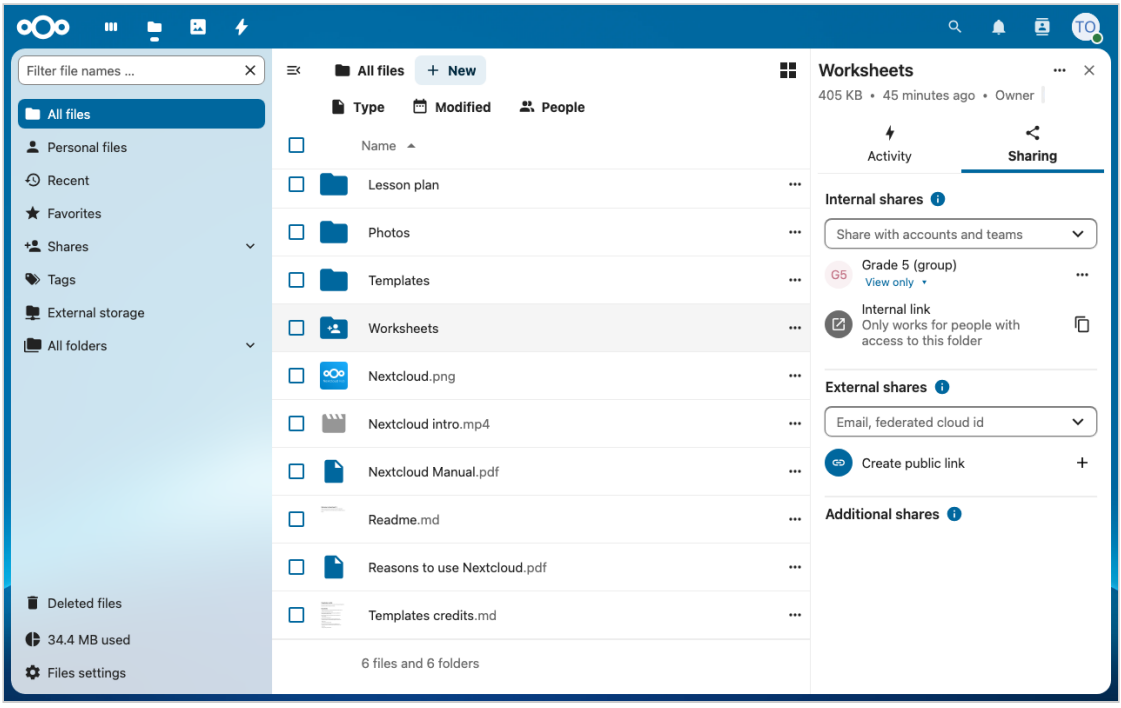


Figure 48. Sharing panel

2. In **Internal shares** ("Share with accounts and teams"), type `Grade 5` and pick the group.
3. Set the permission to **View only**. The group then appears under Internal shares.

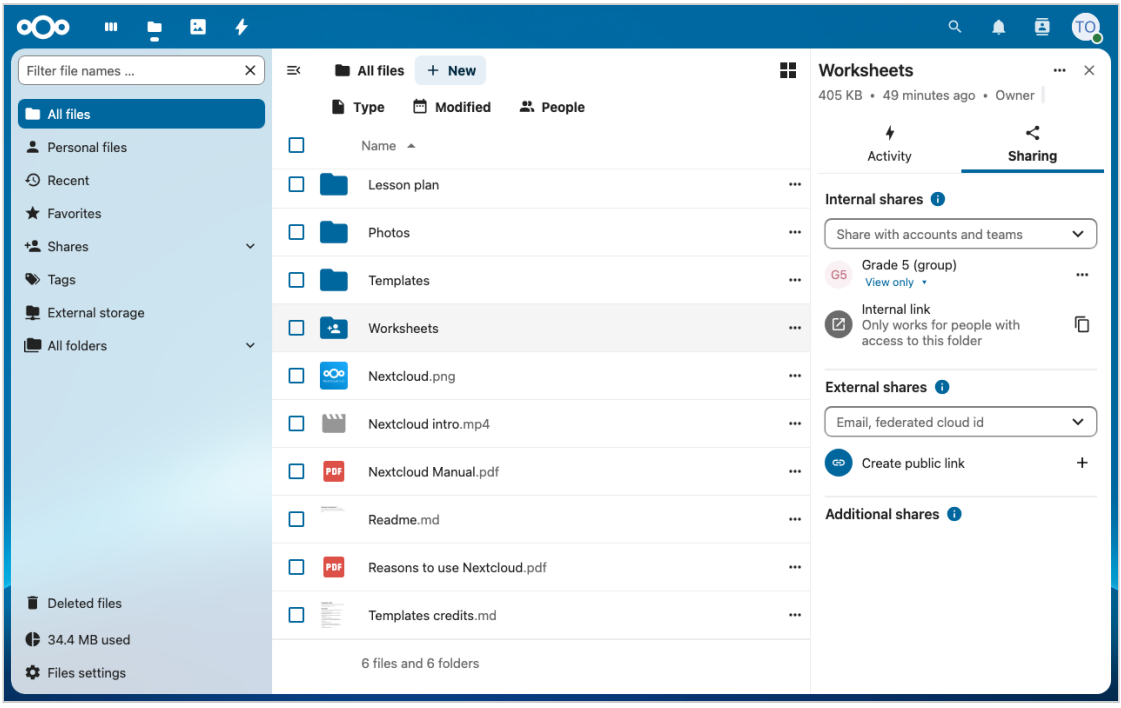


Figure 49. Shared with group

Learners now see the folder but cannot change or delete files.

6.2 Write a document together

1. Click **+ New > New document**.

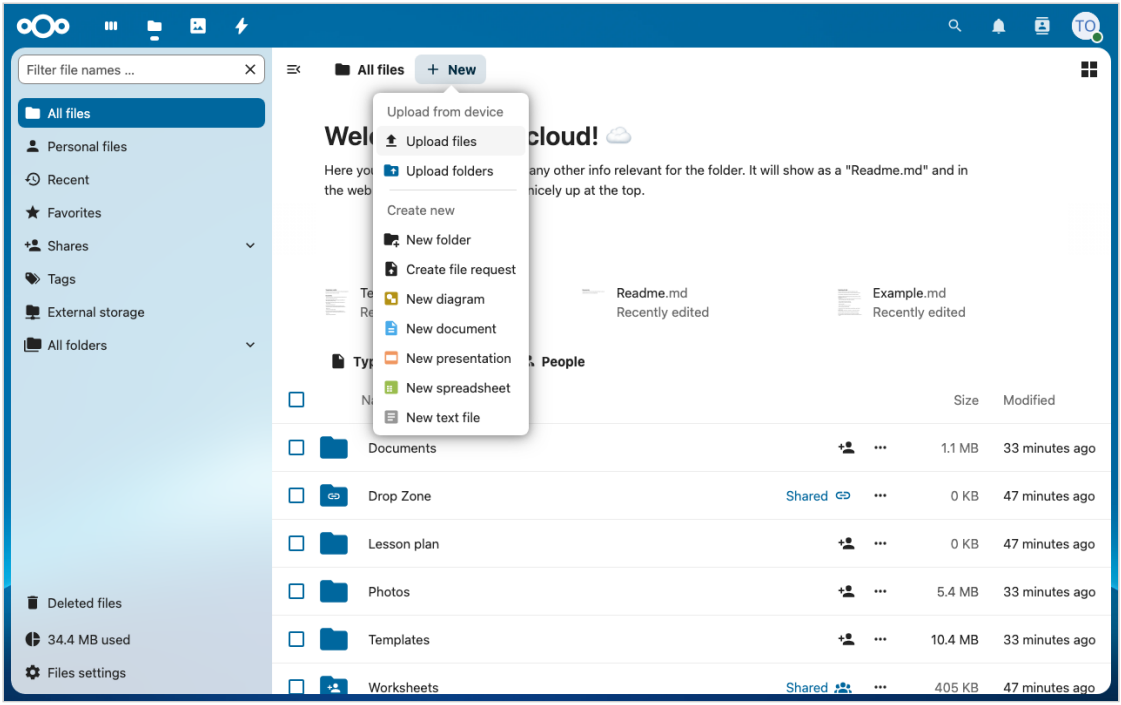


Figure 50. New menu

2. Type a name (`Lesson plan`) and click **Create**.

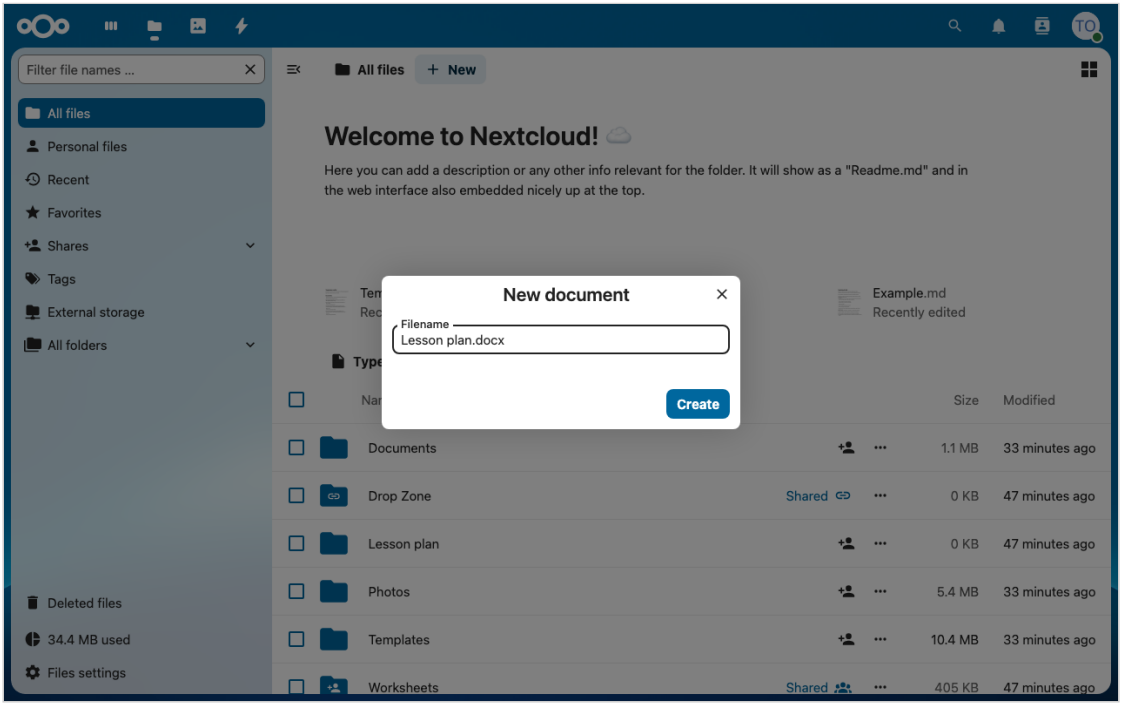


Figure 51. Name

3. Pick a template (Blank is fine) and click **Create**.

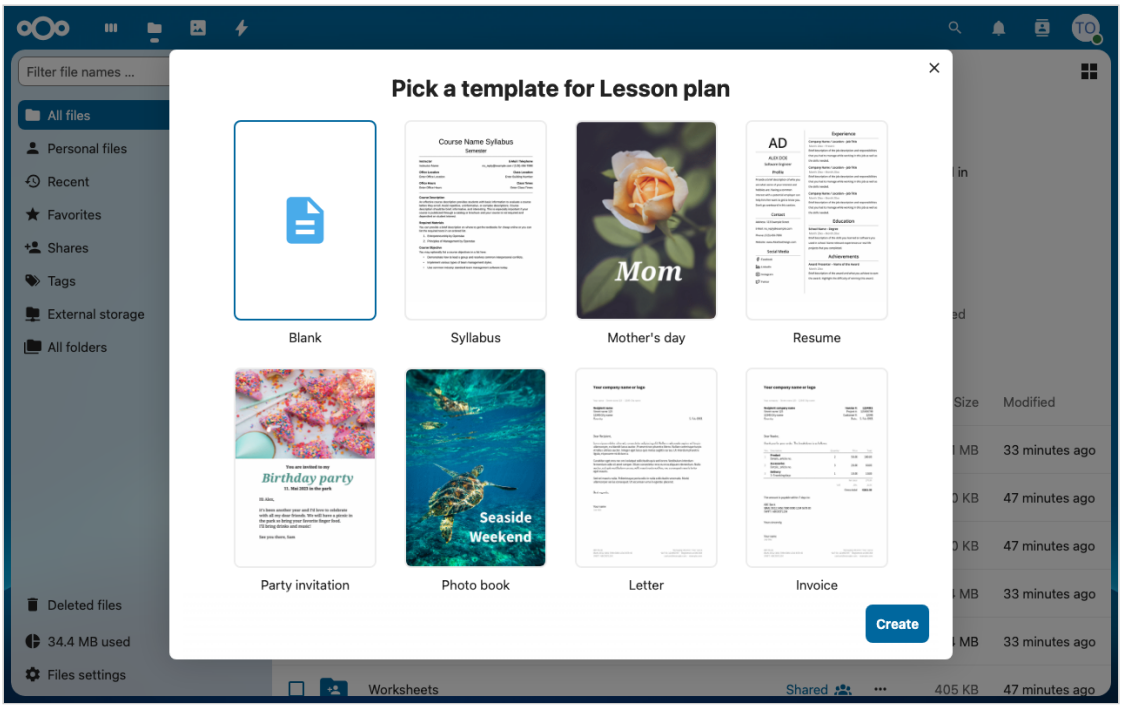


Figure 52. Template

4. Share the file with `teachers` and/or your class group, with editing allowed.

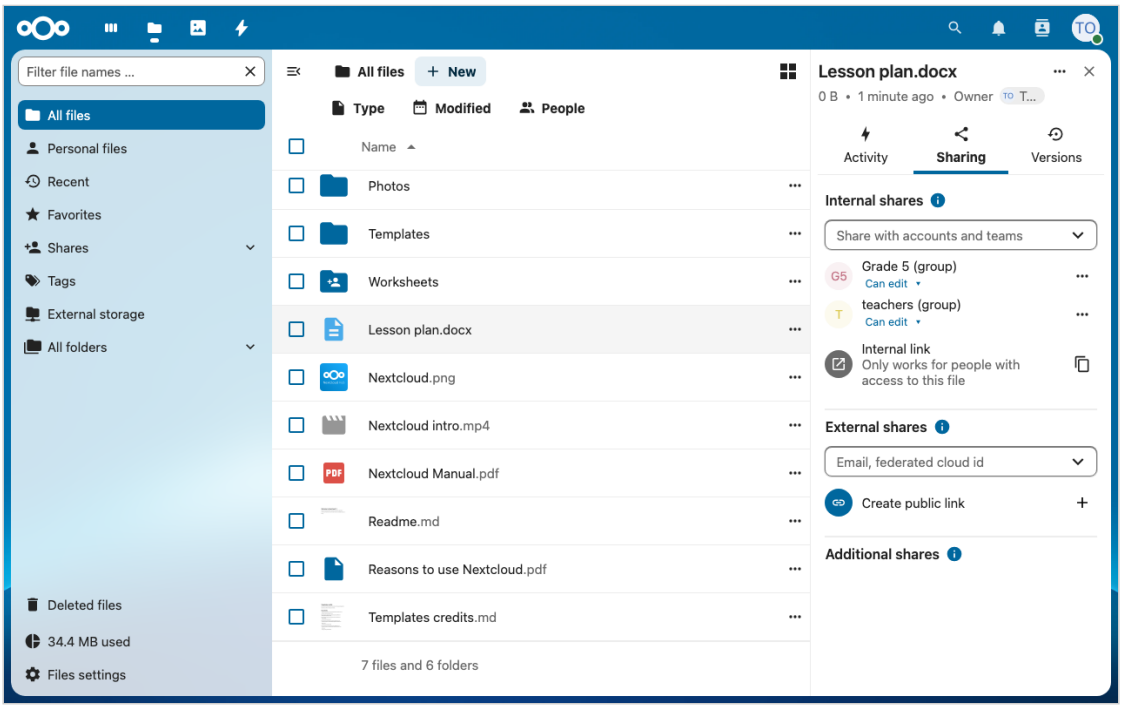


Figure 53. Shared document

5. The document opens in Nextcloud Office. When others open it at the same time, you see their names and cursors.

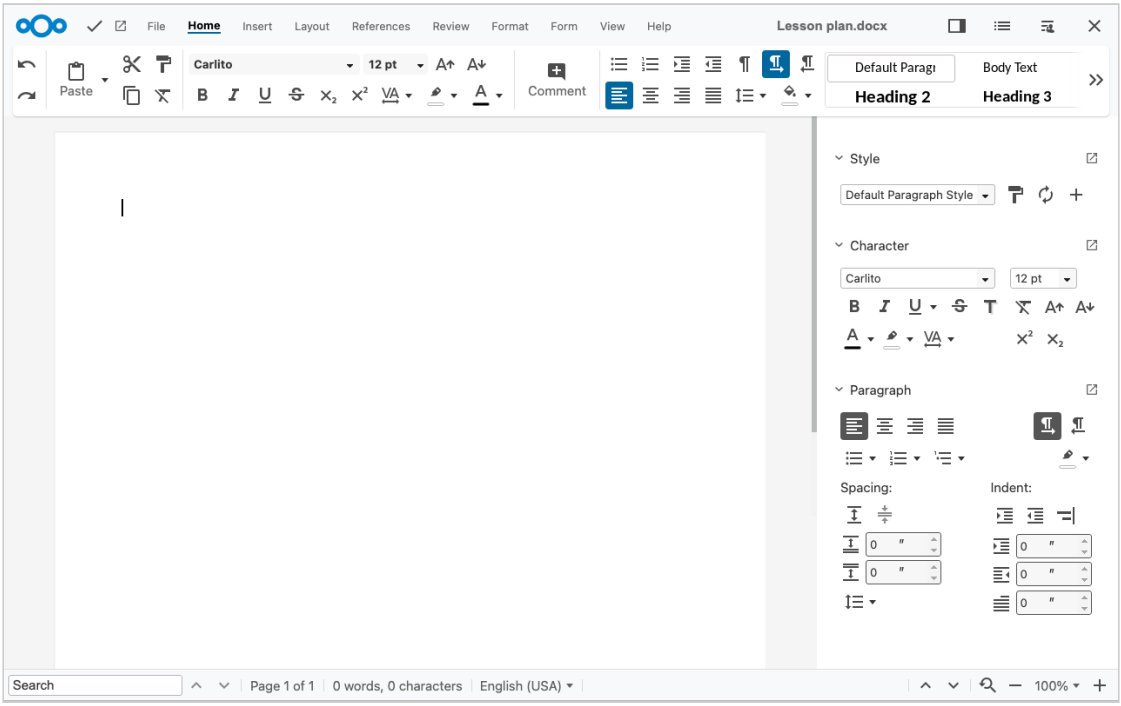


Figure 54. Nextcloud Office

6.3 Collect homework (File drop)

Learners upload work, but cannot see each other's work.

1. Open sharing for folder `Drop Zone`.

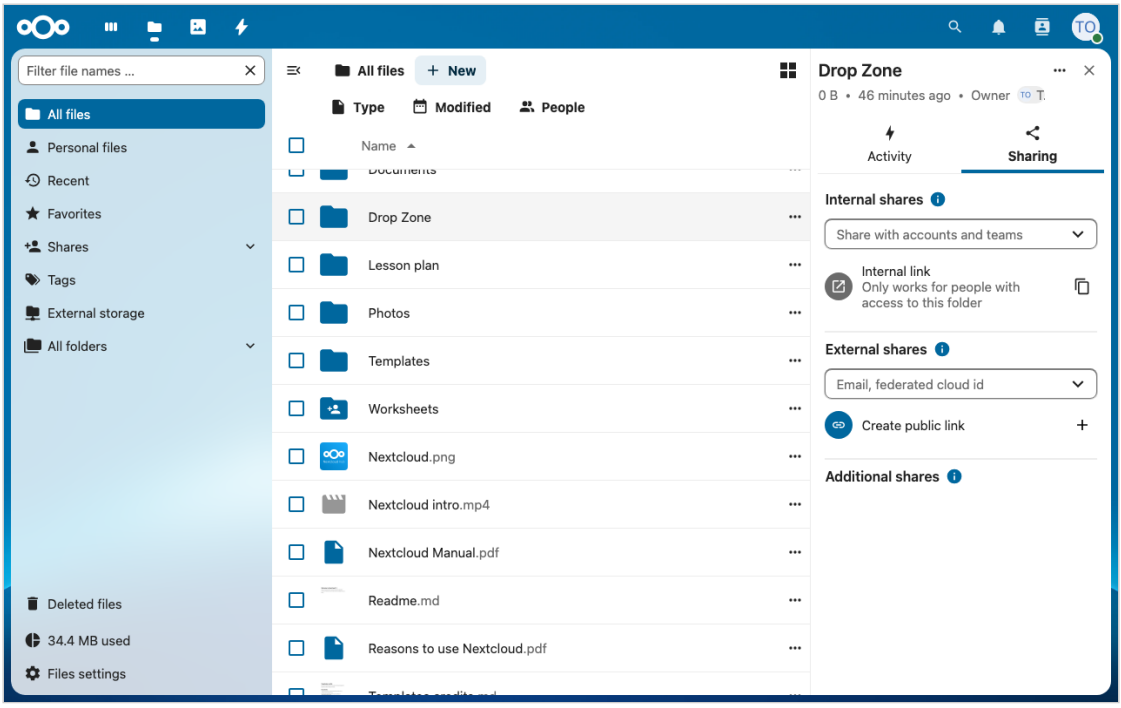


Figure 55. Sharing for Drop Zone

2. Under **External shares**, click **+** next to **Create public link** and choose **File request** (upload only). Copy the link.

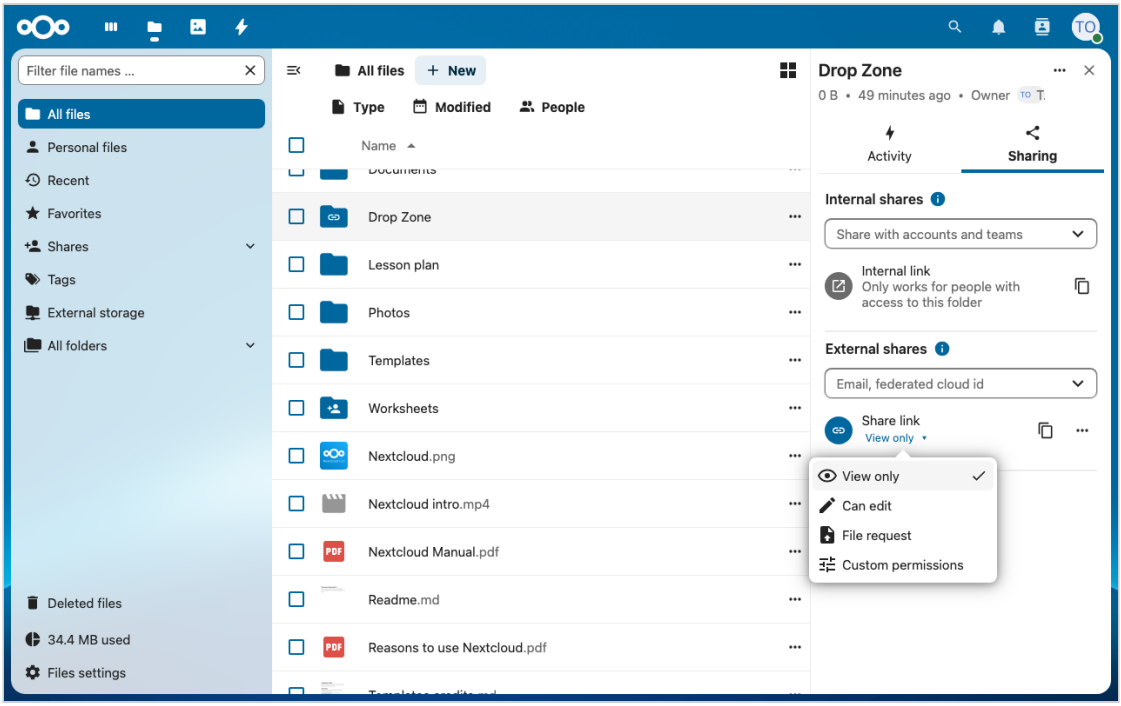


Figure 56. Create link

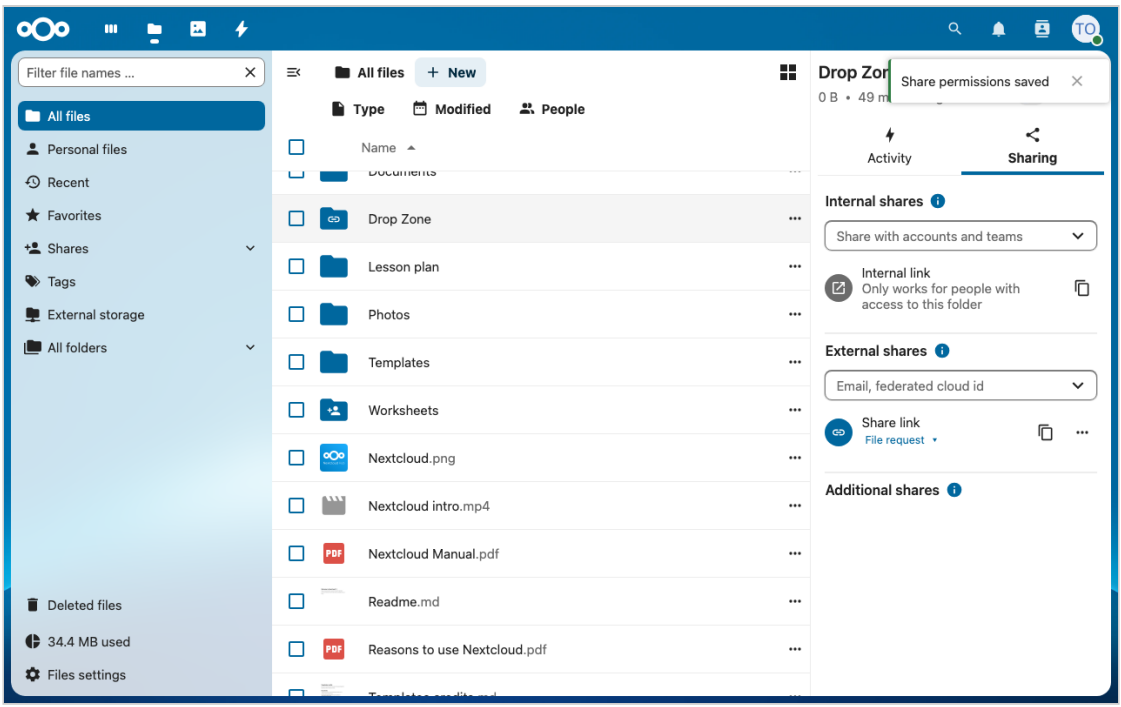


Figure 57. Link created

3. Give learners the link (for example, write it on the board). Uploaded files appear in your `Drop Zone` folder.

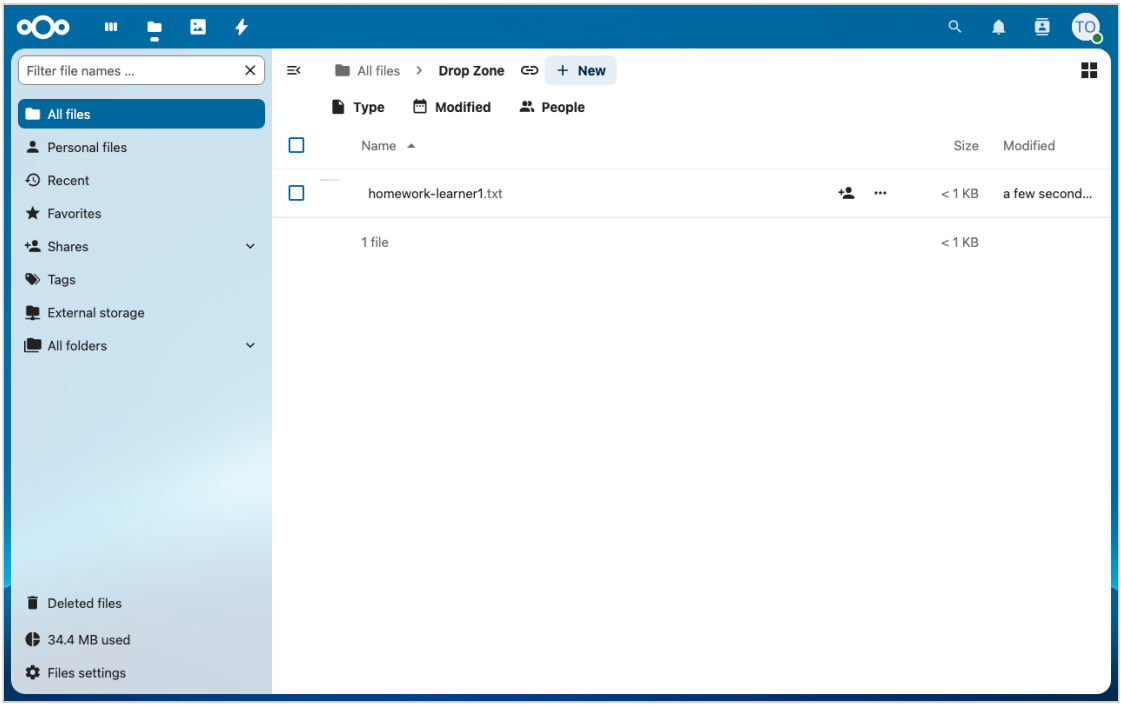


Figure 58. Received homework

7. Nextcloud for learners (try it yourself)

Learners open Nextcloud from the Console, like Kolibri: **IDEA** on the desktop, then **Open** on the Nextcloud card (2.2.1). They have no personal Nextcloud app on the classroom computer, so they always work in the browser.

7.1 Open class files and co-edit

1. Sign in to Nextcloud as `learner1` (private window). Shared folders and files appear in **Files** with the teacher's name.

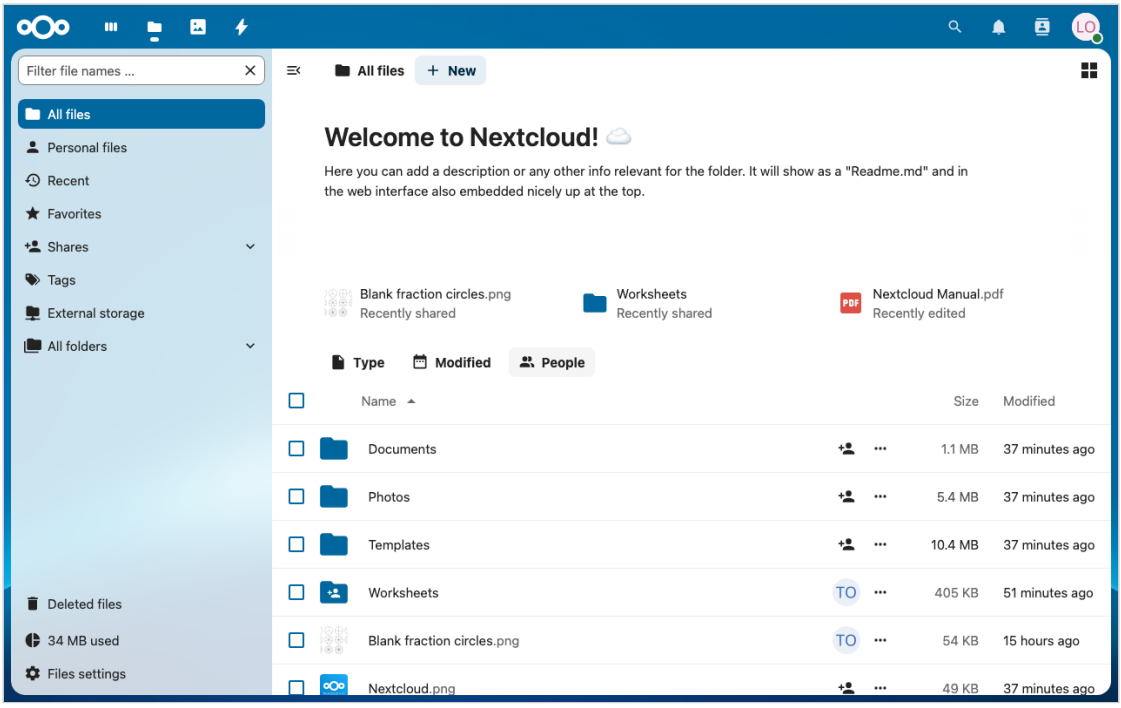


Figure 59. Learner files

2. Open `Worksheets`: the learner can open and download the worksheets, not change them.

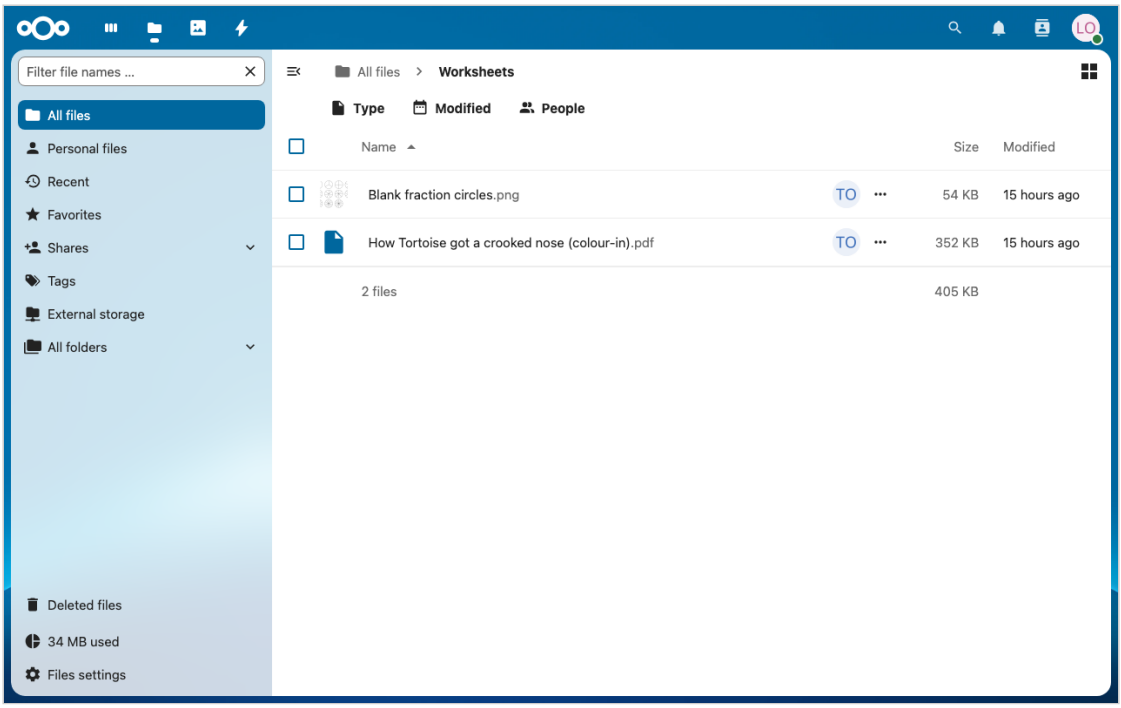


Figure 60. Worksheets

3. Open `Lesson plan.docx`: it opens in Nextcloud Office and the learner can type.

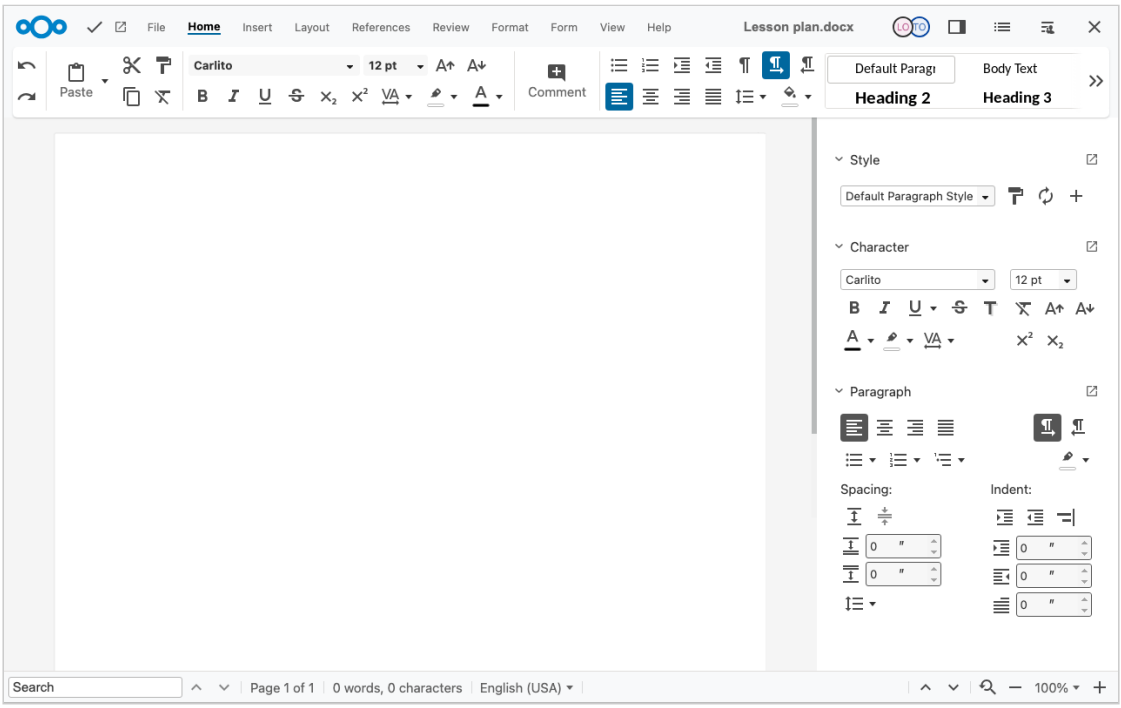


Figure 61. Learner in document

7.2 Hand in homework

1. Open the File drop link from 6.3. The page says **File drop** and shows an **Upload** button; the learner cannot see other files.

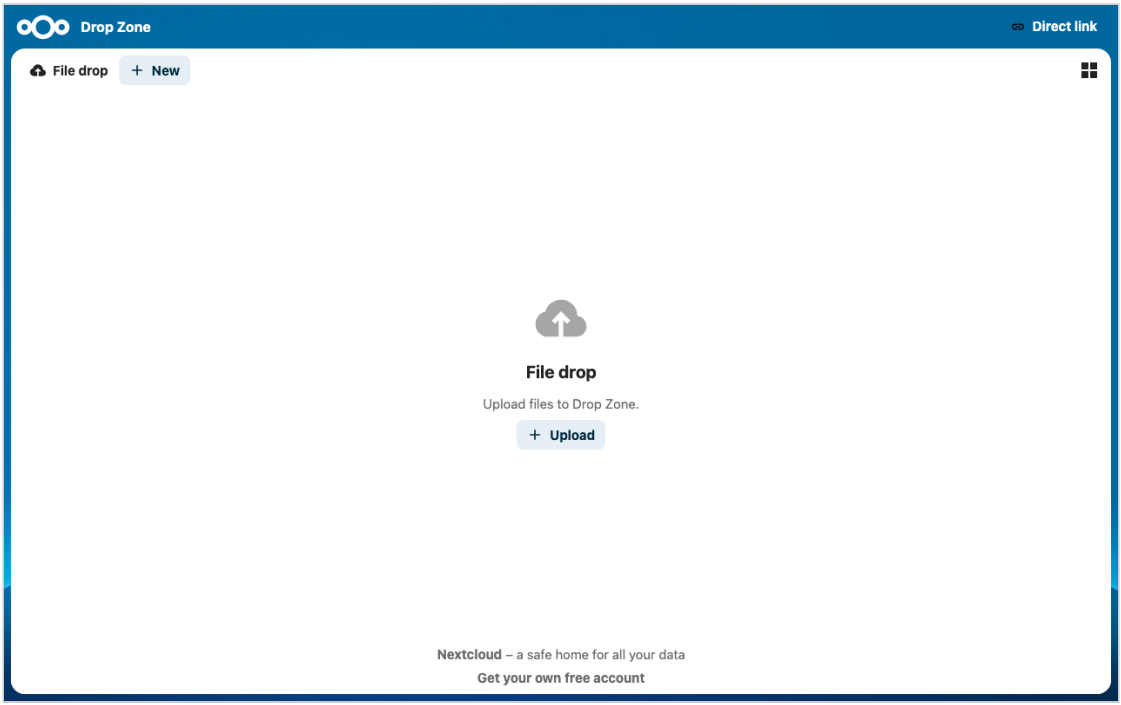


Figure 62. File drop

2. Click **Upload** and choose the file (e.g. a photo of written work). If asked, type your name.

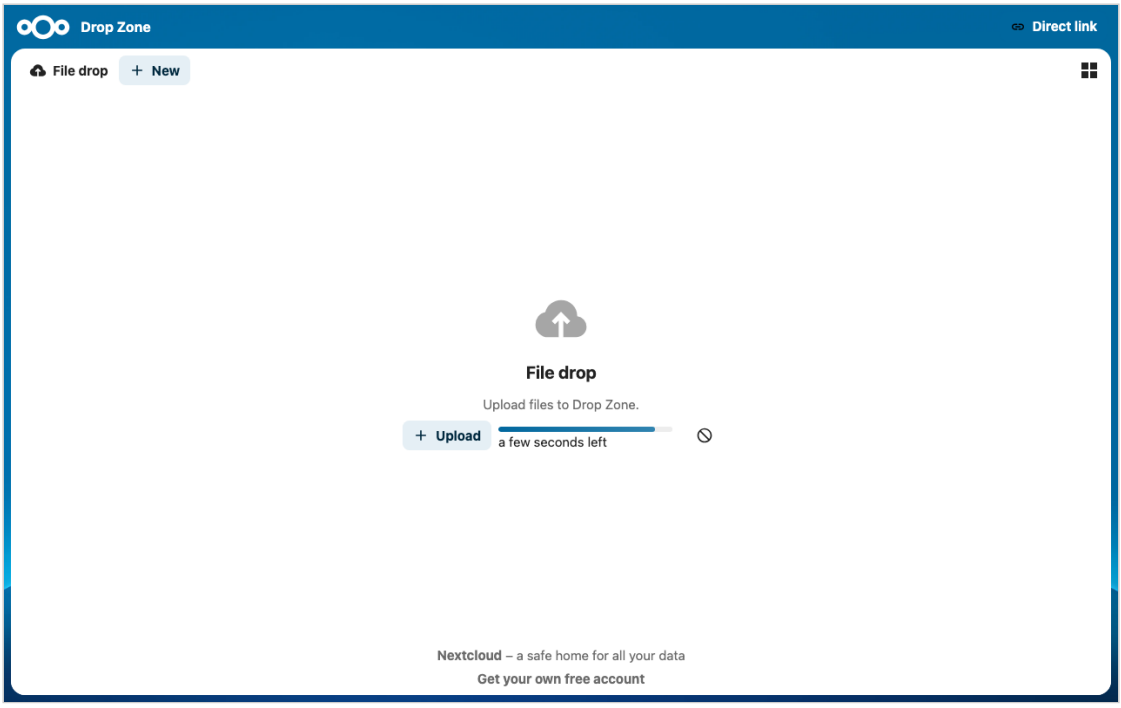


Figure 63. Uploading: the bar shows the progress. When it is gone, the file is in the teacher's folder

The teacher sees the file in `Drop Zone` (6.3).

8. Wikipedia offline (Kiwix)

The demo has **Wikipedia in simple English** (text, no pictures). It works without internet.

8.1 Look something up and use it in class

1. Open Kiwix from the Console (chapter 2). The **Kiwix** library shows the books available.

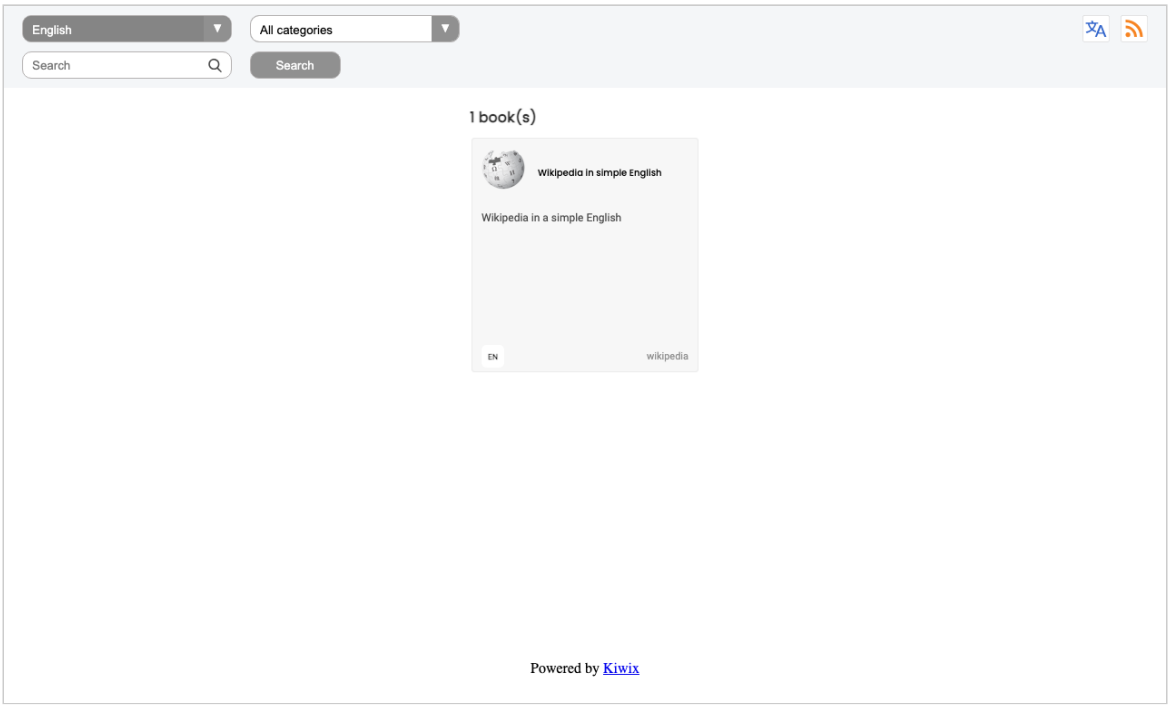


Figure 64. The Kiwix library

2. Click **Wikipedia in simple English**. Its welcome page opens.



Figure 65. Wikipedia home

3. Type in the search box at the top, e.g. `Fraction`. Pick a result.

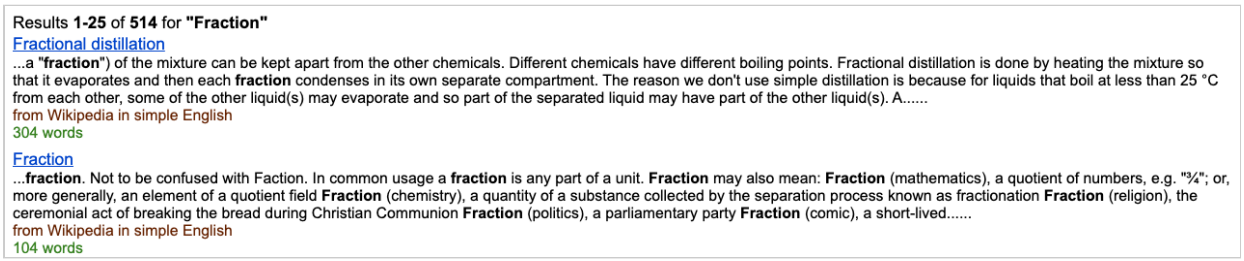


Figure 66. Search results

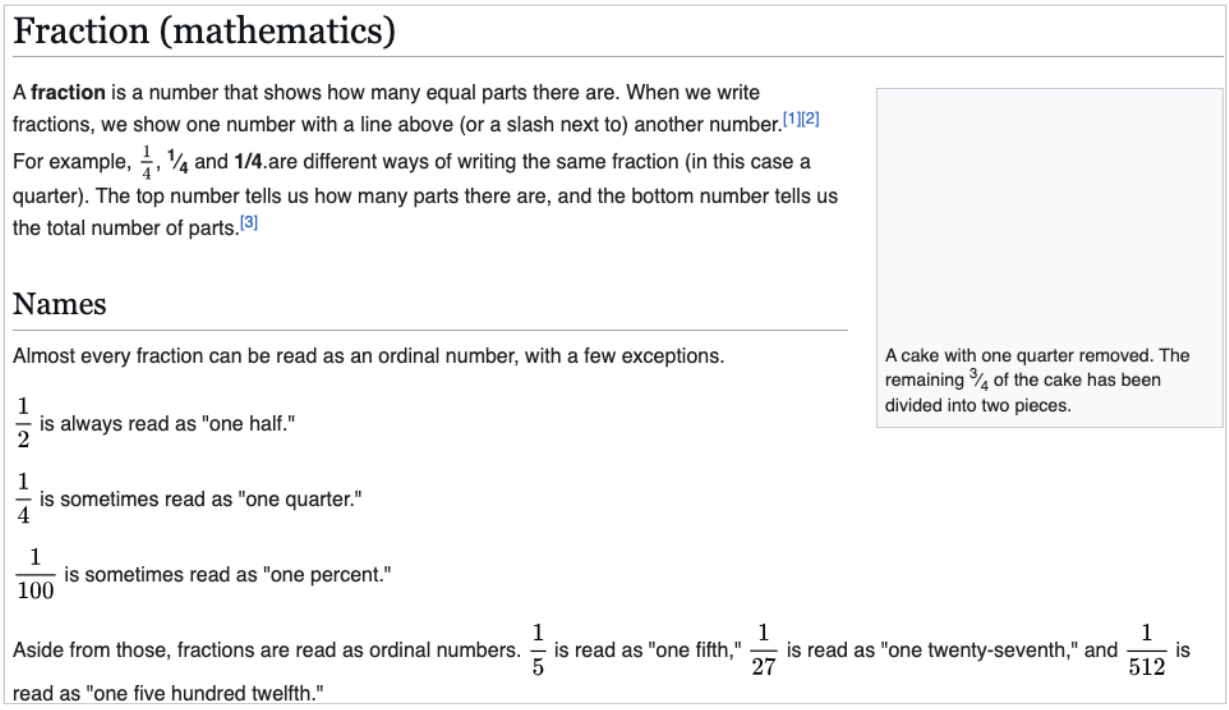
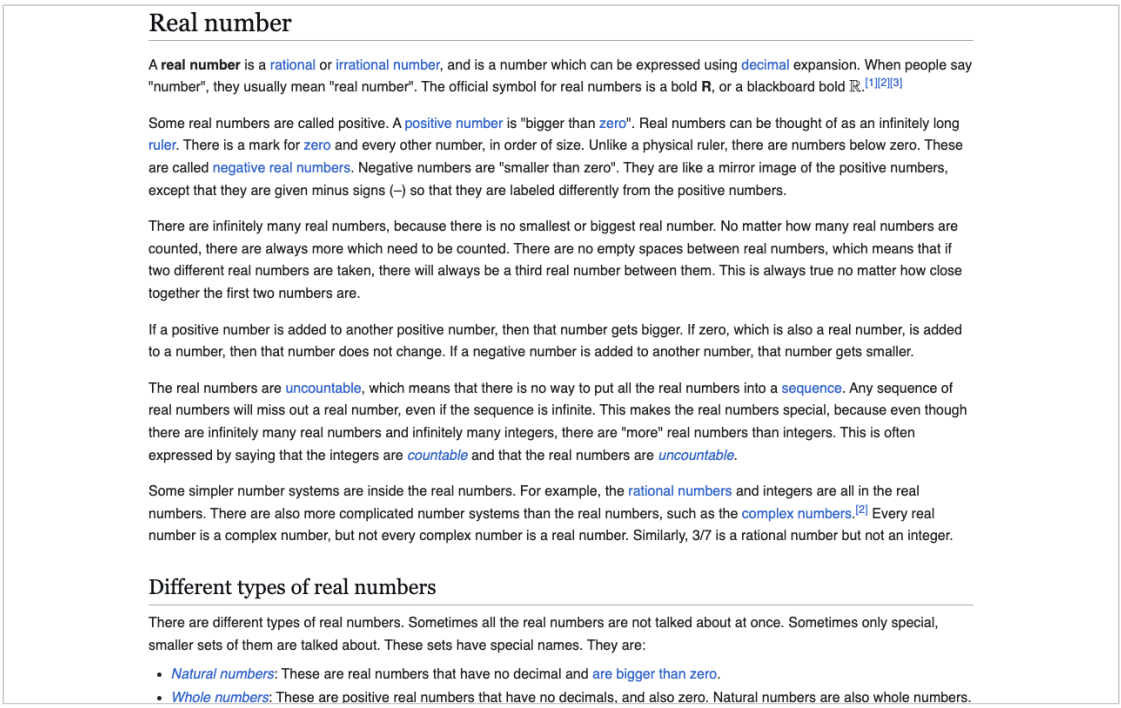


Figure 67. Fraction article

4. Follow blue links to related articles.



Real number

A **real number** is a rational or irrational number, and is a number which can be expressed using decimal expansion. When people say "number", they usually mean "real number". The official symbol for real numbers is a bold **R**, or a blackboard bold $\mathbb{R}$.[1][2][3]

Some real numbers are called positive. A positive number is "bigger than zero". Real numbers can be thought of as an infinitely long ruler. There is a mark for zero and every other number, in order of size. Unlike a physical ruler, there are numbers below zero. These are called negative real numbers. Negative numbers are "smaller than zero". They are like a mirror image of the positive numbers, except that they are given minus signs (–) so that they are labeled differently from the positive numbers.

There are infinitely many real numbers, because there is no smallest or biggest real number. No matter how many real numbers are counted, there are always more which need to be counted. There are no empty spaces between real numbers, which means that if two different real numbers are taken, there will always be a third real number between them. This is always true no matter how close together the first two numbers are.

If a positive number is added to another positive number, then that number gets bigger. If zero, which is also a real number, is added to a number, then that number does not change. If a negative number is added to another number, that number gets smaller.

The real numbers are uncountable, which means that there is no way to put all the real numbers into a sequence. Any sequence of real numbers will miss out a real number, even if the sequence is infinite. This makes the real numbers special, because even though there are infinitely many real numbers and infinitely many integers, there are "more" real numbers than integers. This is often expressed by saying that the integers are *countable* and that the real numbers are *uncountable*.

Some simpler number systems are inside the real numbers. For example, the rational numbers and integers are all in the real numbers. There are also more complicated number systems than the real numbers, such as the complex numbers.[2] Every real number is a complex number, but not every complex number is a real number. Similarly, 3/7 is a rational number but not an integer.

Different types of real numbers

There are different types of real numbers. Sometimes all the real numbers are not talked about at once. Sometimes only special, smaller sets of them are talked about. These sets have special names. They are:

- *Natural numbers*: These are real numbers that have no decimal and are bigger than zero.
- *Whole numbers*: These are positive real numbers that have no decimals, and also zero. Natural numbers are also whole numbers.

Figure 68. Linked article

> **Classroom idea.** give learners 3 questions to answer from one article; they read on their own device.

9. Your own files: the Files Disk

A **Files Disk** is an **IDEA Disk** that the operator prepares in the Console (chapter 12). Its content appears in Nextcloud as the folder **School Files**. Use it to bring in material you downloaded elsewhere, for example worksheets you copied at another school.

> **Only a USB stick?** You can use a normal USB stick without a Files Disk: it appears in Nextcloud as a read-only **USB:** folder (5.5).

Good to know:

- **School Files** is only in **Nextcloud**, not in Kolibri.
- Only teachers see the **School Files** folder. Learners get only the files you share with them.
- **The Backup Disk does not save School Files.** Keep your own copy of important files.

9.1 Open School Files

1. Open Nextcloud from the Console and sign in. In **Files** you see the folder **School Files** (marked as **External storage** in the left menu).

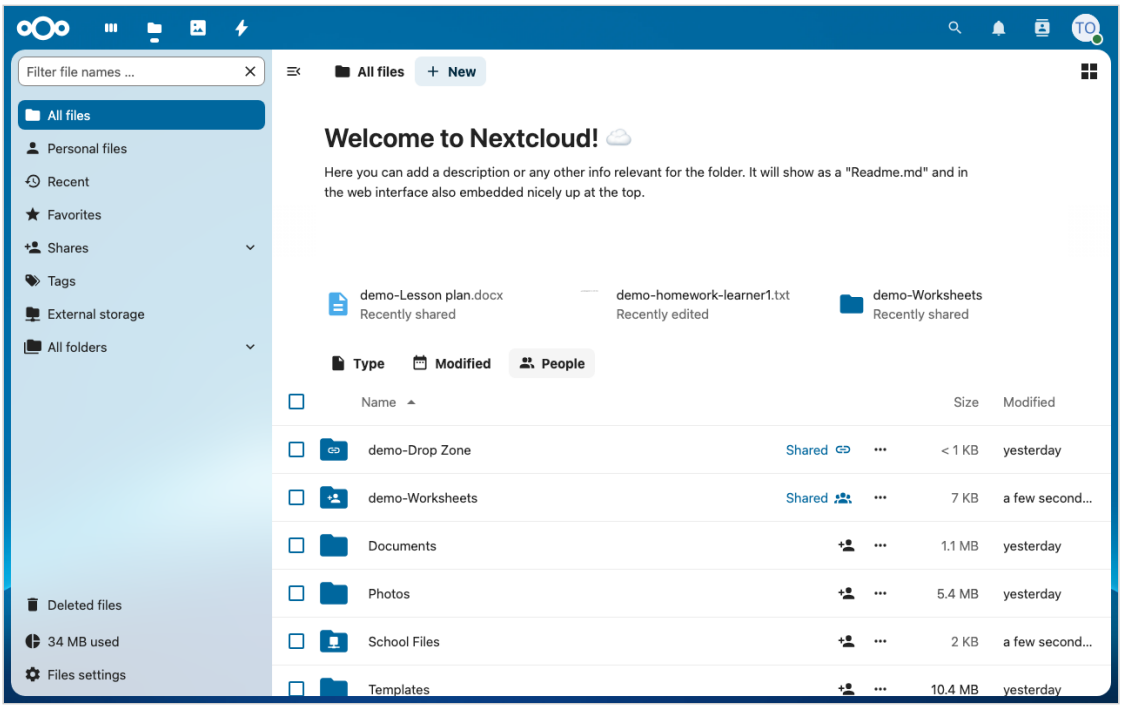


Figure 69. School Files in Nextcloud

2. Open **School Files** to see what is on the disk (here `Fractions-Worksheet-1.pdf`).

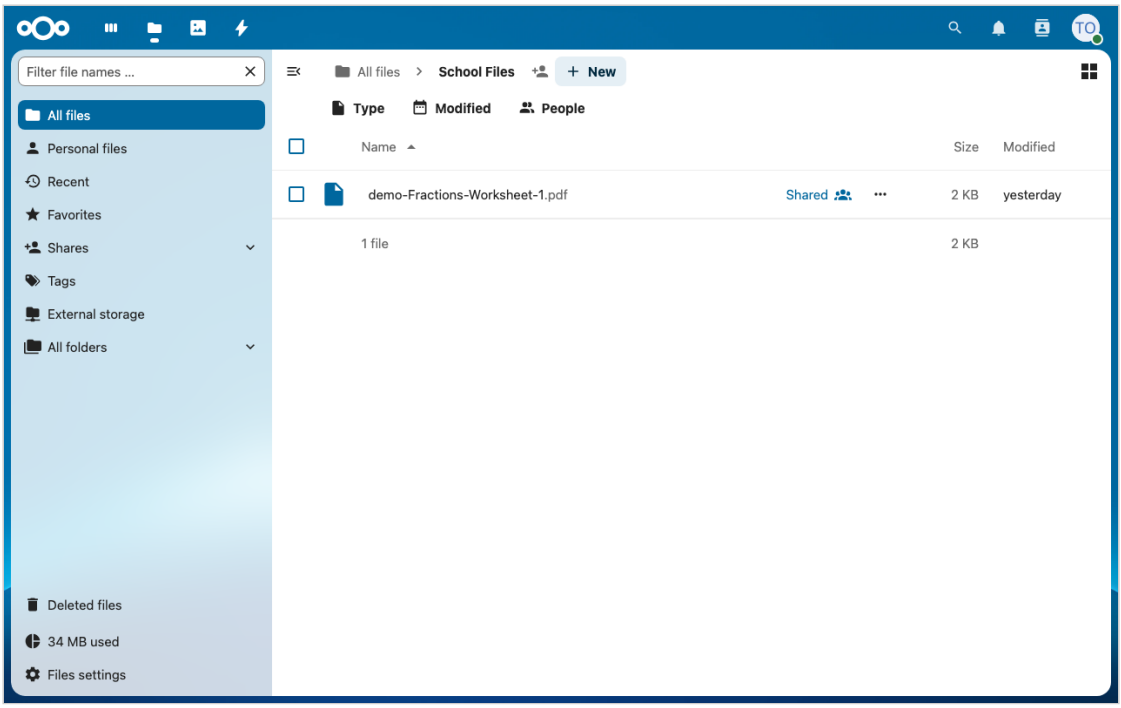


Figure 70. School Files content

9.2 Share a file from School Files with your class

1. On the file's row click **··· (Actions) > Show sharing options**, or click the share icon.
2. In **Internal shares**, type the class group name (here `Grade 5`) and choose it from the list. Keep **View only**. The group then appears under **Internal shares**.

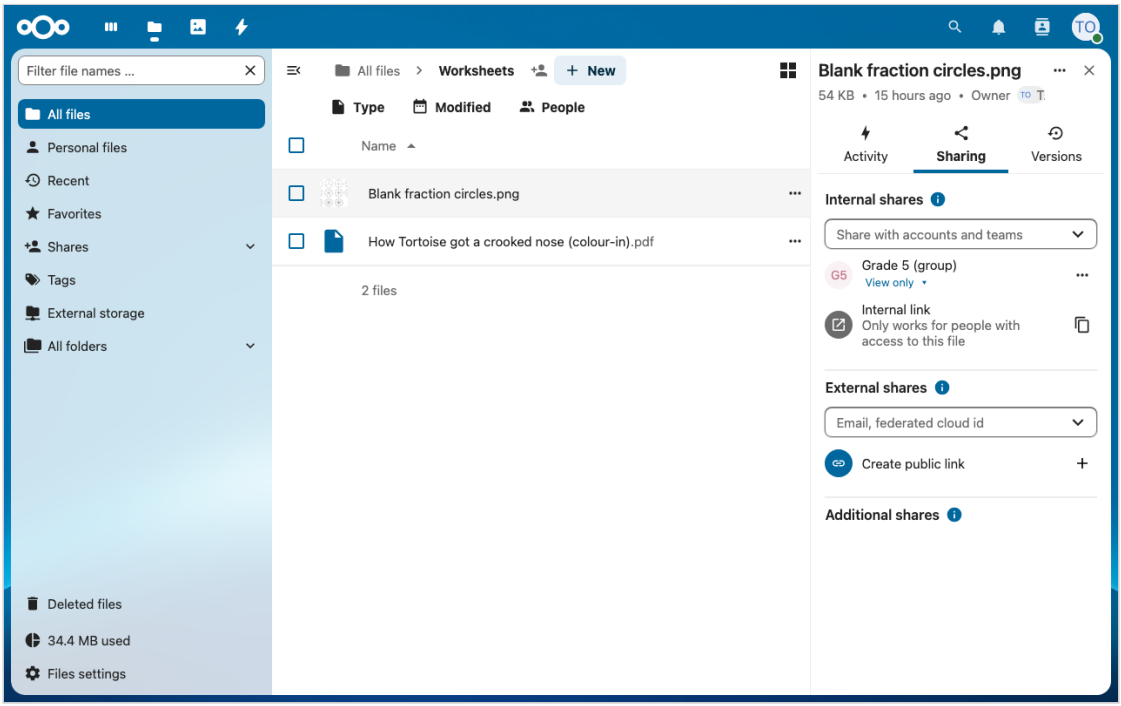


Figure 71. A file shared with Grade 5 (here a worksheet)

3. A learner in that group signs in and finds the file in **All files** and under **Recently shared**.

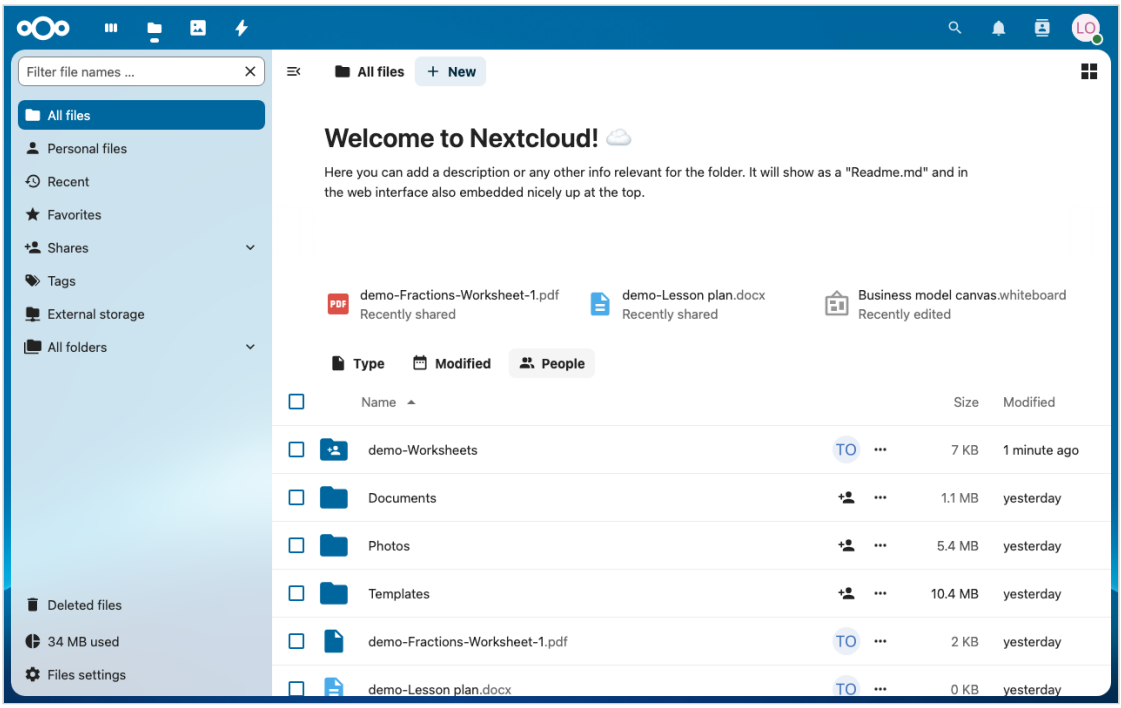


Figure 72. A learner sees the shared file

9.3 Can I put my own files into Kolibri?

Yes, with the **channel builder**: see chapter 5. It turns a folder of your files (on a USB stick, in School Files or in Nextcloud) into a Kolibri channel.

10. Keeping data safe: the Backup Disk

The operator looks after backups (chapter 12). As a teacher you should know:

- A **Backup Disk** is a separate **IDEA Disk** with the **backup** badge in the Console.
- It saves the **apps' data**: Kolibri (classes, learners, progress) and Nextcloud (accounts, and the files inside Nextcloud).
- It does **not** save the **School Files** folder (chapter 9).
- During a backup, the app stops for a short time and then starts again. Backups are planned outside lesson time.
- How often backups run (on request, each time the disk is plugged in, or on a schedule) is set by your IDEA contact.
- If the IDEA box breaks, the operator restores the latest backup onto a spare disk. Only the work done after the last backup is lost.
- Never unplug a disk while a backup is running.

Part B: For the school operator

11. The Console for the operator

This chapter is for the school operator. Do not press **Start**, **Stop** or other action buttons unless your IDEA contact asks you to.

11.1 Operator Login

1. In the Console, click the person icon (top right). The **Operator Login** panel opens with a username and a password field (the eye icon shows or hides the password).

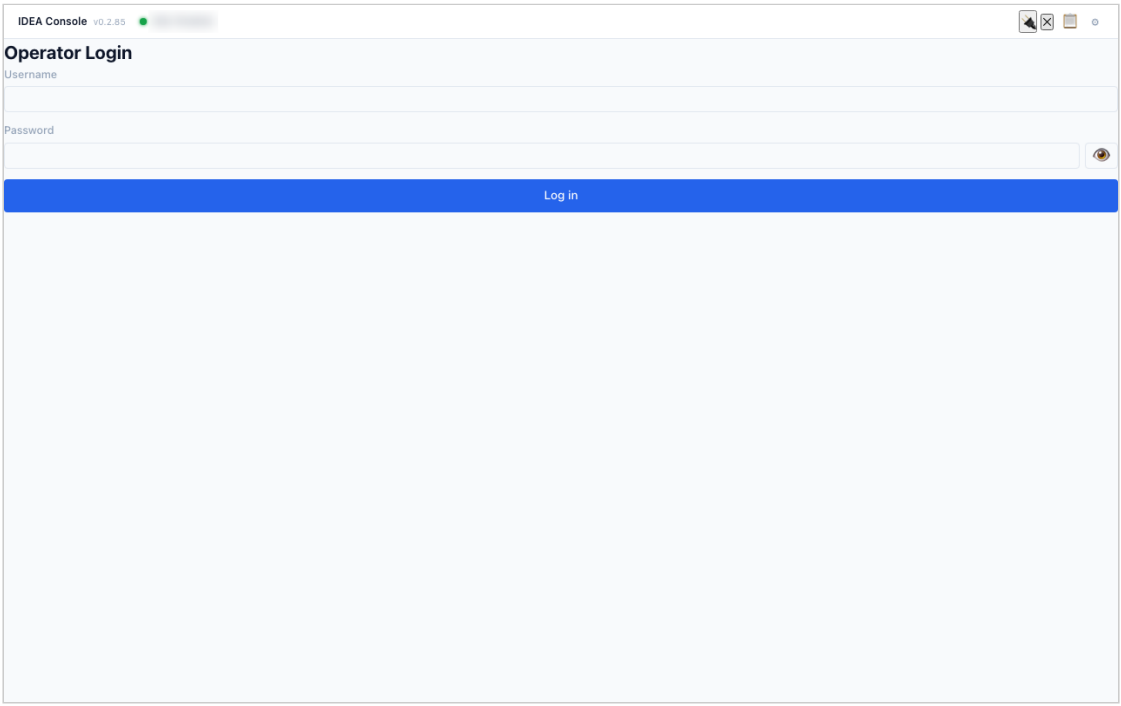


Figure 73. Operator Login

2. Type the operator username and password and click **Log in**. The **Account** panel shows your name, the role **Operator**, **Change Password**, **Manage Operators** and **Log out**.

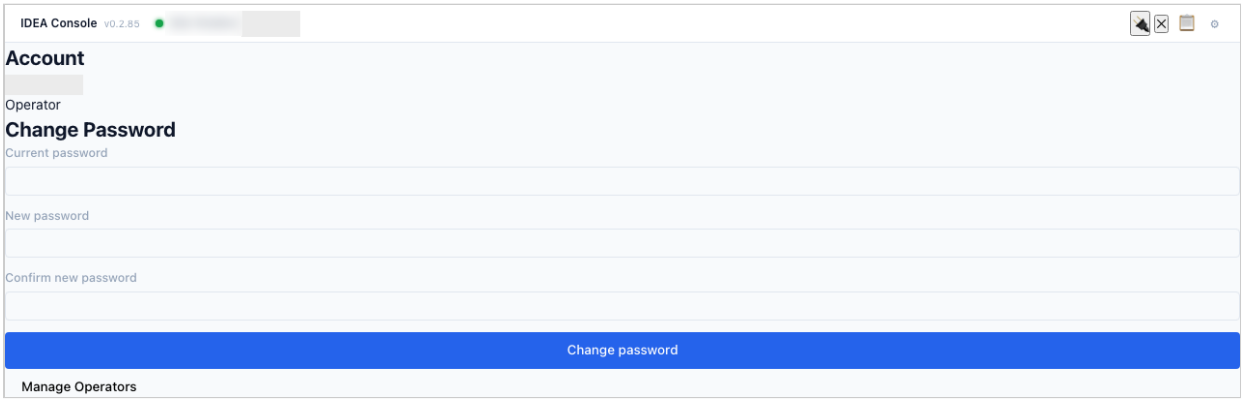


Figure 74. Account panel

3. **Change the default operator password straight away** on a new school box: **Account > Change Password**, type the current password, the new password twice, and click **Change password**. Keep the new password with the school head, not on a sticker on the box.

4. Close the panel with ×. Use **Log out** in the same panel when you are done.

11.2 The Network view: the IDEA box and its disks

1. After login the left side shows **NETWORK**: **ALL APPS**, then the IDEA box with a green dot (online) and its disks.

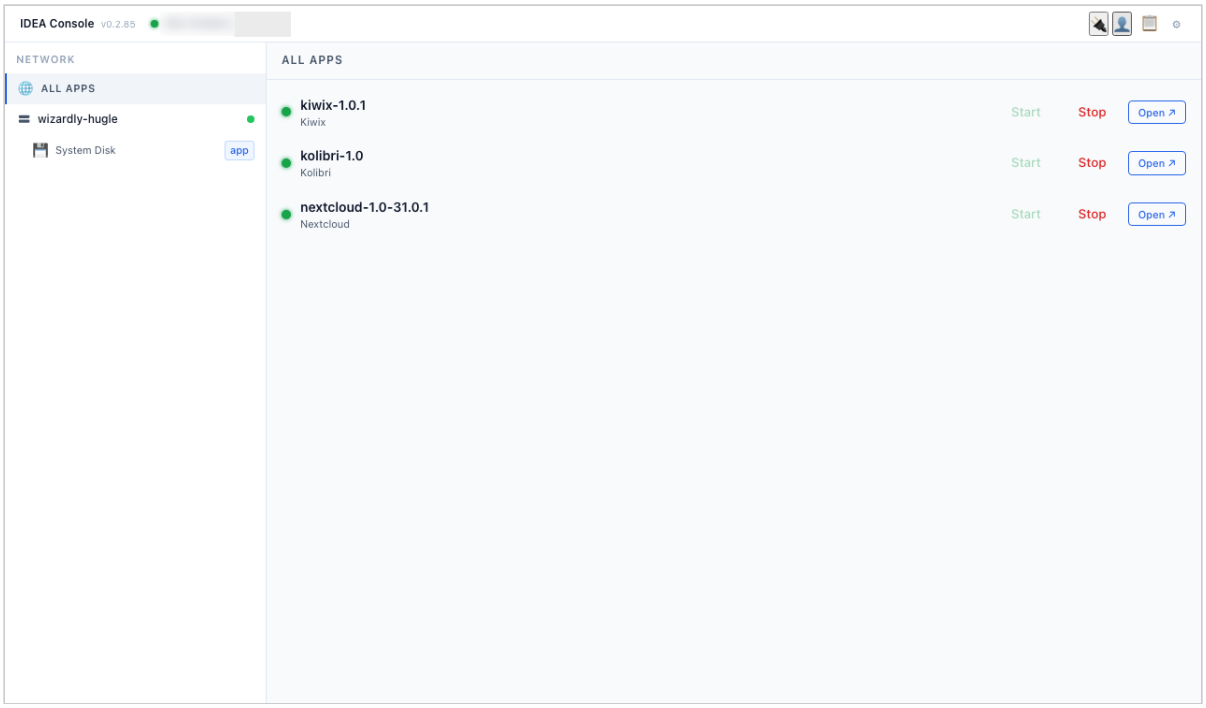


Figure 75. All apps

2. Click the IDEA box name to select it, then click a disk (here **System Disk**). Each disk shows a role badge: **app**, **backup**, **files** or **empty**. The right side lists the apps on that disk.

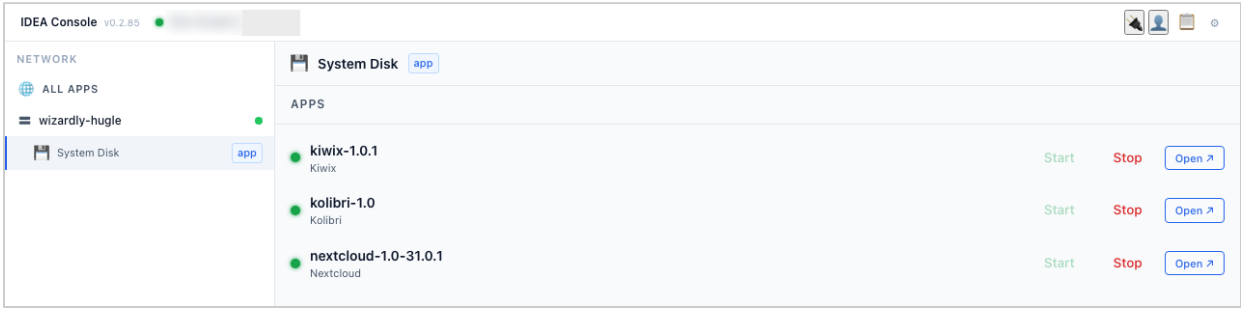


Figure 76. Disk view

11.3 App status

1. Each app row shows the app name and version, a coloured status dot, and the buttons **Start**, **Stop** and **Open ↗**.

2. Click the status dot (**Show details**) to see **STATUS** (for example **Running**), **VERSION**, **PORT**, **CREATED**, **LAST STARTED**, **LAST BACKUP** and the description.

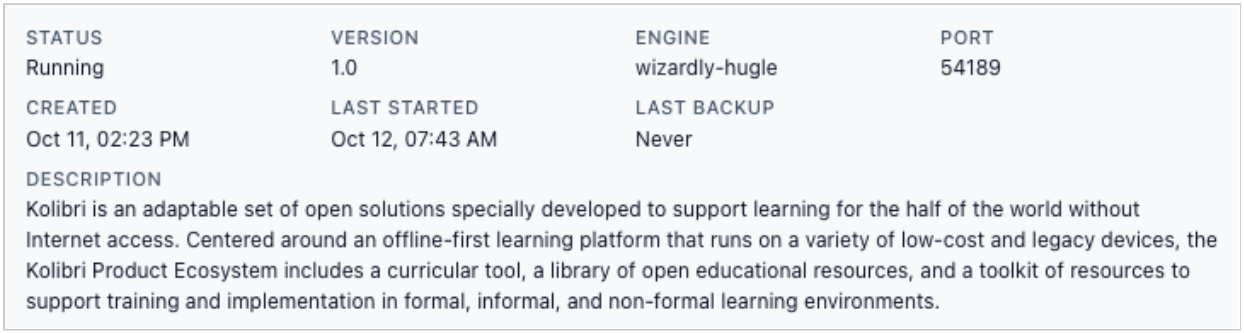


Figure 77. App details

3. To check that an app really works, click **Open ↗**: the app's page must load.
4. Only start or stop an app when your IDEA contact asks you to. Do not touch apps your IDEA contact did not mention.

11.4 Command History

The clipboard icon (top right) lists recent Console commands with ✓ (done) or ✗ (failed) and how long ago. Read it out to your IDEA contact when you report a problem.

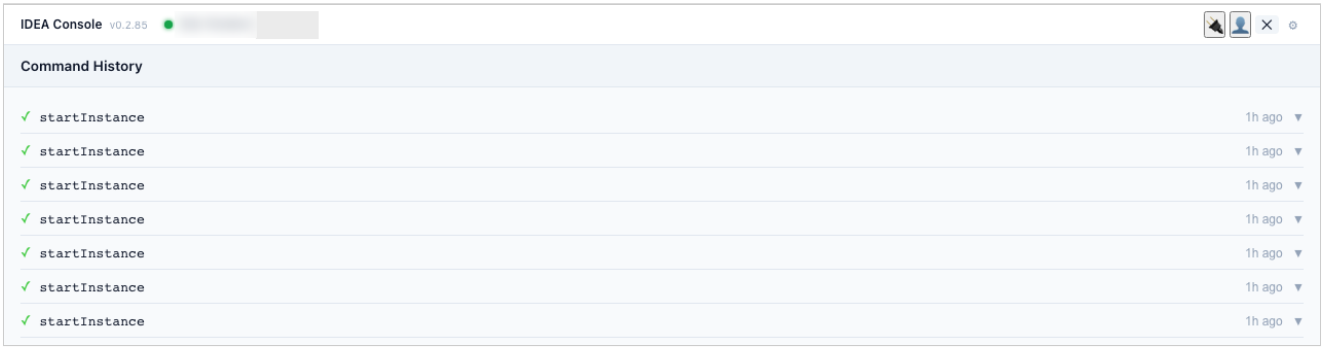


Figure 78. Command History

11.5 Buttons to leave alone

- The small ↺ next to the IDEA box name is **Reboot**: it restarts the whole IDEA box. Never press it during lessons.
- **Erase this disk…**, **Install App**, **Make this a Files Disk**, **Make this a Backup Disk**, **Restore** and **Stop**: only with your IDEA contact on the phone. The System Disk has no Erase button.

12. Disks: Files Disk and Backup Disk

This chapter is for the school operator, with the IDEA contact on the phone.

12.1 A new disk (not yet an IDEA Disk)

1. A new disk appears under the IDEA box with its model and size and an **UNFORMATTED** badge.

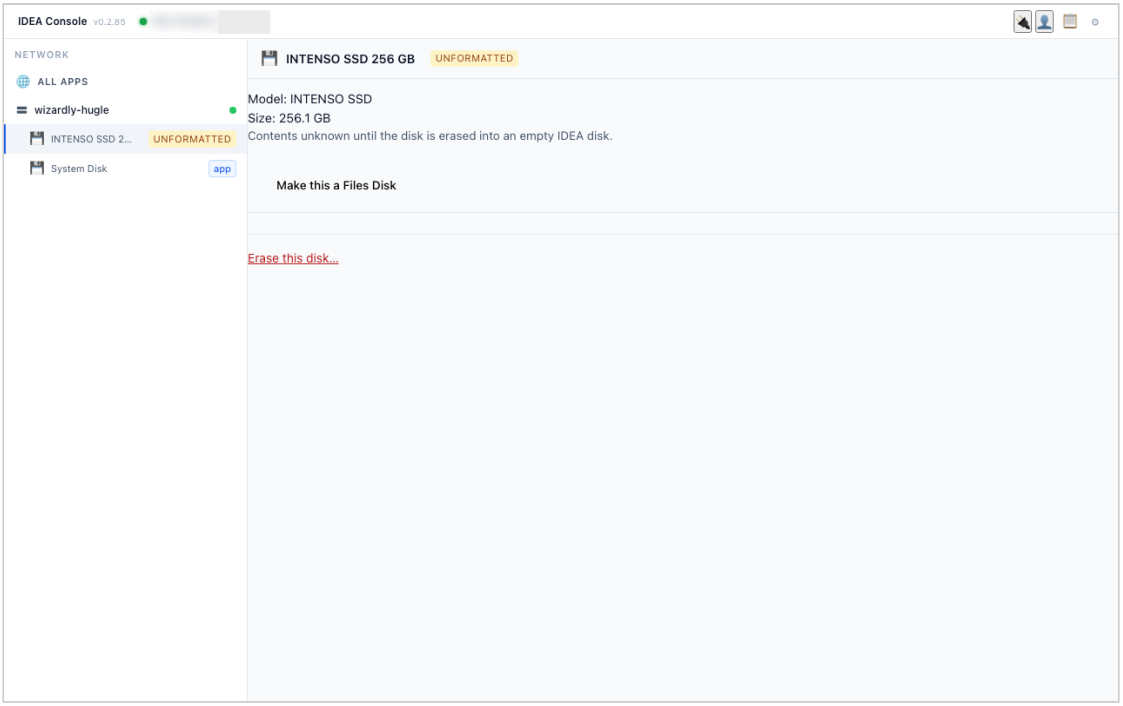


Figure 79. Unformatted disk

2. Its view shows the model, the size and two buttons: **Make this a Files Disk** and **Erase this disk…**.

USB sticks. A teacher's FAT32 or exFAT stick shows as **USB stick: (name) (read-only)**. It needs no erasing: the IDEA box only reads it, and teachers use it in Nextcloud and the channel builder (chapter 5). **Erase this disk…** is still offered, but erasing turns the stick into an IDEA Disk and deletes the teacher's files, so only do it when asked. A stick with a `kolibri` folder is built into channels automatically only if the channel builder is on this box and not stopped: if you stopped it, the stick row says so. Start it again to build.

12.2 Erase a disk

1. Click **Erase this disk…**. The confirmation shows the disk name, model, size and contents, and the red warning "This erases everything on the disk…".
2. Check the **size and contents**: several disks can have the same name (for example **IDEA Disk**). Make sure it is not the Backup Disk.

3. The box above the text field says **TYPE IDEA DISK TO CONFIRM** (the disk name in capitals). Type the name as it is written on the disk, for example `IDEA Disk`. The **Erase this disk** button stays grey until the name is right. Press **Cancel** if anything looks wrong.

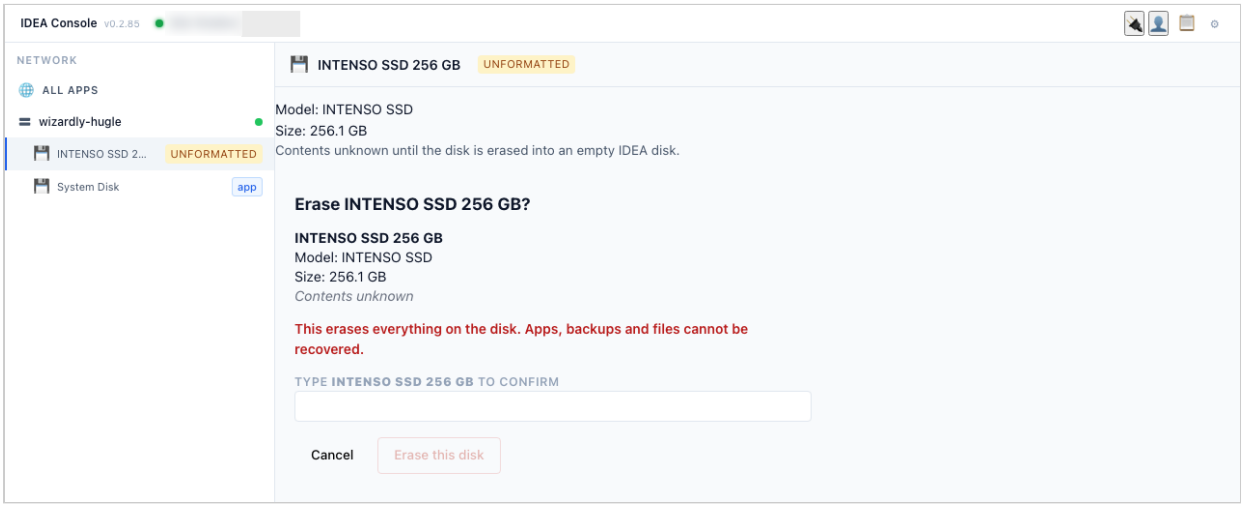


Figure 80. Erase confirmation

4. After the erase, the disk shows as an **IDEA Disk** with the **empty** badge.

> **A Files Disk that Nextcloud is using cannot be erased.** Ask your IDEA contact to stop Nextcloud first.

12.3 Make a Files Disk

1. Select the **IDEA Disk** marked **empty**. The view asks "What would you like to do with this disk?": **Make this a Files Disk**, **Make this a Backup Disk** or **Install App**.

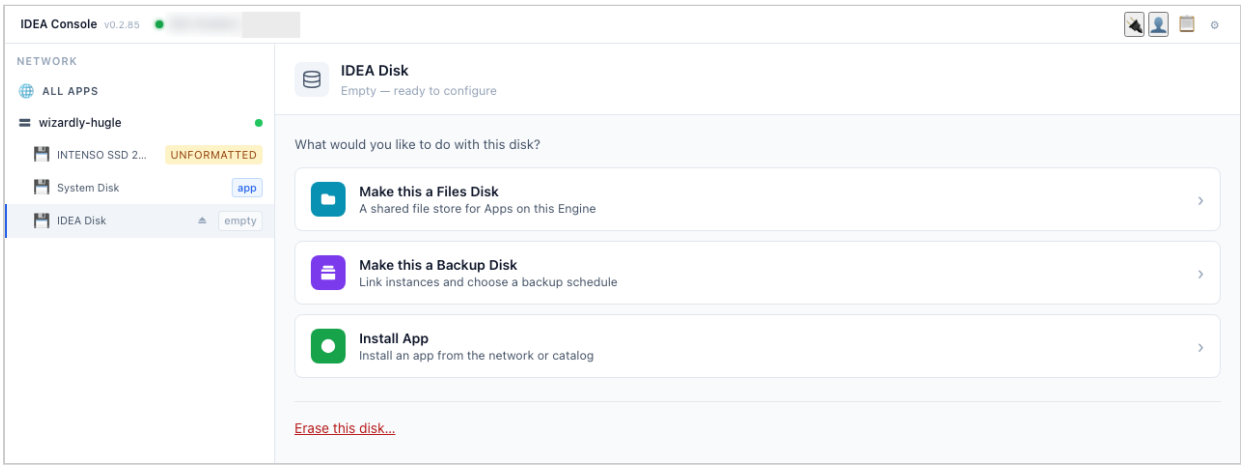


Figure 81. Empty IDEA Disk

2. Click **Make this a Files Disk**. The panel says "Nothing on the disk is erased", with **SHARE NAME** filled in as **School Files**. Click **Make this a Files Disk**.

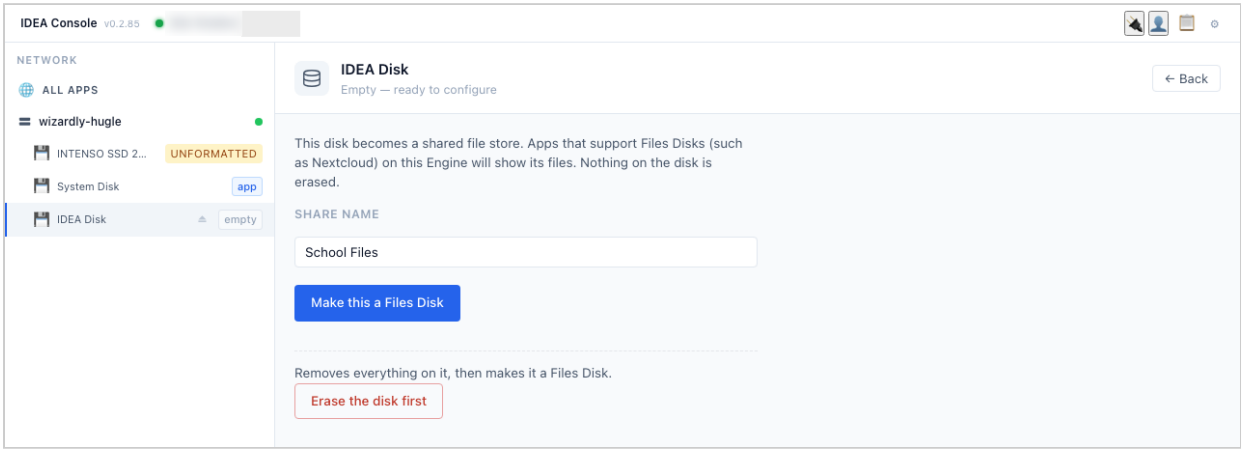


Figure 82. Make this a Files Disk

3. After a short wait the disk shows the **files** badge, "Share name: School Files", the free space, an **Eject** button and **Erase this disk…**. Once your IDEA contact has linked it, the view also says **Available in: Nextcloud**. Teachers then find it in Nextcloud (chapter 9).

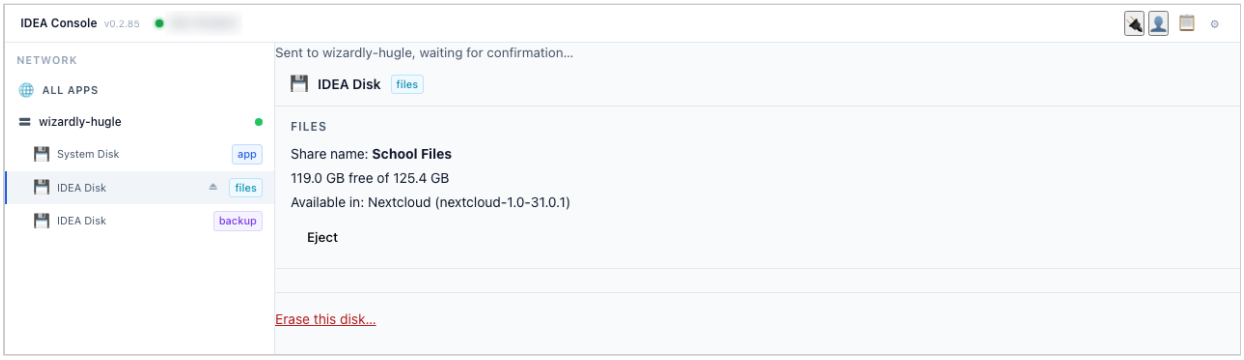


Figure 83. Files Disk available in Nextcloud

12.4 Eject a disk

An IDEA Disk has a small eject icon next to its name in the Network list and an **Eject** button in its disk view. The System Disk has none. Never eject or unplug a disk during a backup.

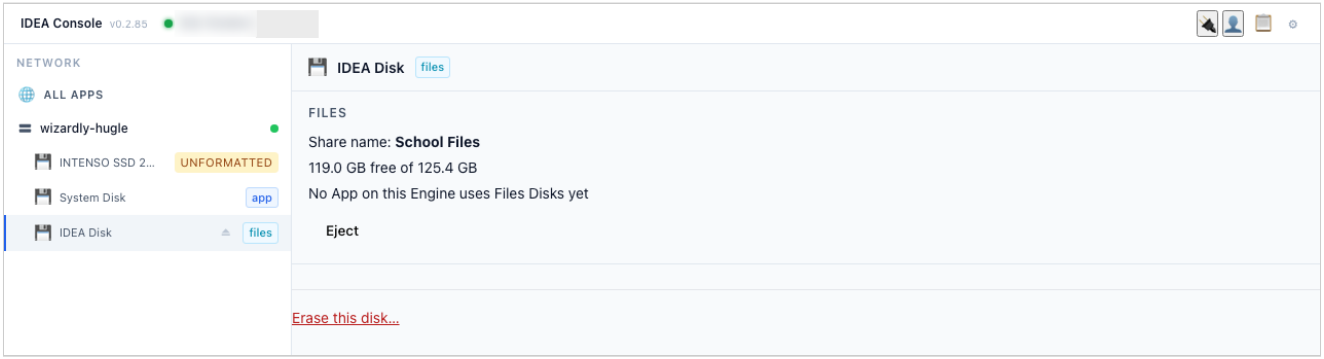


Figure 84. Eject button on a Files Disk

12.5 Make a Backup Disk

1. Select an **IDEA Disk** marked **empty** and click **Make this a Backup Disk**.

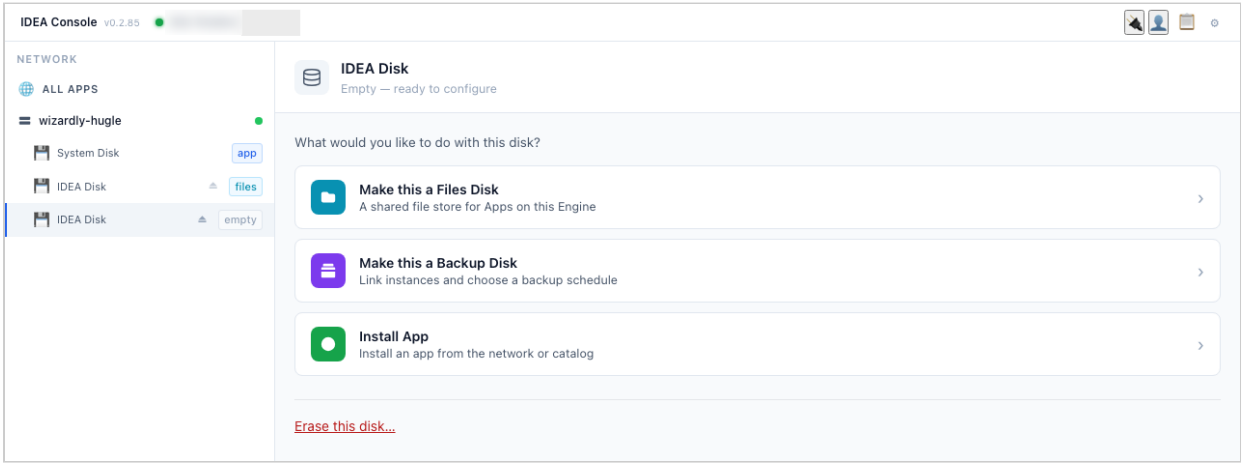


Figure 85. Empty disk options

2. Choose when backups run: **On demand** (only when you press **Back up**), **Immediate** (as soon as the disk is plugged in) or **Scheduled**. Your IDEA contact tells you which.

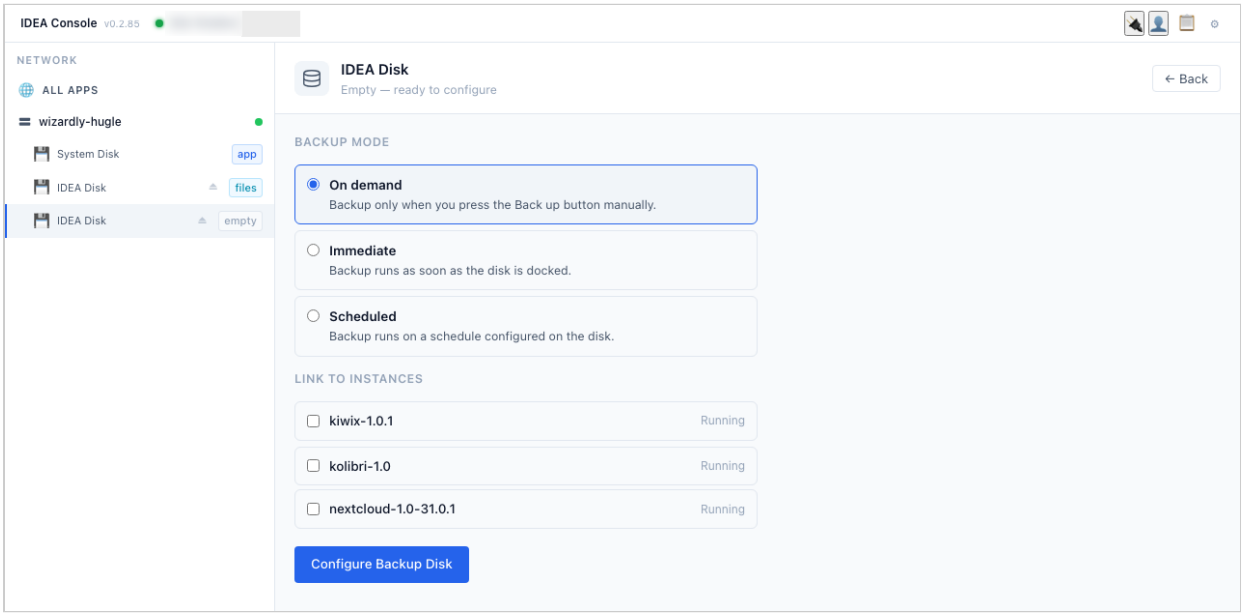


Figure 86. Backup mode

3. Under **LINK TO INSTANCES**, tick only the school apps to protect: Kolibri and Nextcloud.

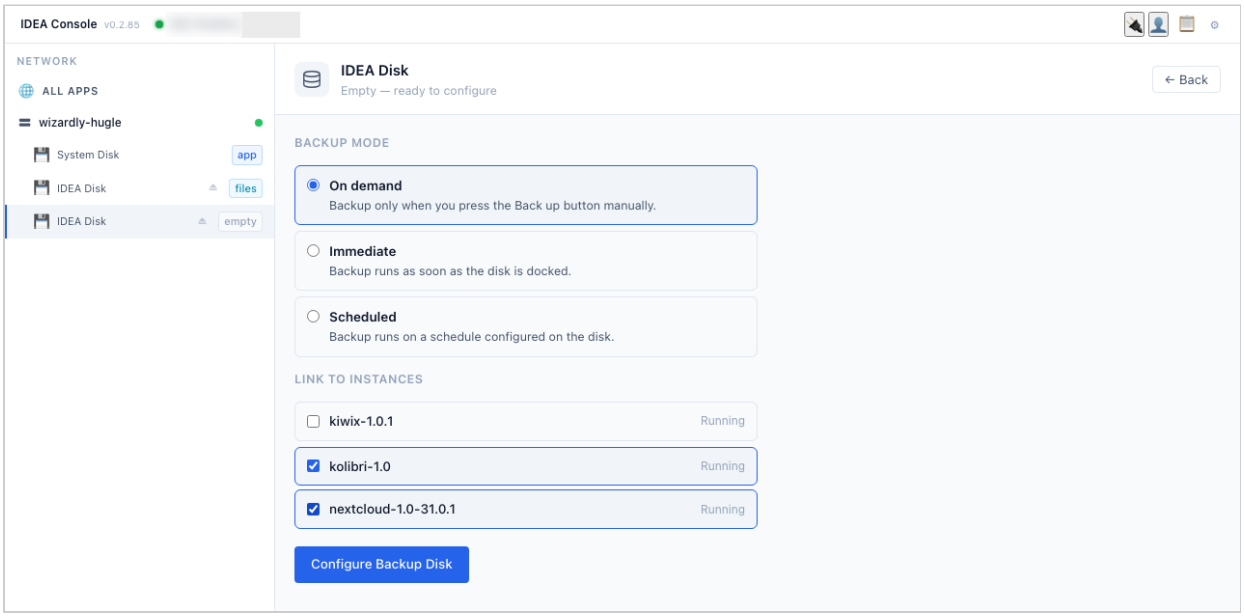


Figure 87. Linked apps

4. Click **Configure Backup Disk**. The disk gets the **backup** badge and a **Backups** list with each linked app.

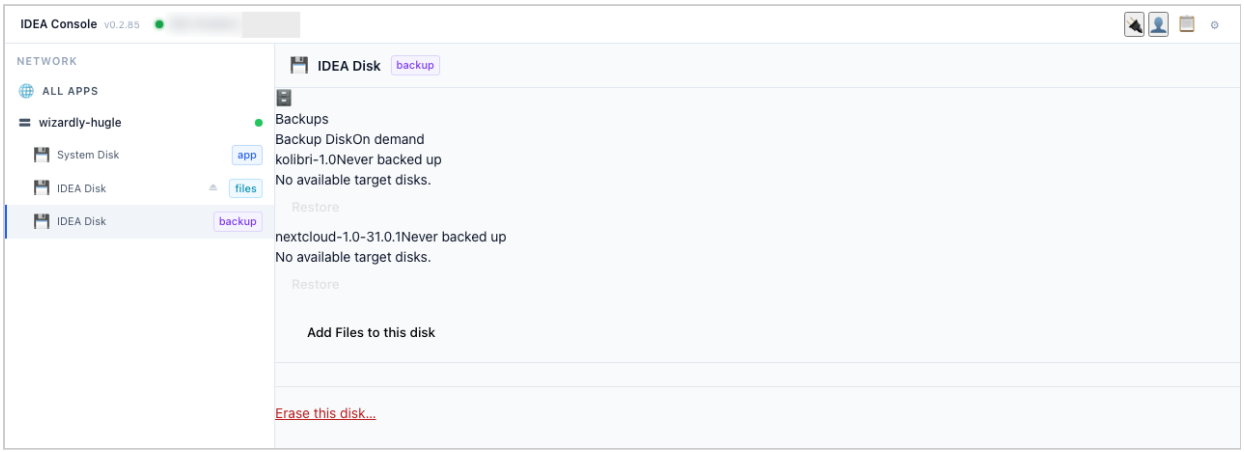


Figure 88. Backup Disk configured

12.6 Run a backup

1. In **ALL APPS**, each linked app row has a **Back up** button next to Start, Stop and Open. Click it for Kolibri. The row shows **Backing up…** with a progress bar, and the **OPERATIONS** box at the top shows the same job.

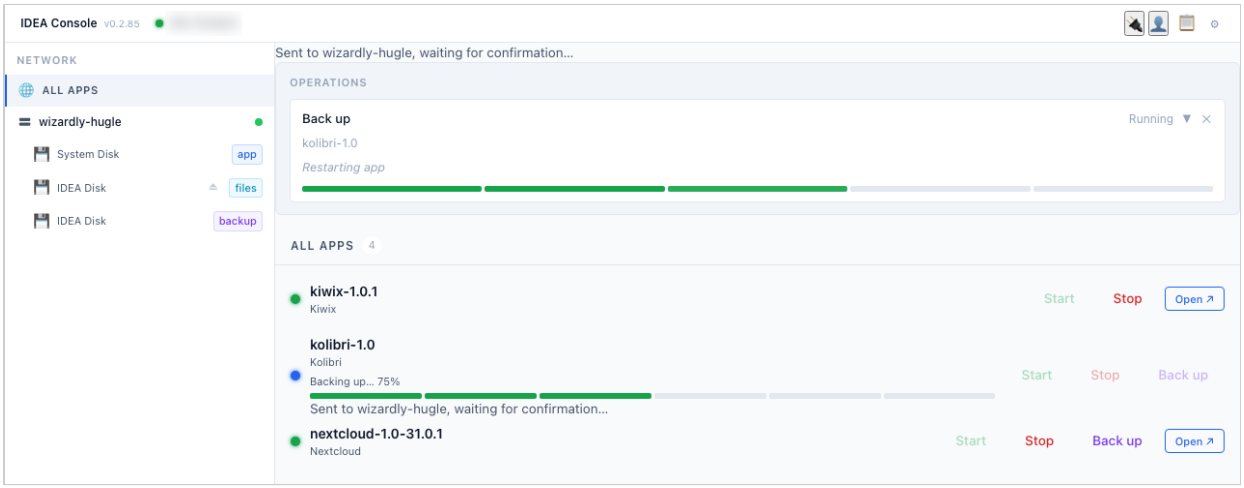


Figure 89. Back up clicked

2. The app stops for a short time while it is copied, then starts again. When the job is finished the OPERATIONS box shows **Done**; close it with ×. A backup takes a few minutes. Do this outside lesson time.
3. When it is done, the Backup Disk's **Backups** list shows **Last backup** with the date and time for each app. Repeat for Nextcloud.

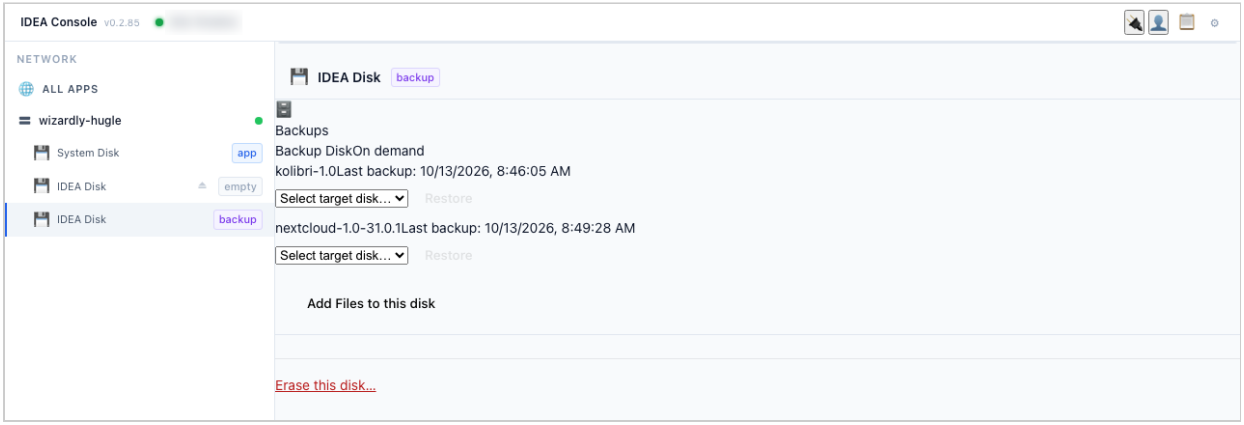


Figure 90. Last backup for both apps

12.7 Restore after a breakdown

You need a **spare disk** (an empty IDEA Disk) to restore onto.

1. Open the **Backup Disk**. Each app in the **Backups** list shows **Last backup**, a **Select target disk…** list and a **Restore** button (grey until a target is chosen).

2. In **Select target disk…** choose the spare disk. Only spare disks are offered here. Click **Restore**.

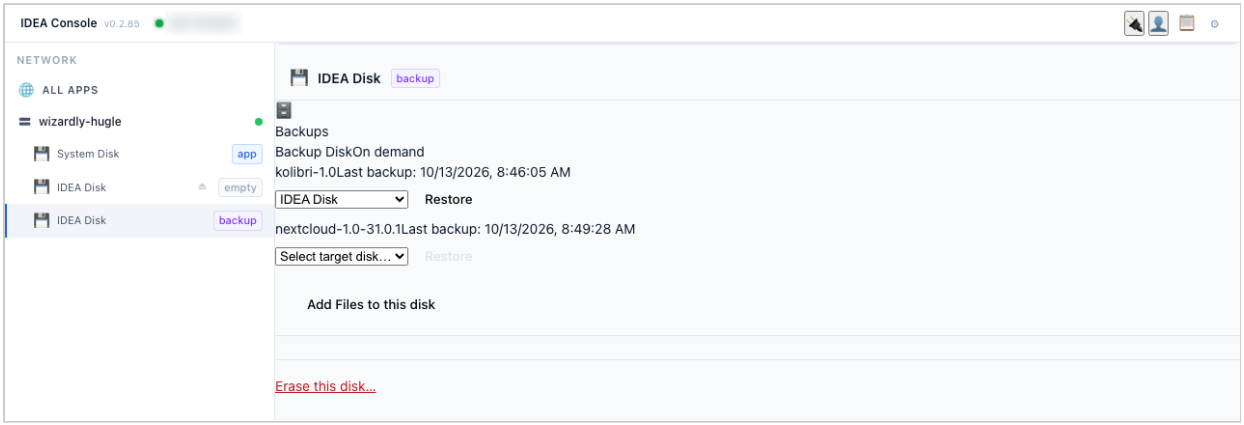


Figure 91. Target chosen

3. The Console asks: "Are you sure? This will overwrite **kolibri-1.0** on **IDEA Disk**." Click **Confirm Restore**.

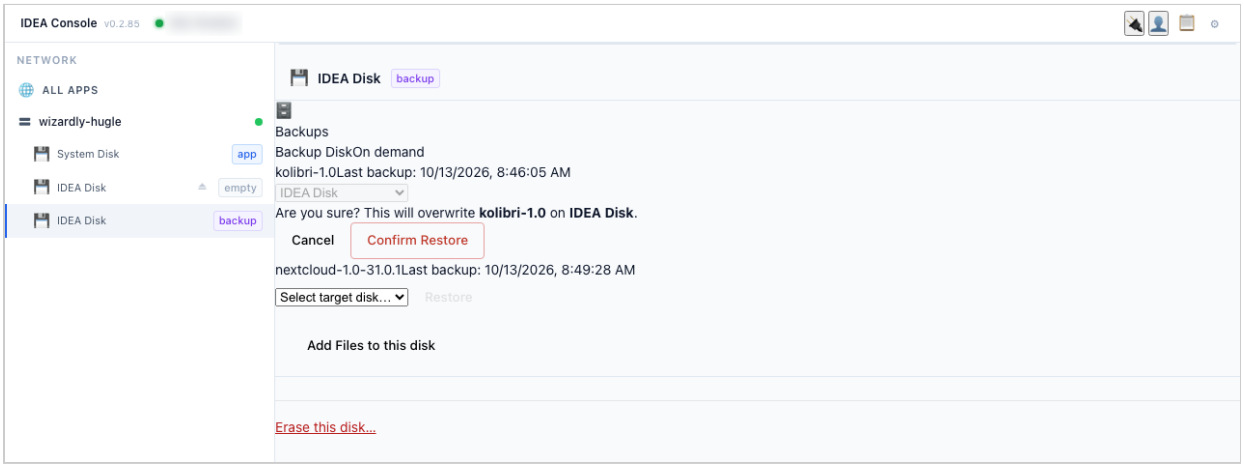


Figure 92. Confirm Restore

4. After about a minute the spare disk shows the **app** badge and lists the restored app.

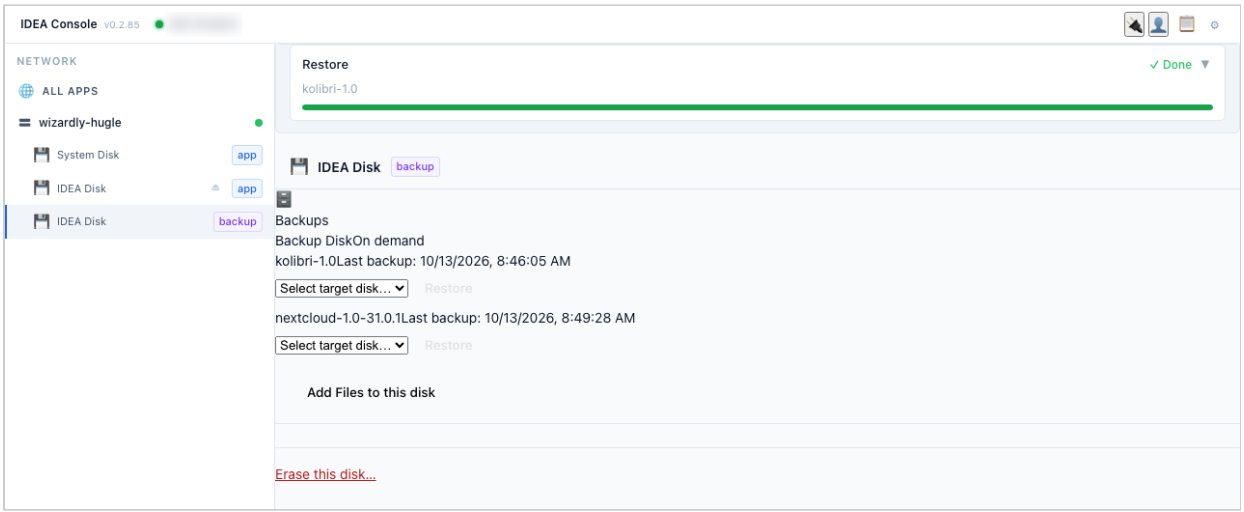


Figure 93. Restore done

5. The restore **moves** the app: it is no longer listed on its old disk, and the restored app is **stopped** (orange dot, **Start** button). Until you start it, the app cannot be opened. A restore is for a real breakdown, not for practice on a working school box.

6. Click **Start** on the restored app's row. After about 30 seconds it runs again and the row shows **Open ↗**.

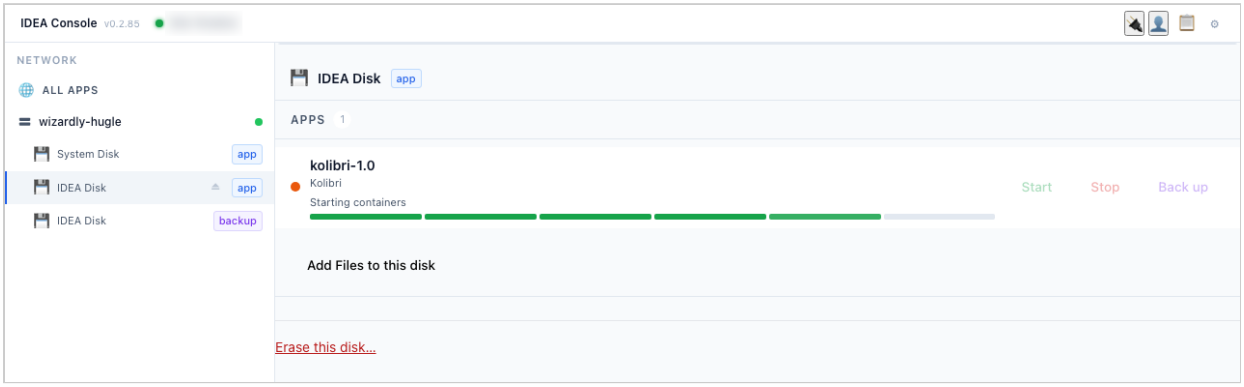


Figure 94. Start clicked

7. Open the app from its card on the **Apps** page (as in Figure 2) and sign in as the teacher (coach).

8. Check that the data is back: classes, learners, lessons, quizzes and reports, as at the last backup.

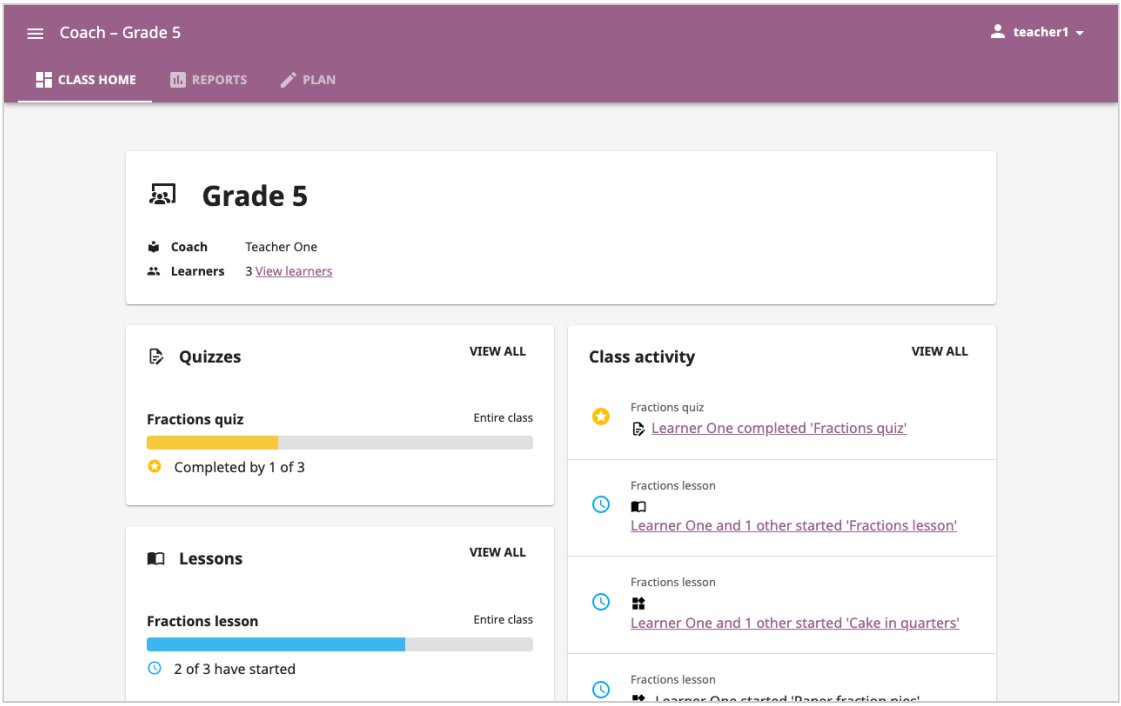


Figure 95. Class after restore

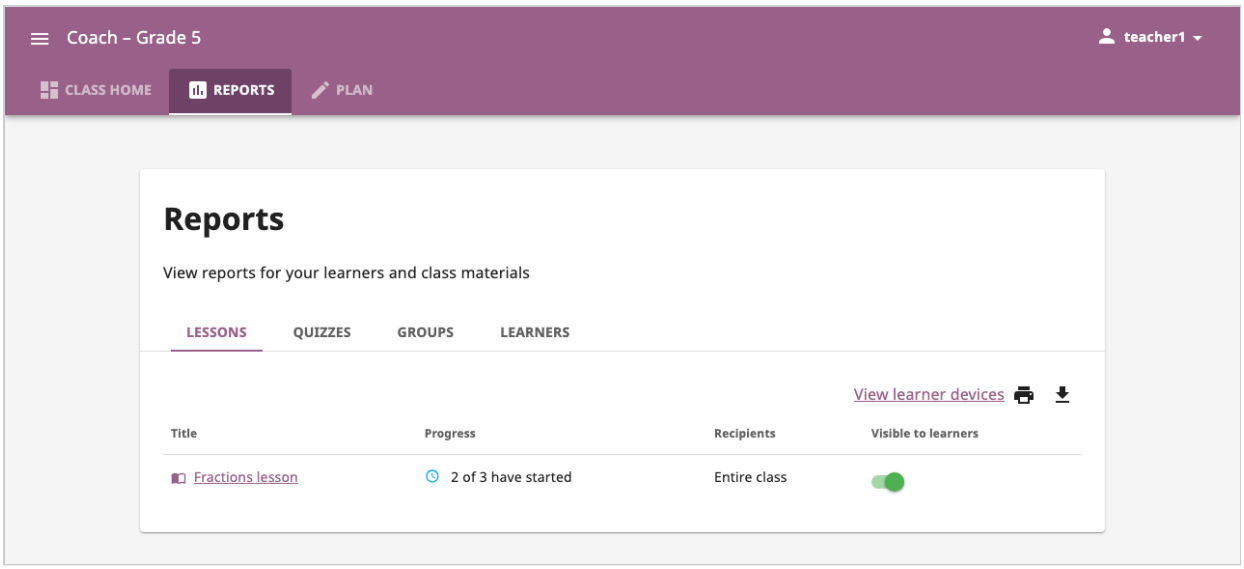


Figure 96. Reports after restore

9. Later, your IDEA contact can move the app back to the System Disk and prepare the spare disk again.

Important. Anything done after the last backup is lost. Make a new backup of the restored app as soon as it runs.

13. Student computers: reset a card from the Console

This chapter is for the school operator, with the IDEA contact on the phone (or the IDEA contact working remotely from the Console).

Every student computer (Raspberry Pi 400 or Pi 500) starts from a **microSD card** with the same IDEA student image. When a computer goes wrong (it no longer starts, the screen looks different, the card is full or a student changed settings), you do not repair it. You **reset the card**: the IDEA box writes a fresh copy of the student image onto it. This takes about 5 to 10 minutes.

Good to know:

- **Resetting wipes the card.** Anything saved on the computer itself (the local Documents folder) is gone. Students' work in **Nextcloud** stays: that is why students save there (chapter 7).
- Every card is the same, so a card works in **any** student computer, Pi 400 or Pi 500.
- The student computers **do not update themselves**. New software comes only with a new, tested image version, which you get by resetting the card.
- You need a **USB card reader** plugged into the IDEA box. Any card reader works.

13.1 Before you start

1. Switch the student computer off and take the microSD card out. Note the computer's name from its sticker (for example `idea-pc-3f2a`).
2. Put the card into the card reader, and plug the card reader into a free USB port of the IDEA box.
3. Open the Console and sign in with **Operator Login** (chapter 11).

13.2 Find the card

1. Click the IDEA box in the list on the left. Scroll down to **CLIENT CARDS**. It shows the client image the box will write, and every card in a card reader: the reader's name, the card's size and what is on it now.

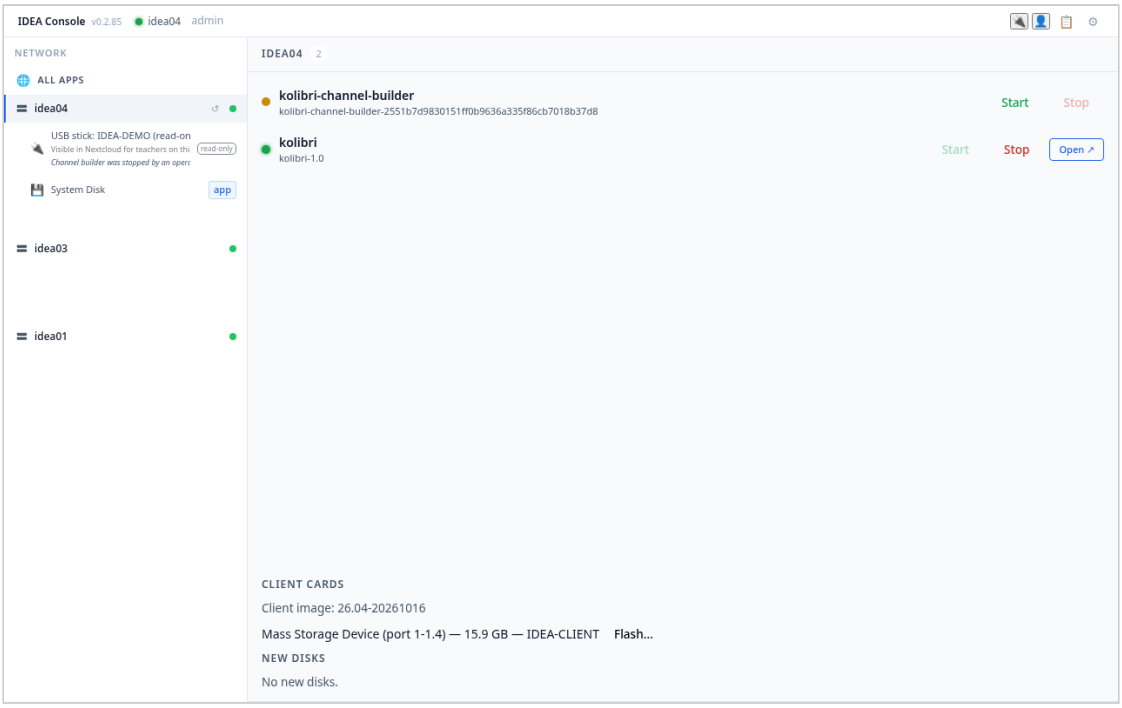


Figure 97. Client cards: one card in a reader, with Flash...

2. Check that the **size** matches your card. If the card is not listed, see 13.5.

> **Never pick an IDEA Disk.** The Files Disk, the Backup Disk and app disks are never offered here, but always check the name and size before you continue.

13.3 Reset (flash) the card

1. The card belongs to the IDEA box you flash it from: its **IDEA** shortcut tries that box first. Flash it from the IDEA box of the classroom it will be used in.
2. Click **Flash...** on the card's row.
3. Read the panel **FLASH CLIENT CARD**. It repeats the card, its size, what is on it now and the image. It says in red that everything on the card will be wiped. If anything looks wrong, click **Back**.

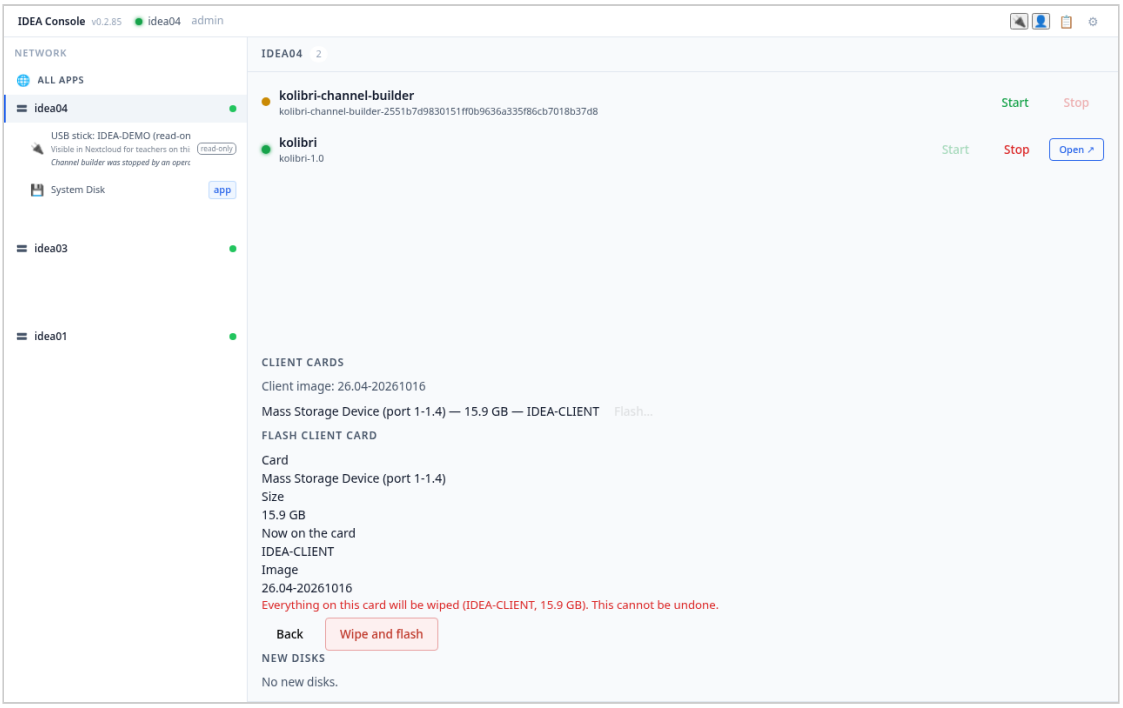


Figure 98. Flash client card: check the card, then Wipe and flash

4. Click **Wipe and flash**. The Console shows the progress in steps, for example **Step 3 of 6: Writing**.

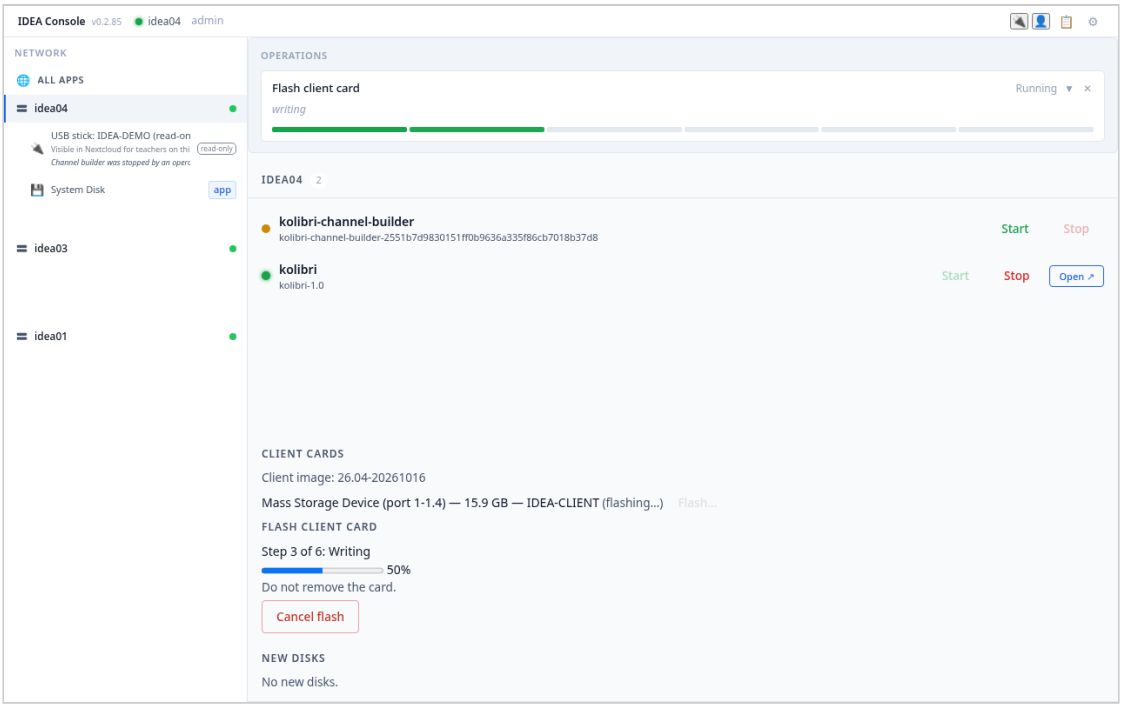


Figure 99. Flash in progress

5. Wait for **Done – safe to remove**. Do not pull the card reader out before that. Then click **Close**.

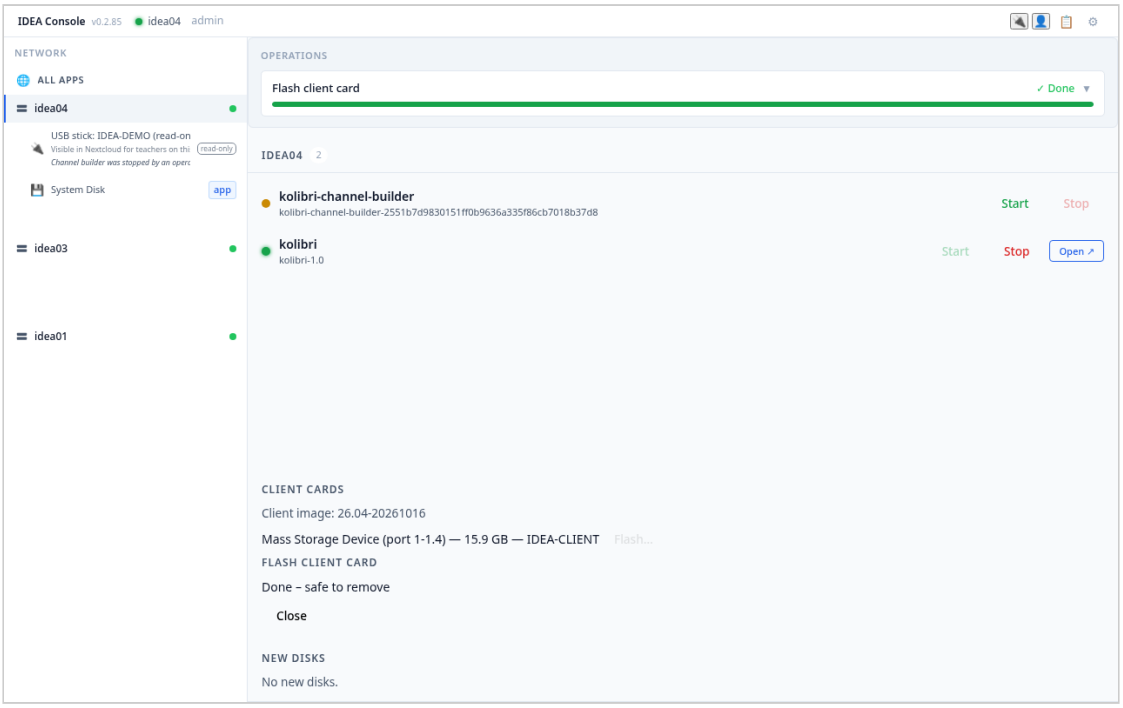


Figure 100. Done – safe to remove

6. Take the card out of the reader and put it back into the student computer. Switch the computer on.

13.4 After the reset

1. The first start takes a few minutes longer: the card sets itself up and names the computer after its Pi (the name on the sticker).
2. The computer opens the student desktop by itself, with no password. The desktop has one shortcut, **IDEA**.

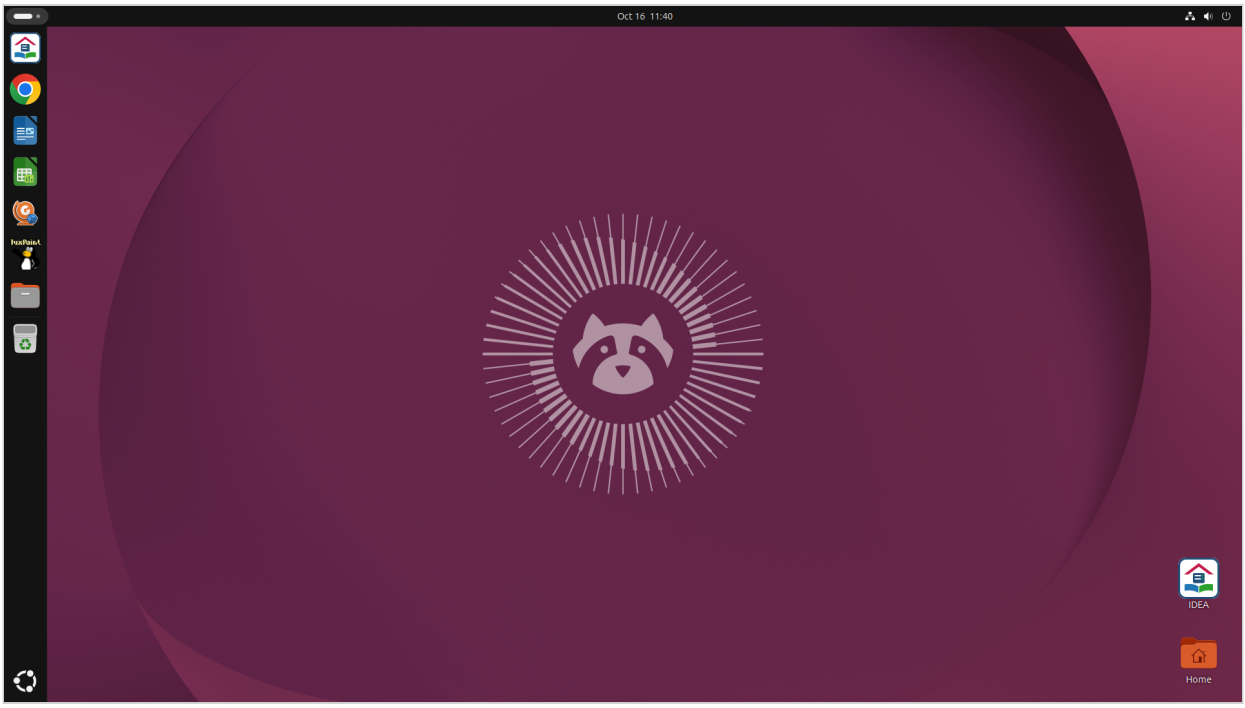


Figure 101. The student desktop after the first start, with the IDEA shortcut

3. Double-click **IDEA**. Chrome opens the start page, finds the IDEA box and opens the Console (chapter 2). Check that **Open** on the Kolibri card works.

13.5 Questions

Question	Answer
The card is not listed.	Check that the card is pushed fully into the reader and that the reader is plugged into **this** IDEA box. Try another USB port.
The Flash button is not there or the Console refuses.	The IDEA box never writes to an IDEA Disk, the system disk, or a disk that is in use. Check that you picked the card and not another disk. Otherwise call your IDEA contact.
The flash stopped with an error.	Flash again. If it fails again, the card is probably worn out: use a new card.
The computer does not start after the reset.	Try the card in another student computer. If it starts there, the computer may need a bootloader update: call your IDEA contact.
Can I install an extra program on one computer?	No. It would be lost at the next reset. Ask your IDEA contact to add it to the next image version.
The IDEA shortcut says No IDEA server found.	Check the network connection. Type the name on the IDEA box's sticker in the field on the start page (chapter 2). If it keeps happening, the card may belong to an IDEA box that has been switched off or moved: flash it again from the right box.
Can my IDEA contact do this remotely?	Yes. Put the card in the reader and tell them. They press Flash from the Console, and you put the card back when the Console says it is done.

Appendix: demo accounts

These are the accounts of the demo school **IDEA Demo School**, used for the screenshots. In Kolibri, learners have no password: they sign in with only their username. Every other demo password follows the IDEA default: the account name followed by `911!` (for example `teacher1911!`). It is easy to remember and the same in every school, so change it before real use (3.1, 11.1). The same usernames are used in Kolibri and Nextcloud.

Username	Password	App	Role	Used in
admin	admin911!	Kolibri	Kolibri admin of IDEA Demo School	3.2, 3.3, 5.2
teacher1	teacher1911!	Kolibri	Coach of class Grade 5	3, 12.7
learner1	none (username only)	Kolibri	Learner in Grade 5	4
learner2	none (username only)	Kolibri	Learner in Grade 5	4
learner3	none (username only)	Kolibri	Learner in Grade 5	3.3
teacher1	teacher1911!	Nextcloud	Teacher (groups Grade 5, teachers)	5.5, 6, 9
teacher2	teacher2911!	Nextcloud	Teacher (group teachers)	6.2
learner1	learner1911!	Nextcloud	Learner (group Grade 5)	7, 9.2
learner2	learner2911!	Nextcloud	Learner (group Grade 5)	7.1

Demo content: Kolibri school **IDEA Demo School**, class **Grade 5**, one lesson per channel of the IDEA-DEMO disk (**Science: plants, animals, earth and sky**, **Reading: African Storybook stories**, **Maths: fractions and multiplication**), plus the **Fractions lesson** and **Fractions quiz** you make in chapter 3; Nextcloud folders **Worksheets** and **Drop Zone**, document **Lesson plan**. The pictures on the IDEA-DEMO disk are from Zimbabwe for now; see `CREDITS.txt` on the disk for every file's source and licence.

Appendix: activating the demo setup

The demo setup fills Kolibri on an IDEA box with example data. It is for demo IDEA boxes and training only. Do not use it on a school's real IDEA box: it adds **IDEA Demo School** and makes it the main Kolibri school on that box.

What it creates in Kolibri:

- the school **IDEA Demo School**
- the class **Grade 5**
- the admin account `admin`, the teacher `teacher1` (coach of Grade 5) and the learners `learner1` to `learner3`. Learners have no password: they sign in with only their username.

The accounts are listed in Appendix: demo accounts. Lessons are not created: the teacher makes them, as in chapter 3.

Who turns it on. Your IDEA contact. It is not a button in the Console. It is a script that comes with the Kolibri app (`demo/provision-demo.sh` in the Kolibri app).

How it works:

1. The IDEA contact stops Kolibri on the IDEA box.
2. They run the demo script once, on the Kolibri app's data.
3. They start Kolibri again.
4. Kolibri now has IDEA Demo School, Grade 5 and the demo accounts.

You can run the script again safely. It deletes nothing and keeps passwords that were already changed.

Channels. The Maths, Grade 5 Science and Early Reading channels come from the **IDEA-DEMO** disk, through the channel builder (chapter 5).